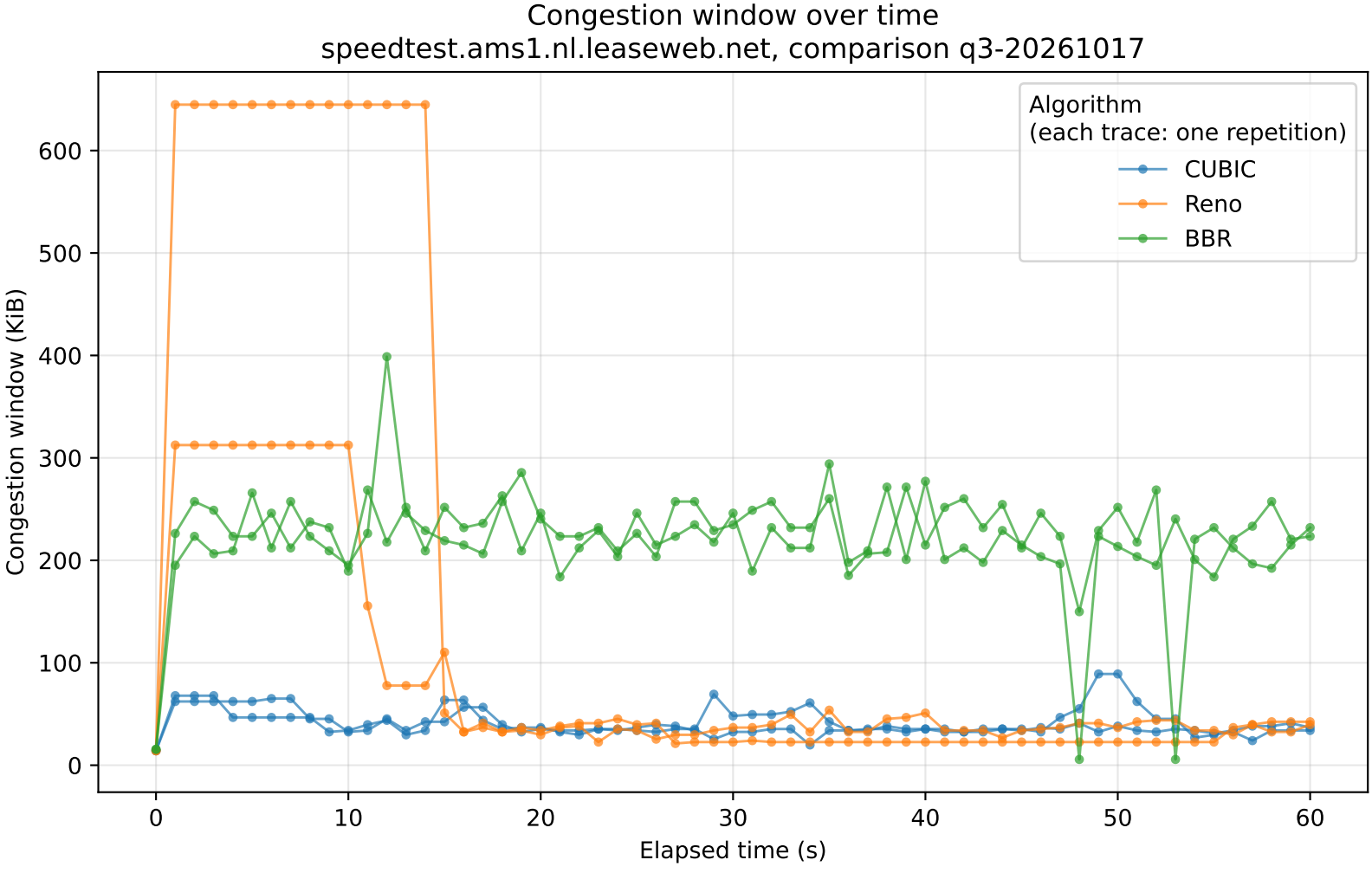
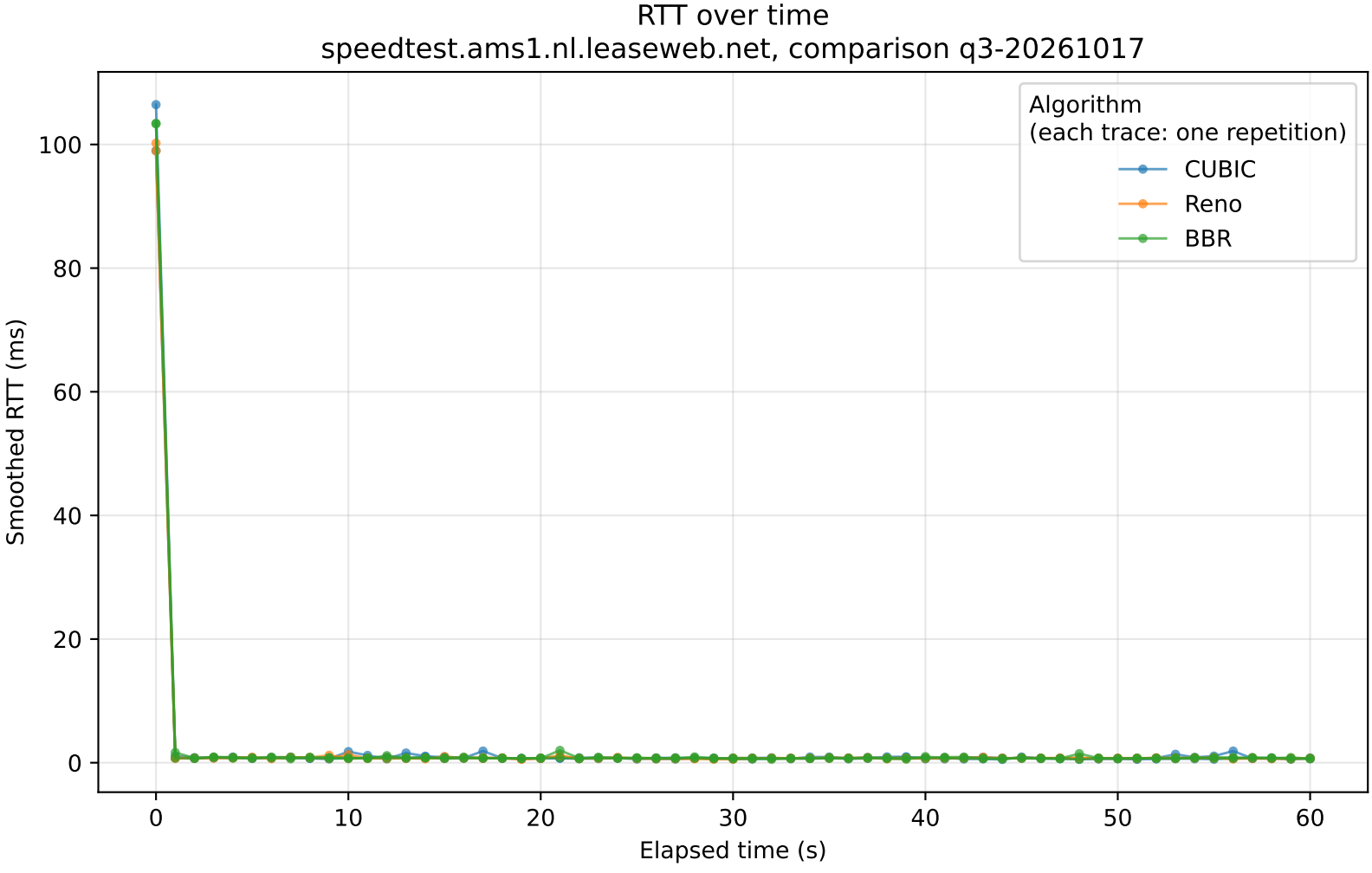
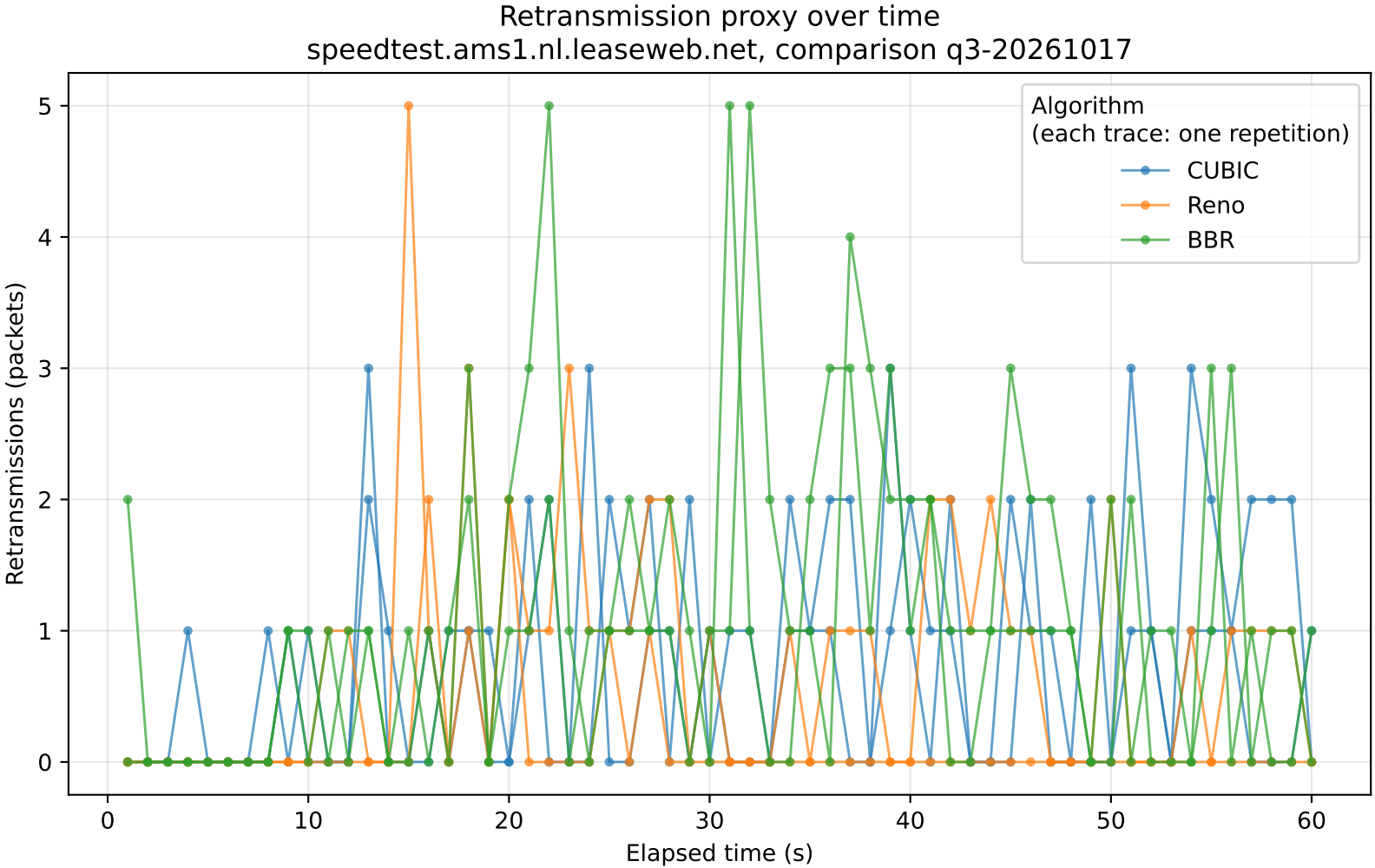
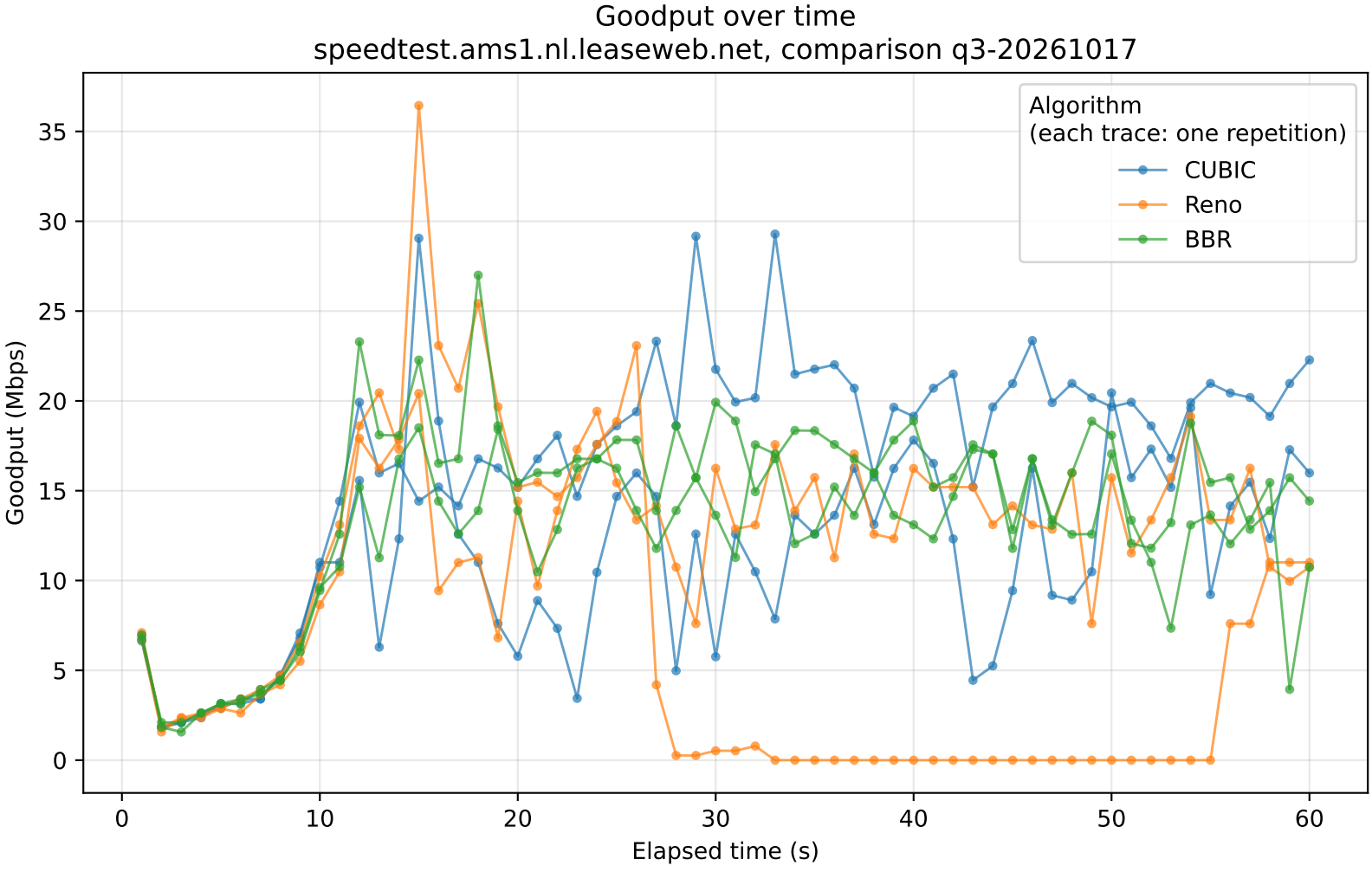
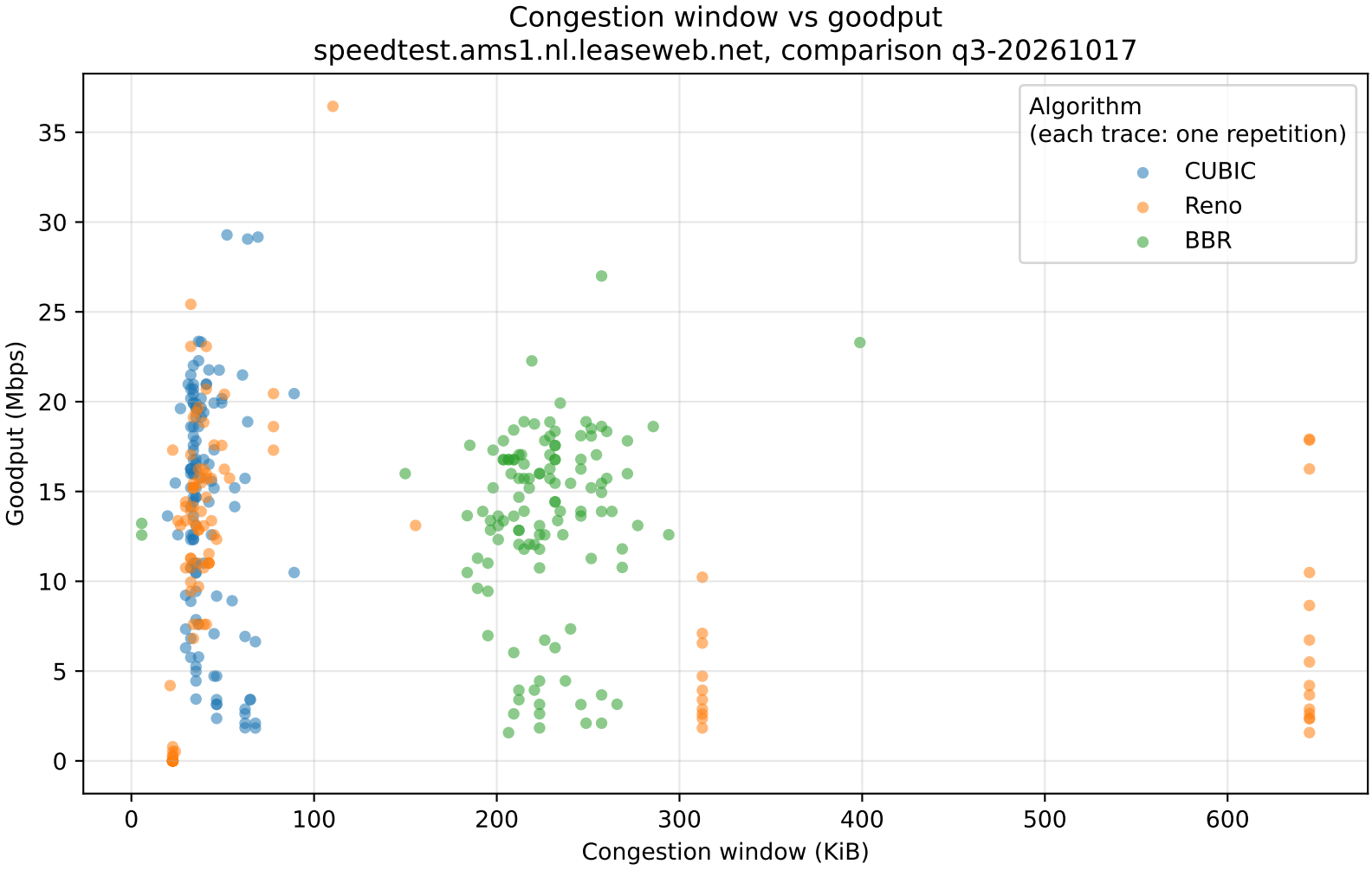
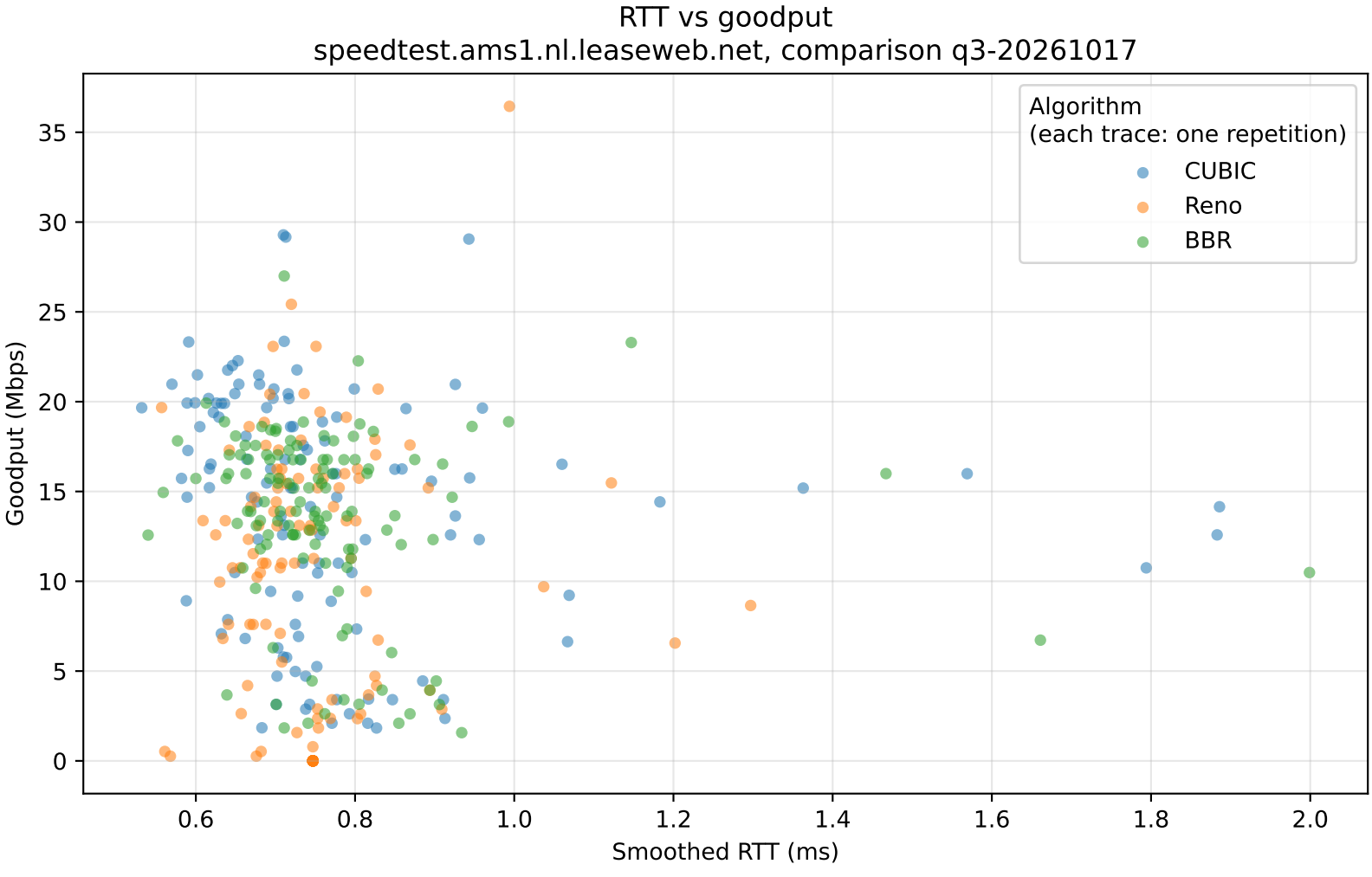

Congestion window over time
speedtest.ams1.nl.leaseweb.net, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
Reno
BBR
Congestion window (KiB)
0
100
200
300
400
500
600
Elapsed time (s)
0
10
20
30
40
50
60

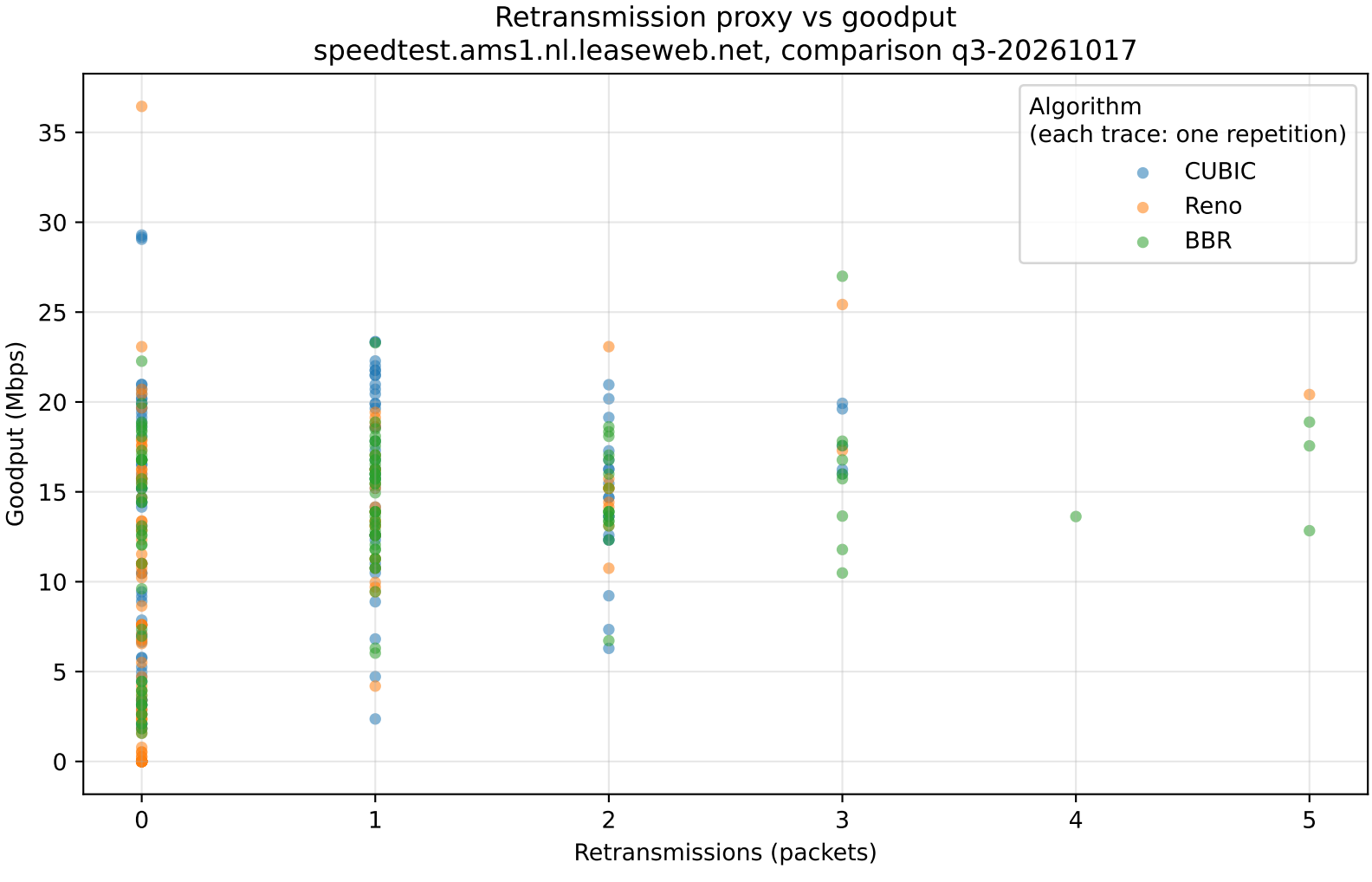

RTT over time
speedtest.ams1.nl.leaseweb.net, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
Reno
BBR
Smoothed RTT (ms)
0
20
40
60
80
100
0
10
20
30
40
50
60
Elapsed time (s)

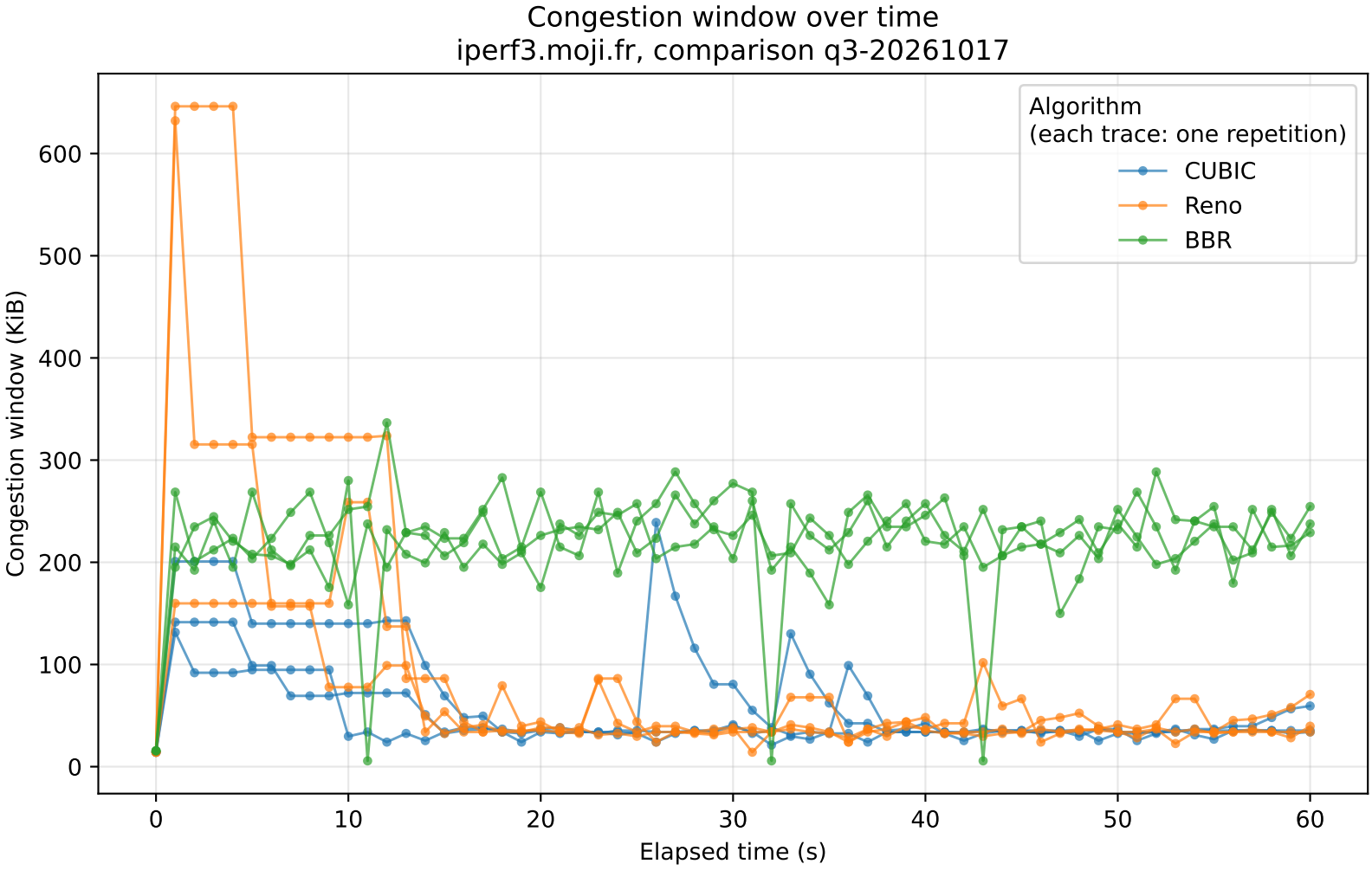

Retransmission proxy over time
speedtest.ams1.nl.leaseweb.net, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
Reno
BBR
Retransmissions (packets)
Elapsed time (s)
0
1
2
3
4
5
0
10
20
30
40
50
60

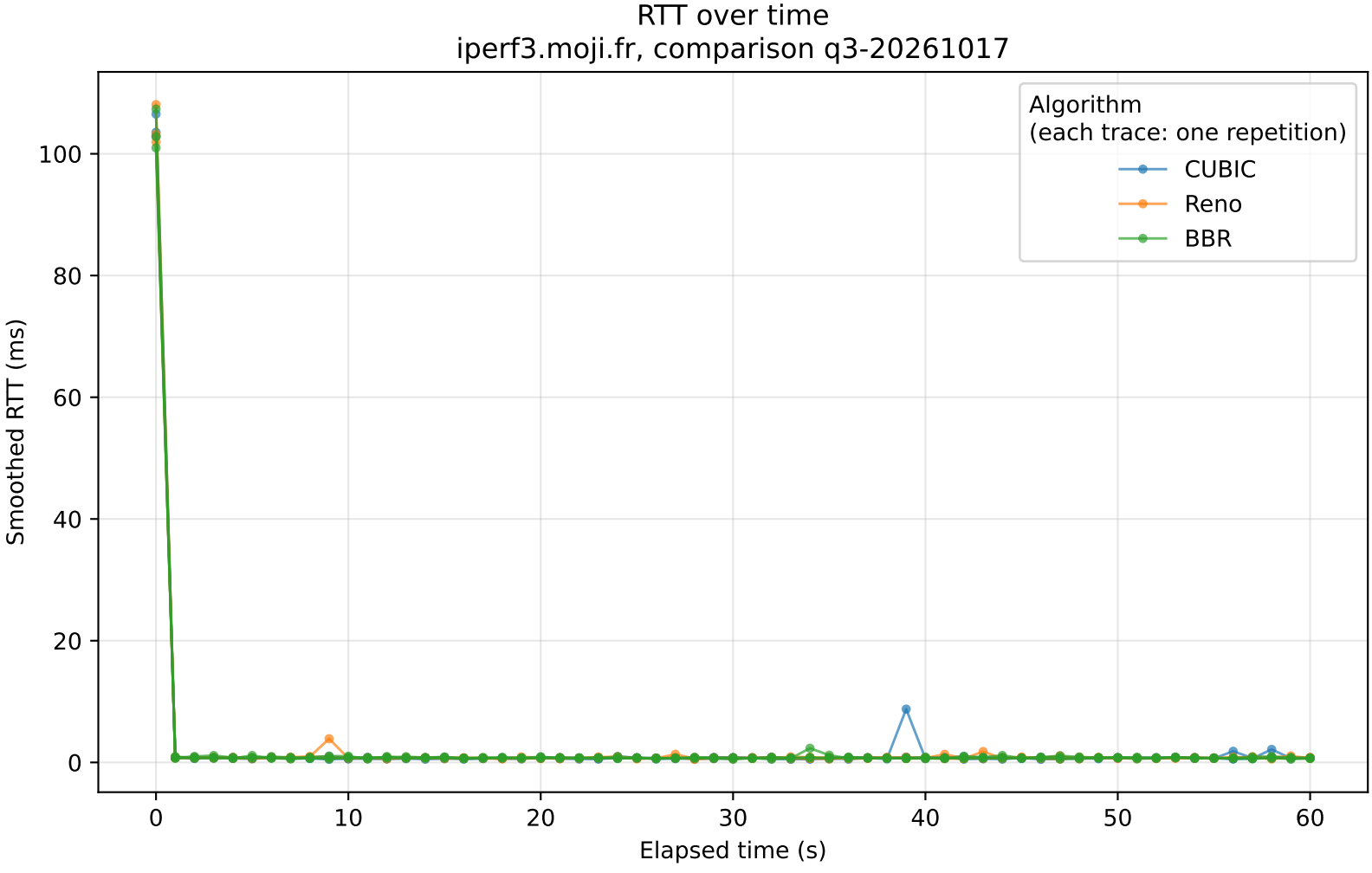

Goodput over time
speedtest.ams1.nl.leaseweb.net, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
Reno
BBR
Goodput (Mbps)
Elapsed time (s)
0
5
10
15
20
25
30
35
0
10
20
30
40
50
60

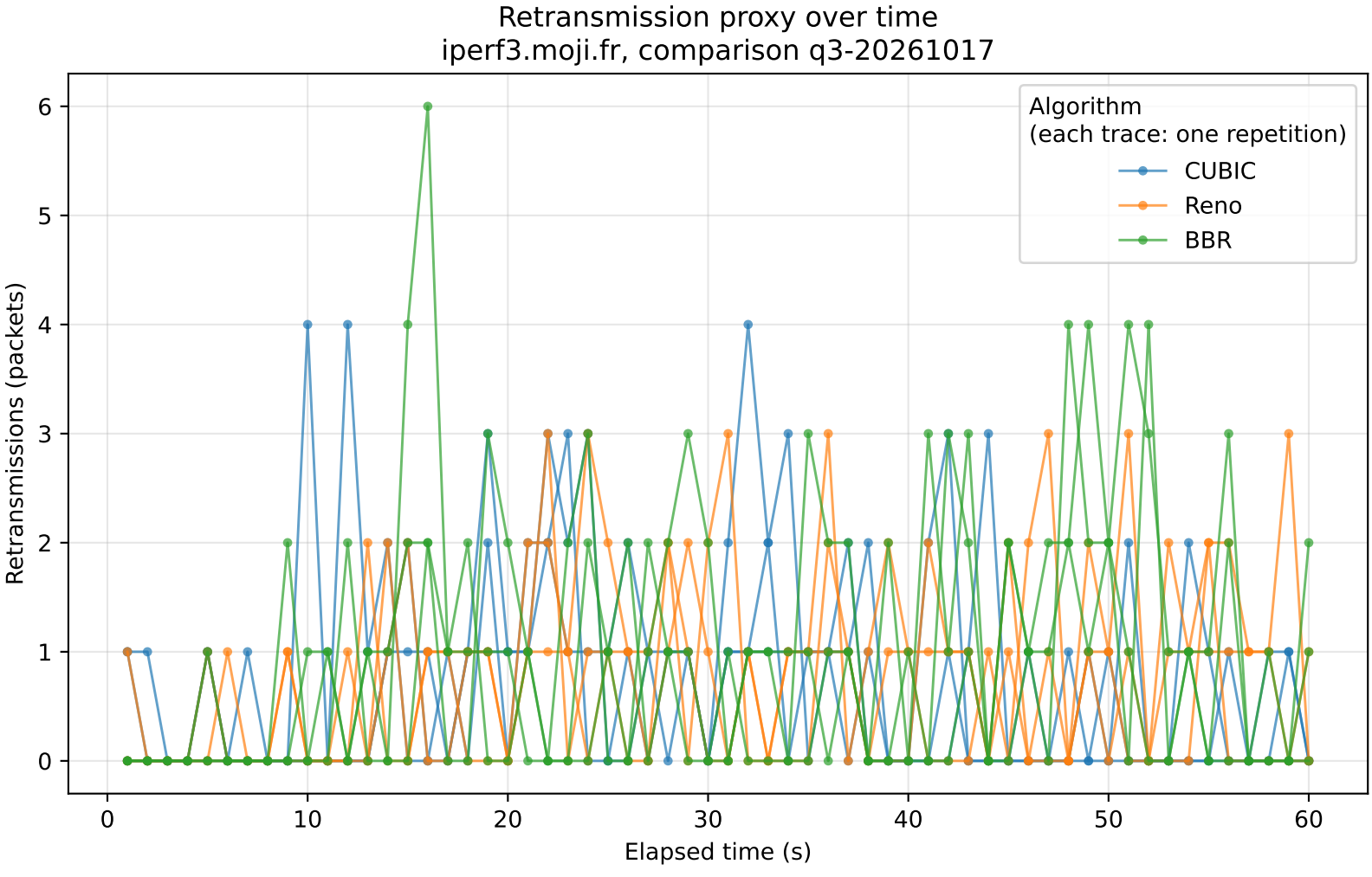

Congestion window vs goodput
speedtest.ams1.nl.leaseweb.net, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
Reno
BBR
Goodput (Mbps)
0
5
10
15
20
25
30
35
0
100
200
300
400
500
600
Congestion window (KiB)

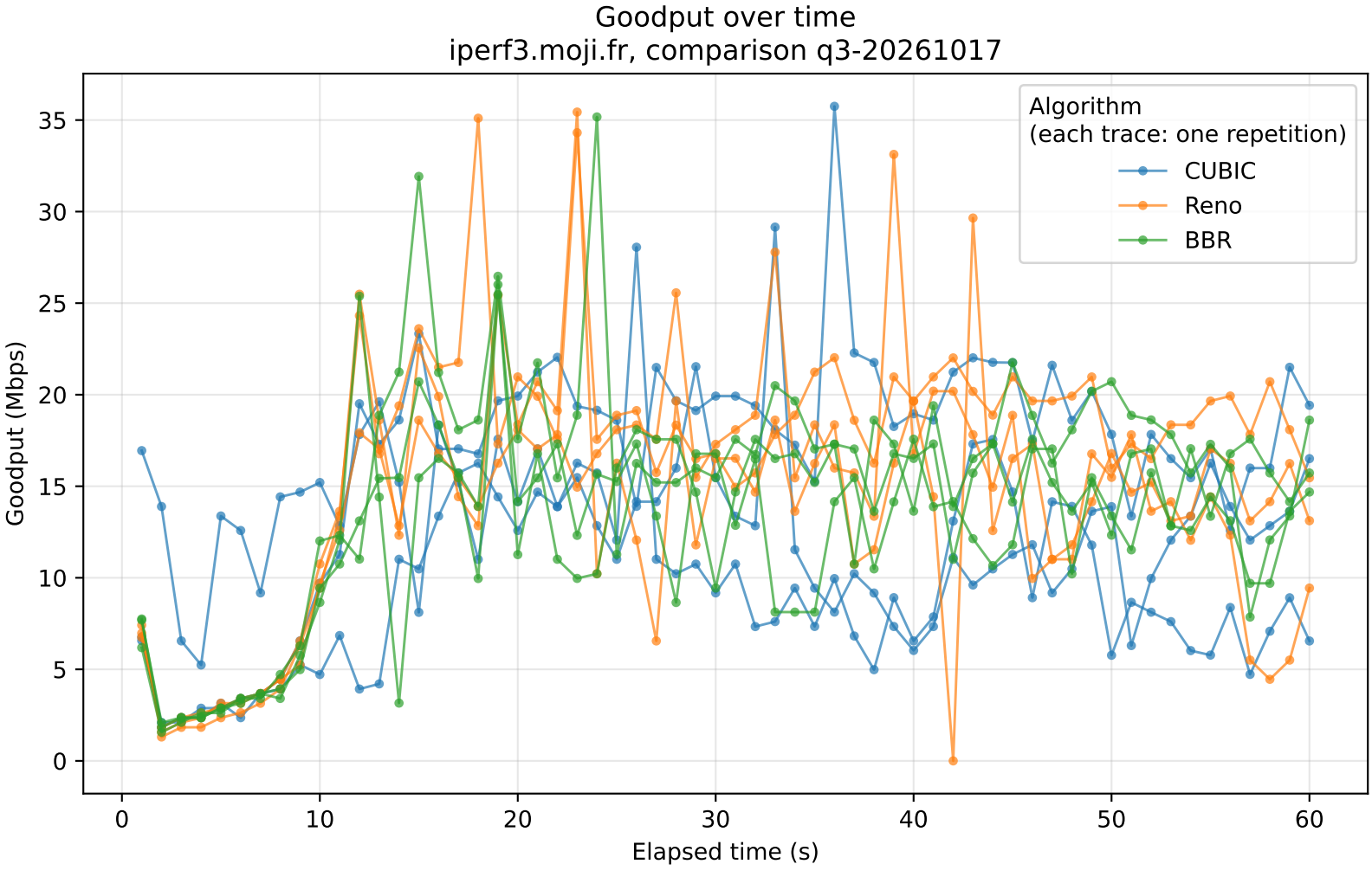

RTT vs goodput
speedtest.ams1.nl.leaseweb.net, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
Reno
BBR
Goodput (Mbps)
0
5
10
15
20
25
30
35
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
Smoothed RTT (ms)

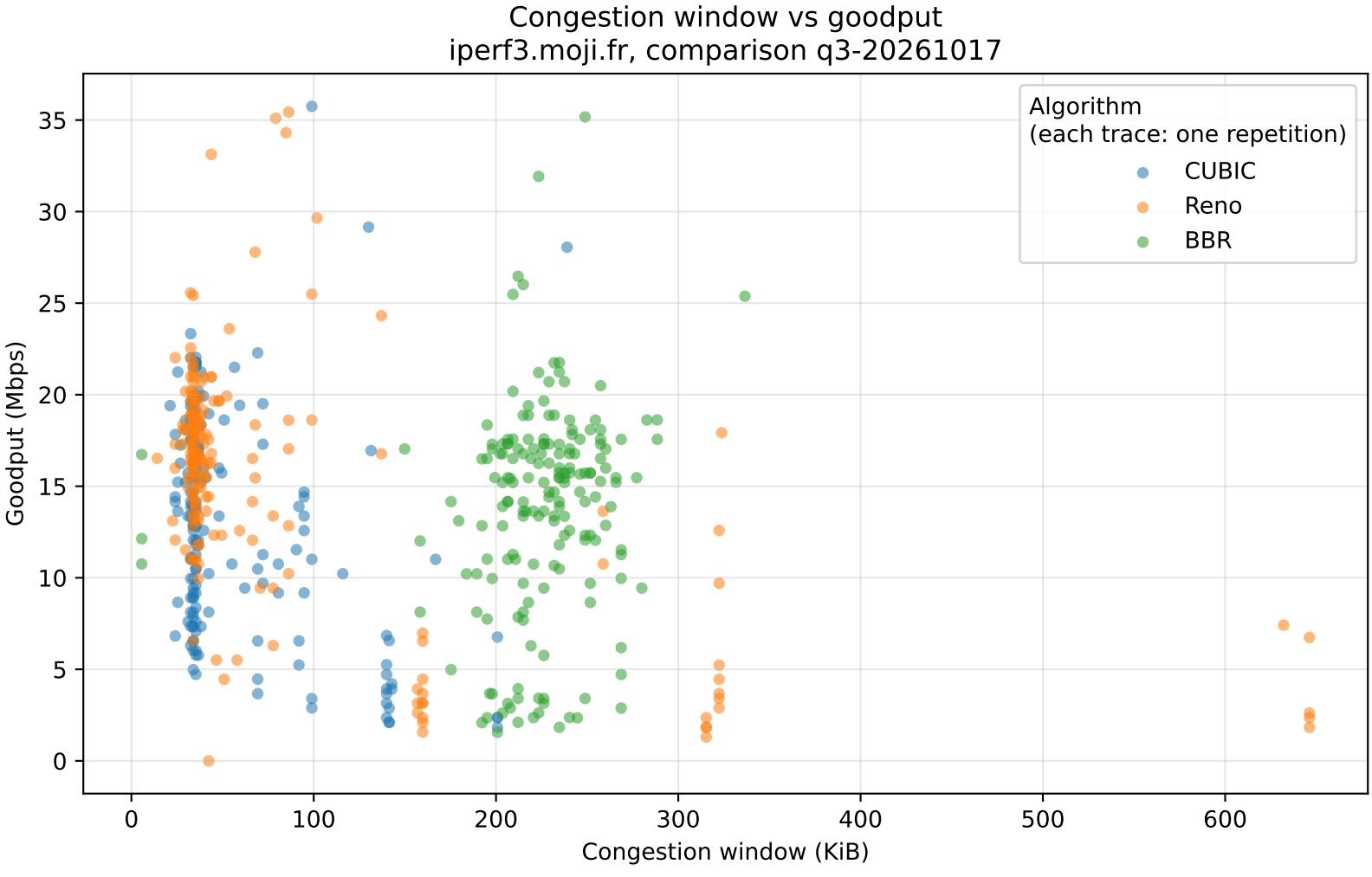

Retransmission proxy vs goodput
speedtest.ams1.nl.leaseweb.net, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
Reno
BBR
Goodput (Mbps)
0
5
10
15
20
25
30
35
Retransmissions (packets)
0
1
2
3
4
5

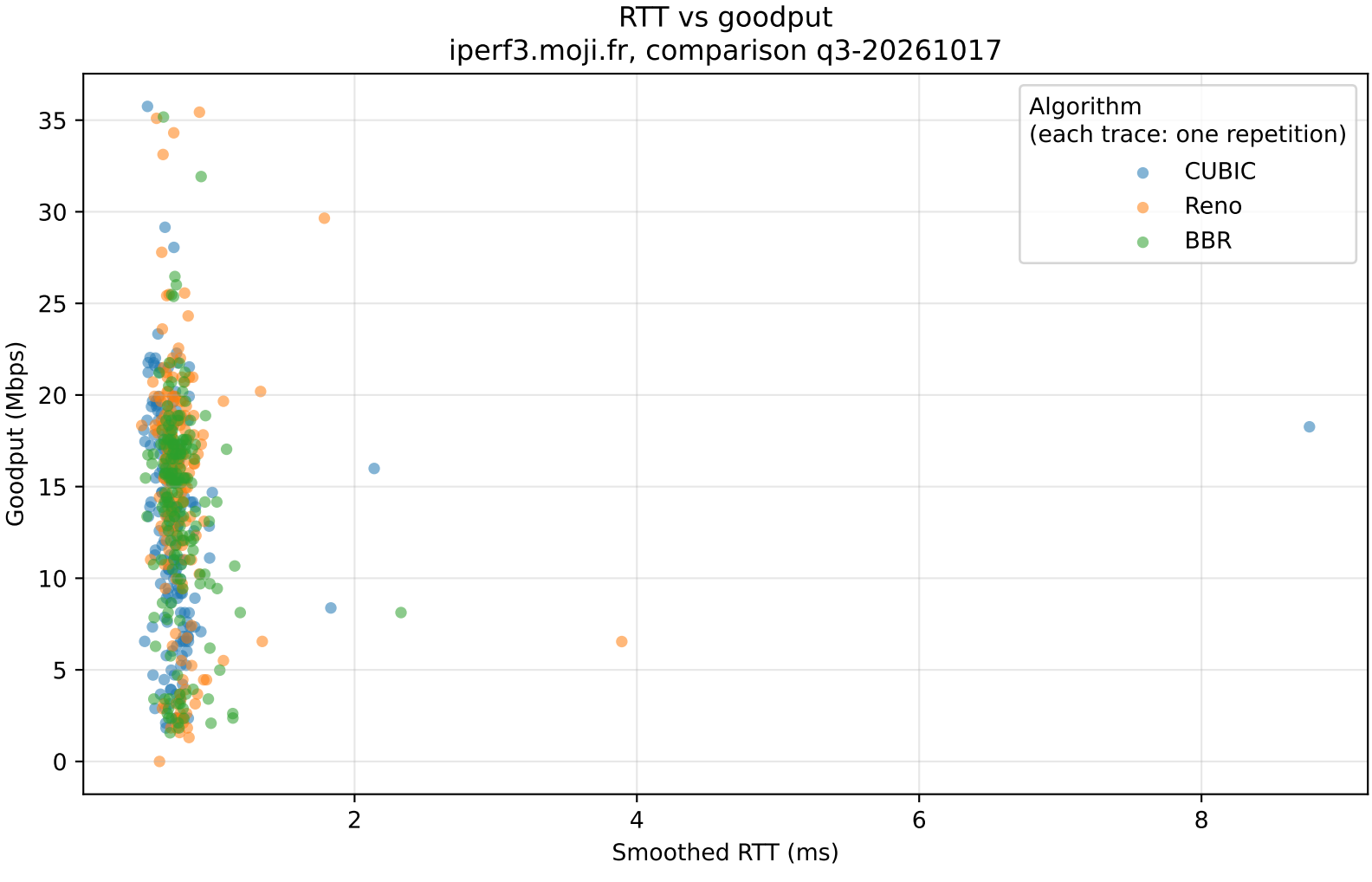

Congestion window over time
iperf3.moji.fr, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
Reno
BBR
Congestion window (KiB)
0
100
200
300
400
500
600
Elapsed time (s)
0
10
20
30
40
50
60

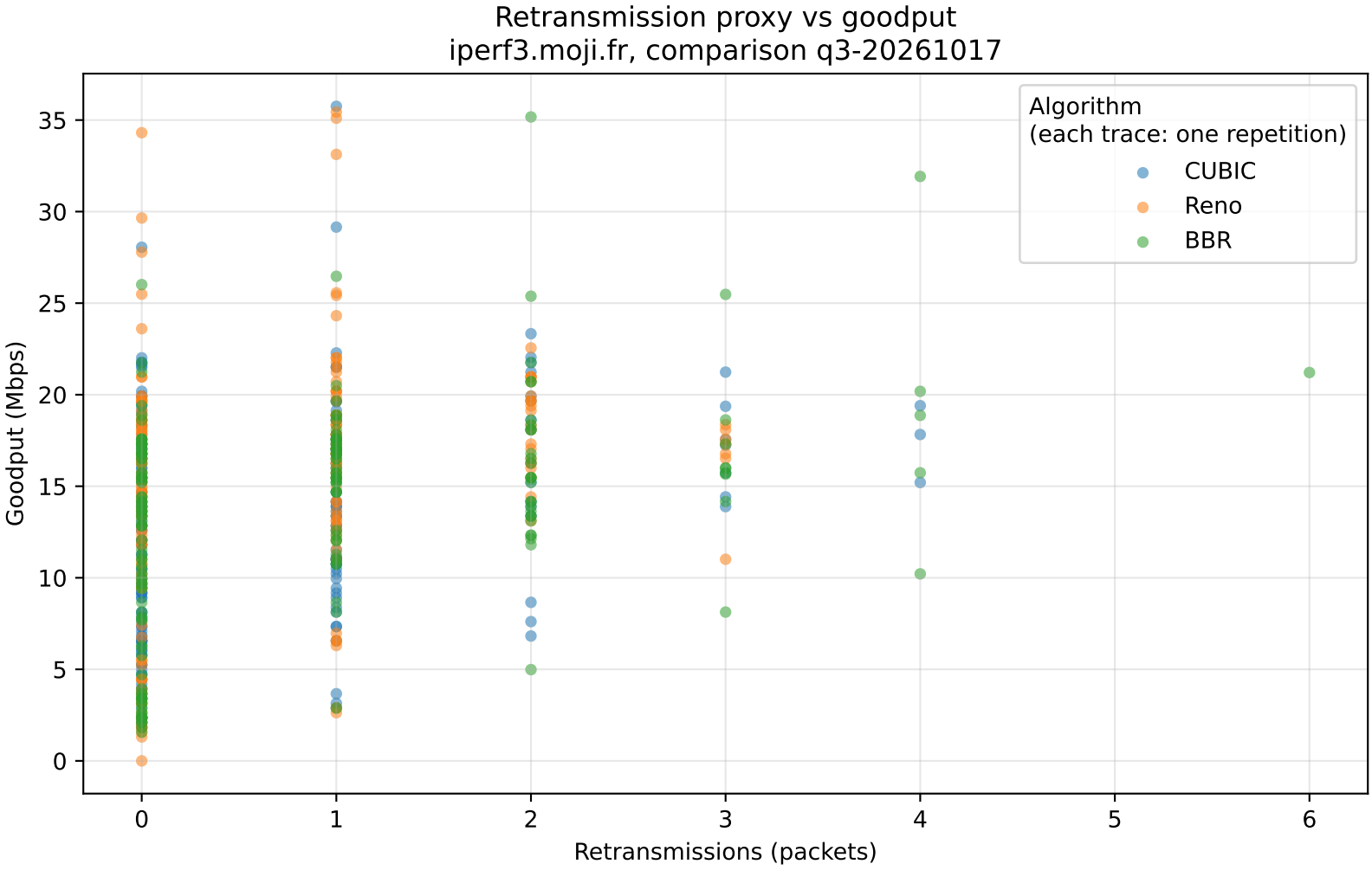

RTT over time
iperf3.moji.fr, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
Reno
BBR
Smoothed RTT (ms)
0
20
40
60
80
100
0
10
20
30
40
50
60
Elapsed time (s)

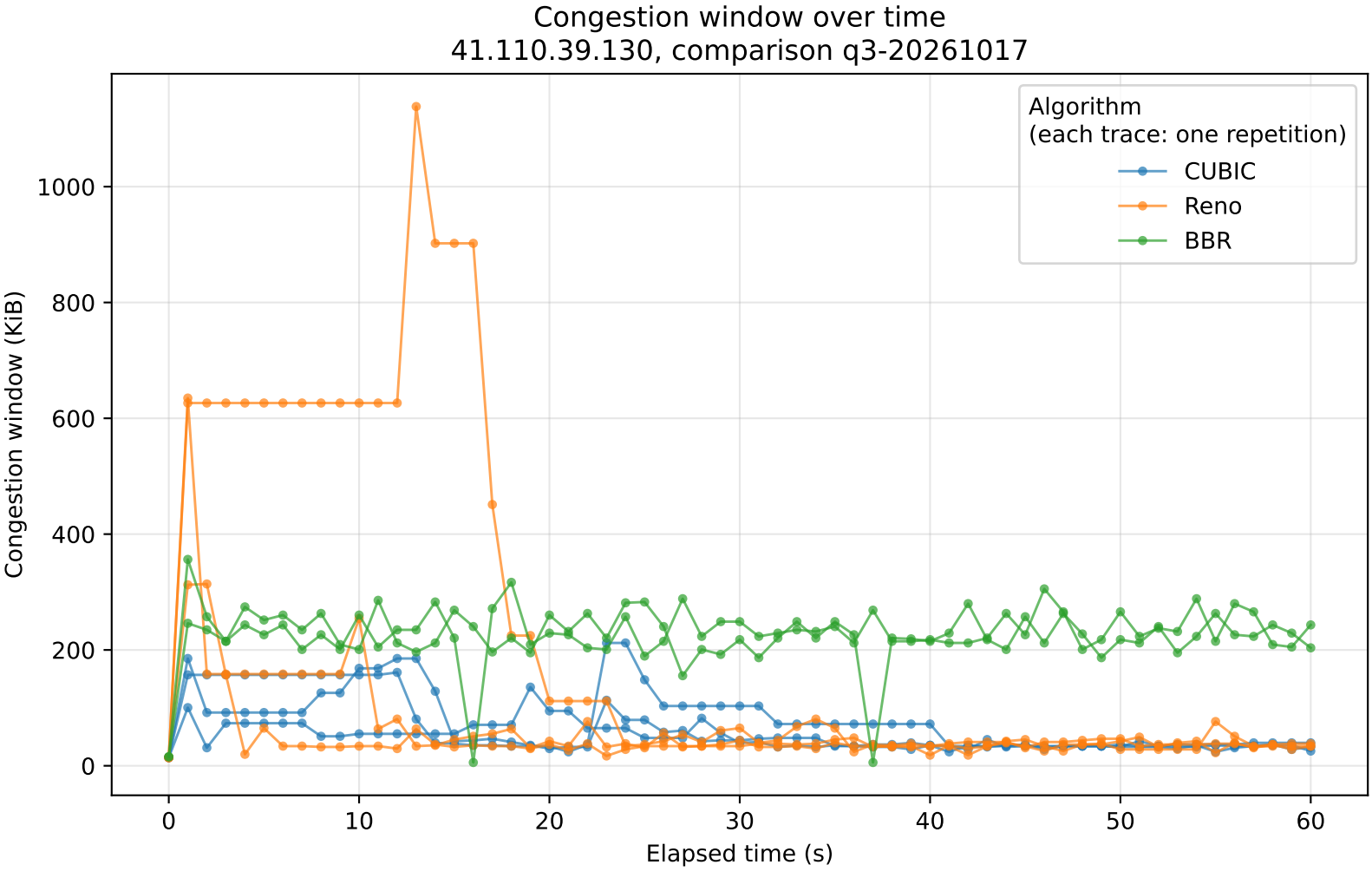

Retransmission proxy over time
iperf3.moji.fr, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
Reno
BBR
Retransmissions (packets)
Elapsed time (s)
0
1
2
3
4
5
6
0
10
20
30
40
50
60

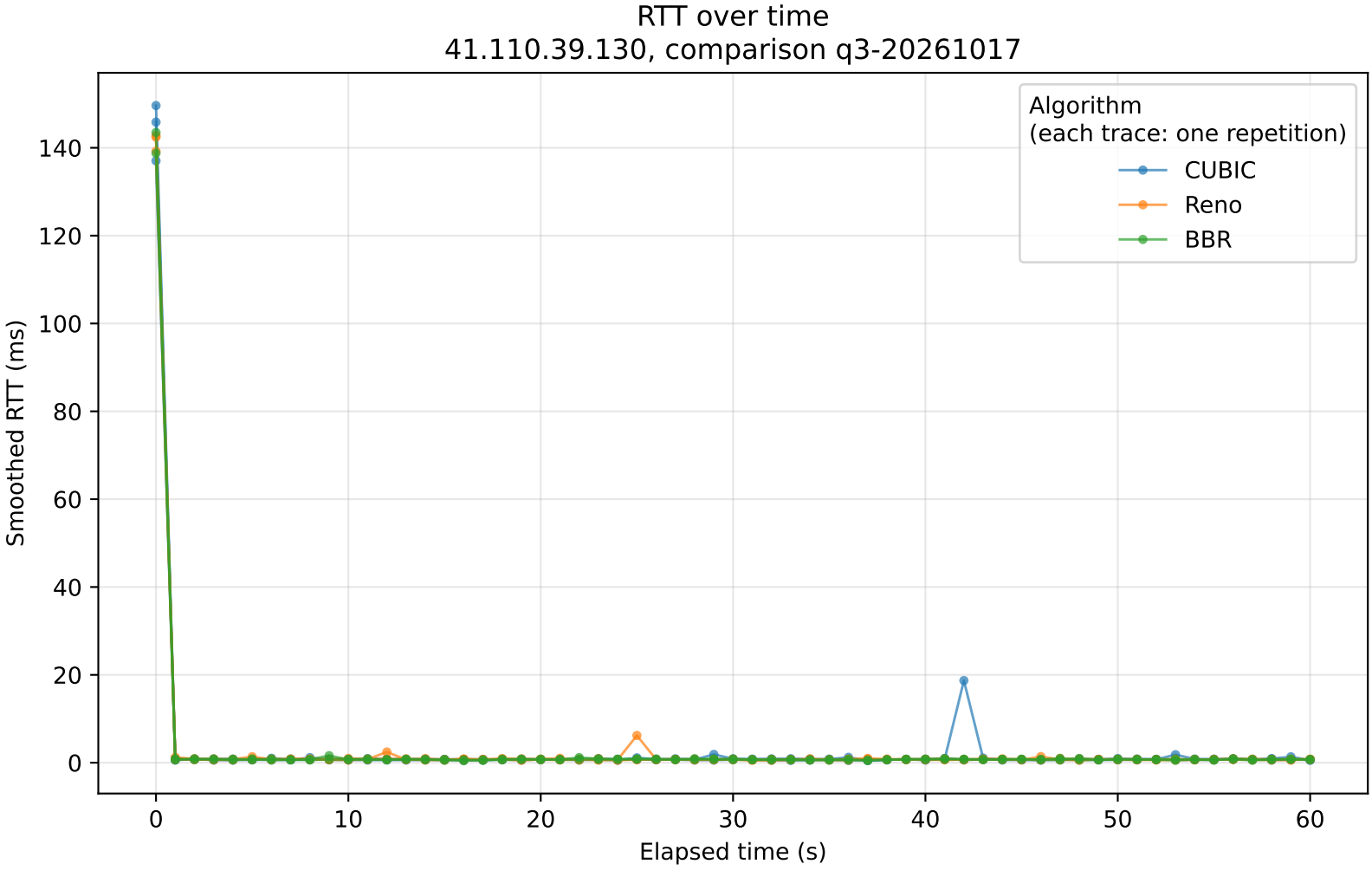

Goodput over time
iperf3.moji.fr, comparison q3-20261017
Goodput (Mbps)
Elapsed time (s)
Algorithm
(each trace: one repetition)
CUBIC
Reno
BBR
0
5
10
15
20
25
30
35
0
10
20
30
40
50
60

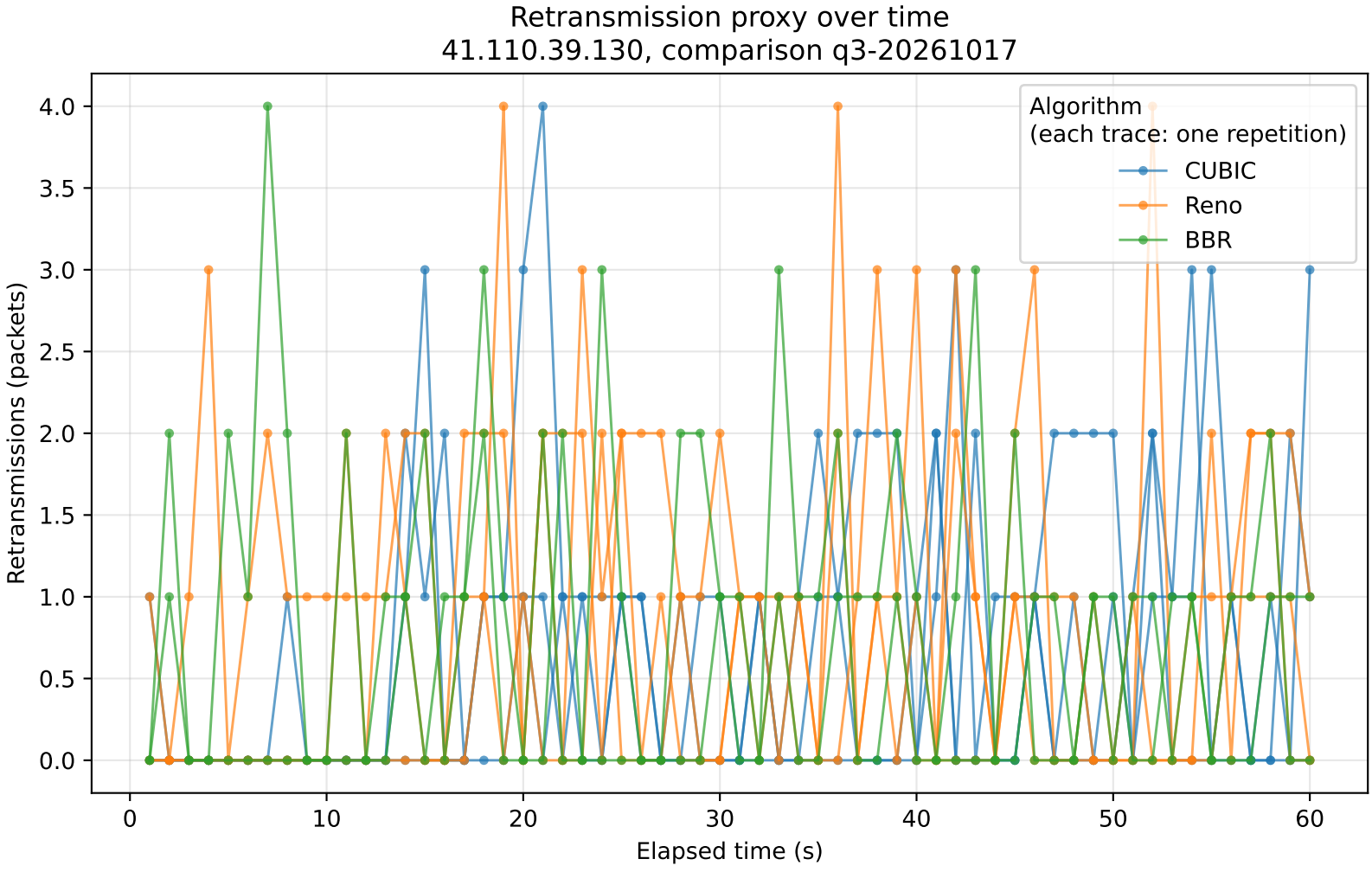

Congestion window vs goodput
iperf3.moji.fr, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
Reno
BBR
Goodput (Mbps)
0
5
10
15
20
25
30
35
0
100
200
300
400
500
600
Congestion window (KiB)

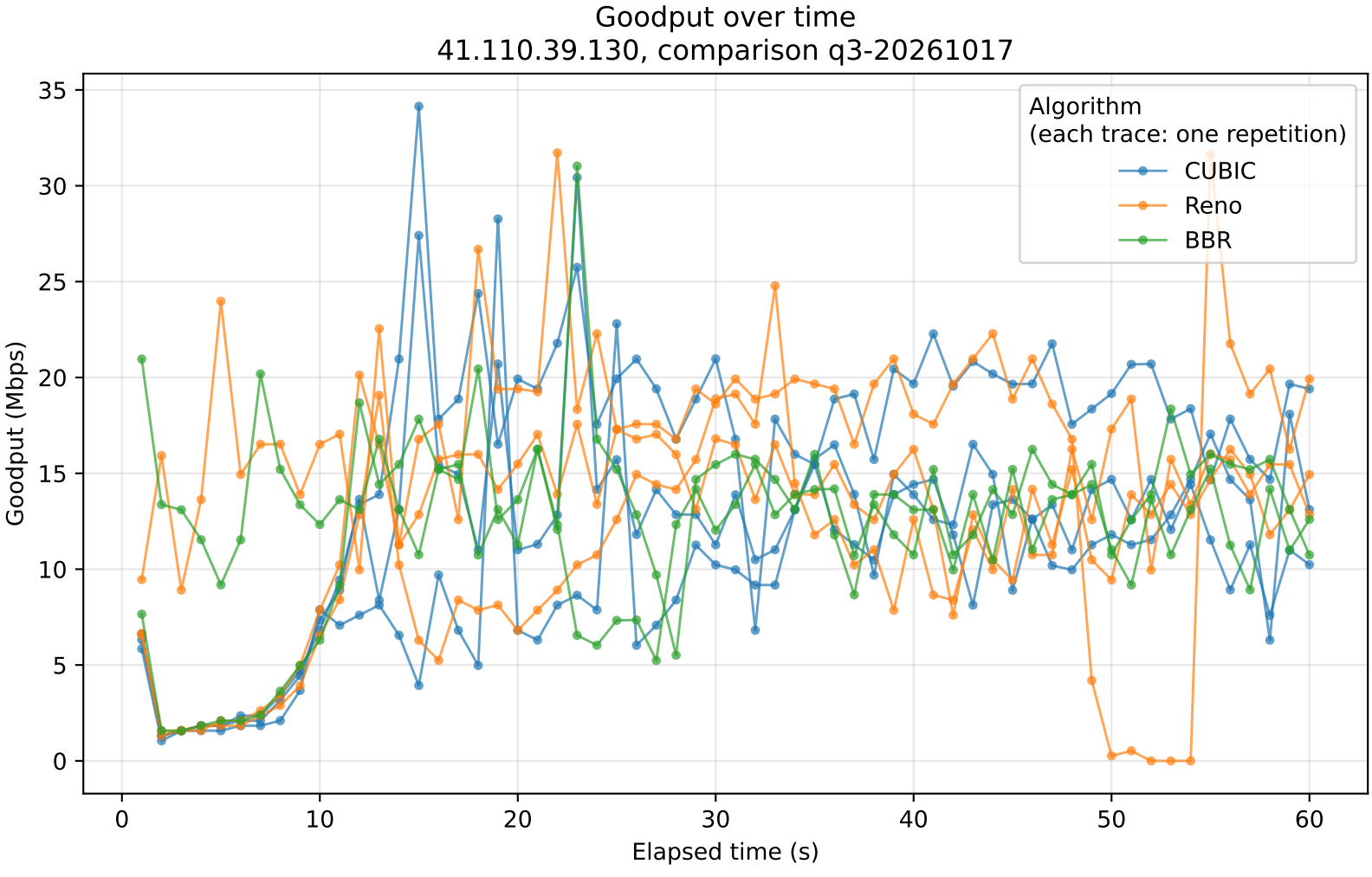

RTT vs goodput
iperf3.moji.fr, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
Reno
BBR
Goodput (Mbps)
Smoothed RTT (ms)
0
5
10
15
20
25
30
35
2
4
6
8

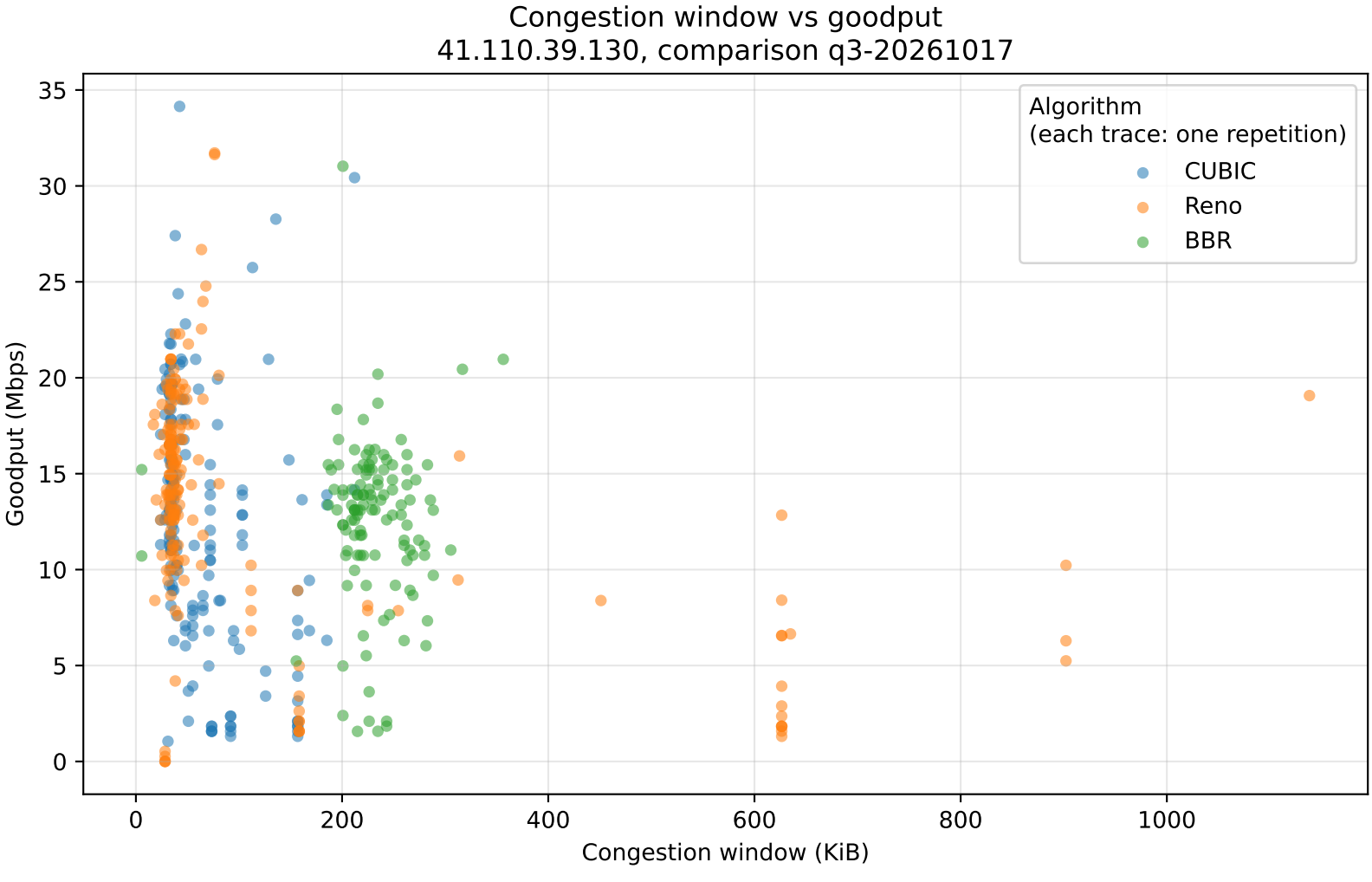

Retransmission proxy vs goodput
iperf3.moji.fr, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
Reno
BBR
Goodput (Mbps)
Retransmissions (packets)
0
5
10
15
20
25
30
35
0
1
2
3
4
5
6

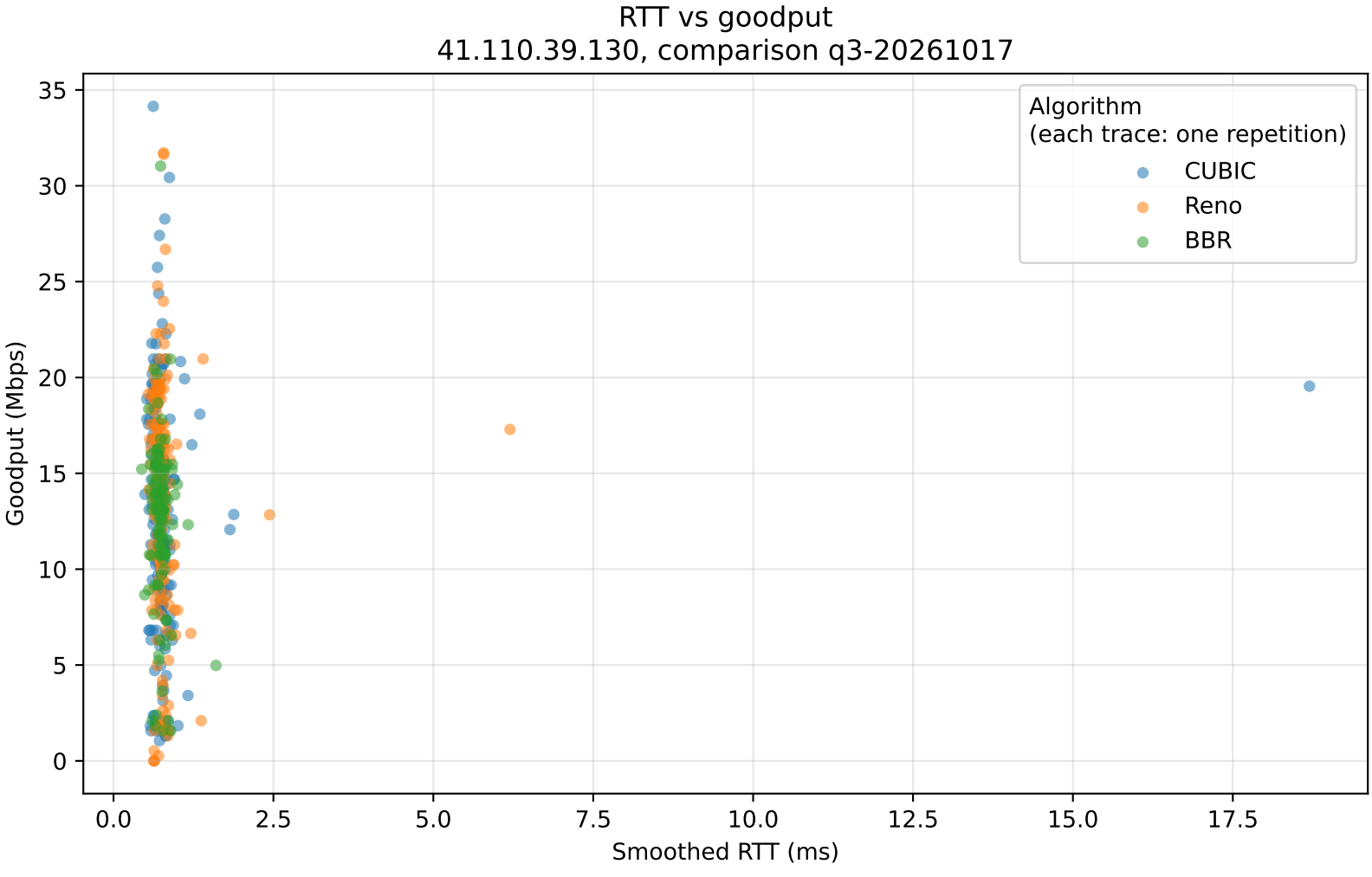

Congestion window over time
41.110.39.130, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
Reno
BBR
Congestion window (KiB)
0
200
400
600
800
1000
Elapsed time (s)
0
10
20
30
40
50
60

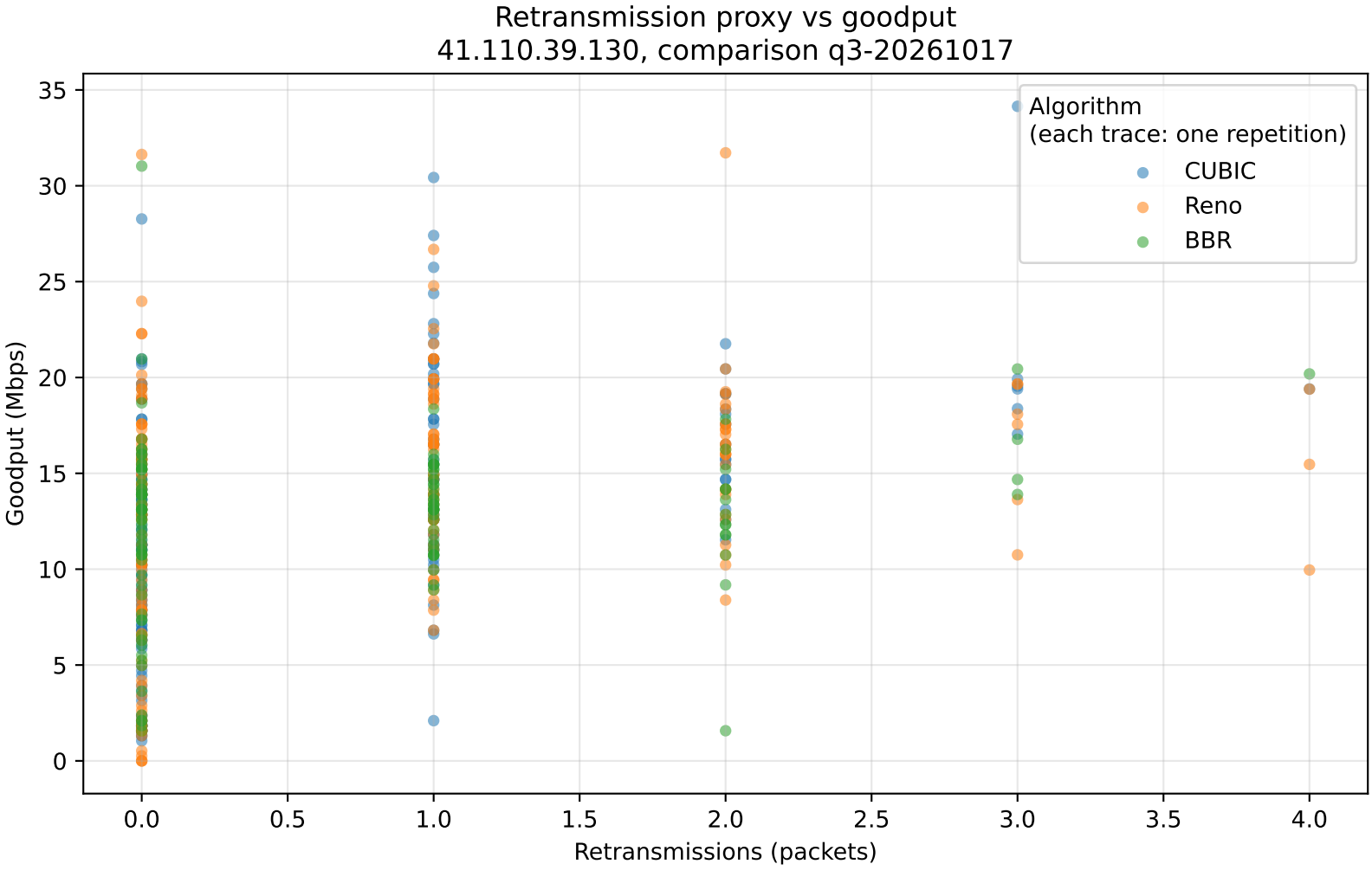

RTT over time
41.110.39.130, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
Reno
BBR
Smoothed RTT (ms)
0
20
40
60
80
100
120
140
0
10
20
30
40
50
60
Elapsed time (s)

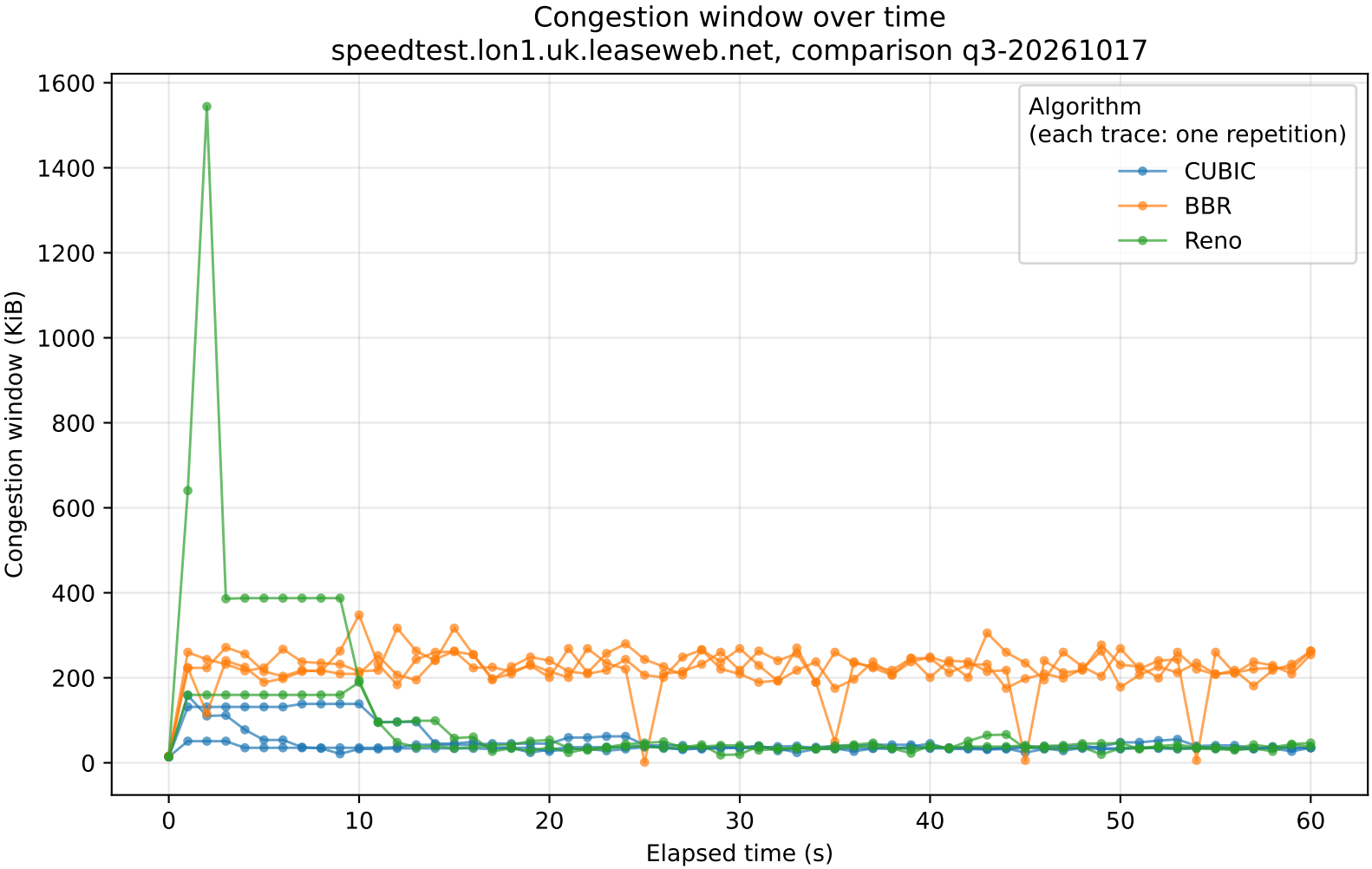

Retransmission proxy over time
41.110.39.130, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
Reno
BBR
Retransmissions (packets)
Elapsed time (s)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0
10
20
30
40
50
60

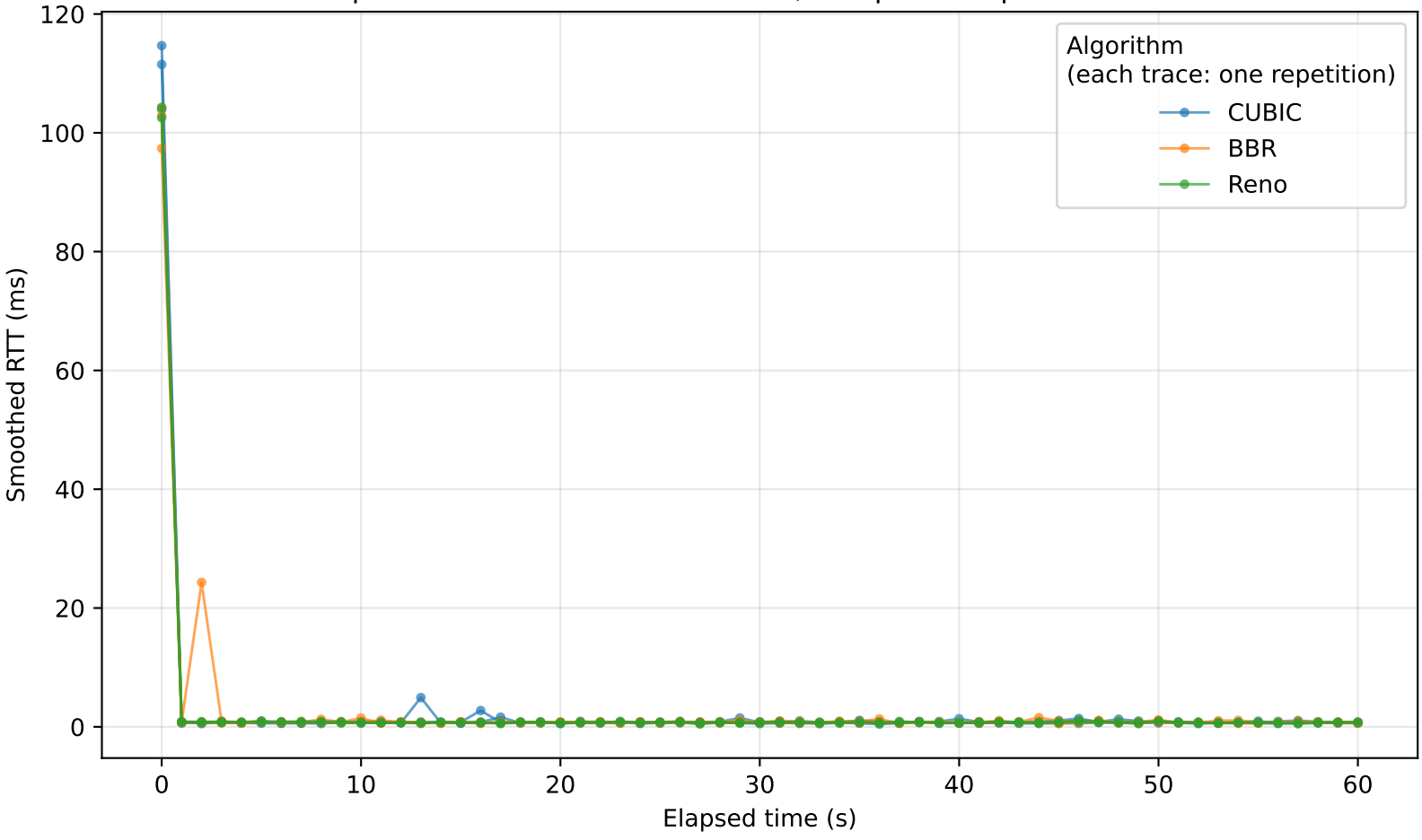

Goodput over time
41.110.39.130, comparison q3-20261017
Goodput (Mbps)
Elapsed time (s)
Algorithm
(each trace: one repetition)
CUBIC
Reno
BBR
0
5
10
15
20
25
30
35
0
10
20
30
40
50
60

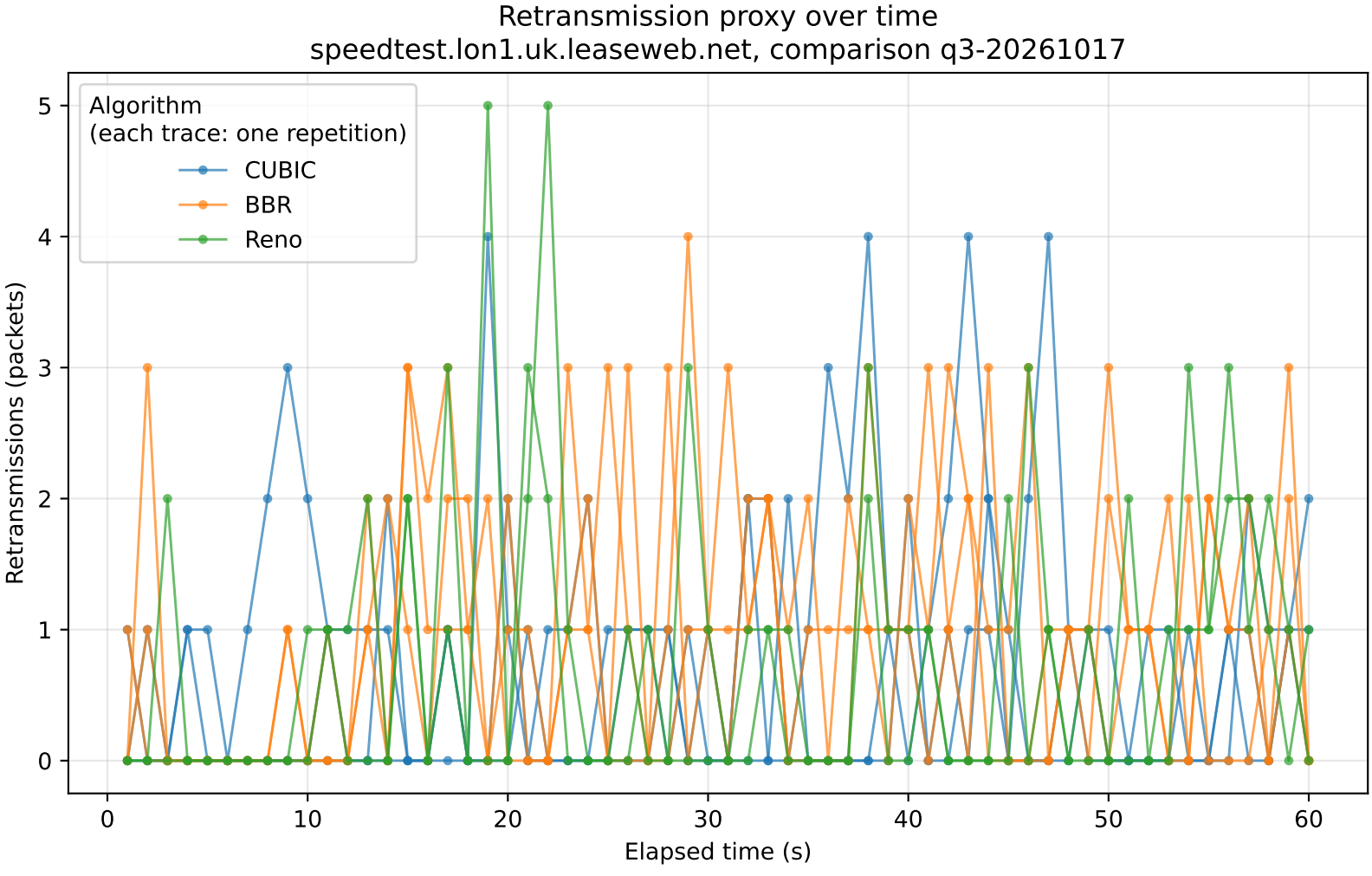

Congestion window vs goodput
41.110.39.130, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
Reno
BBR
Goodput (Mbps)
Congestion window (KiB)
0
5
10
15
20
25
30
35
0
200
400
600
800
1000

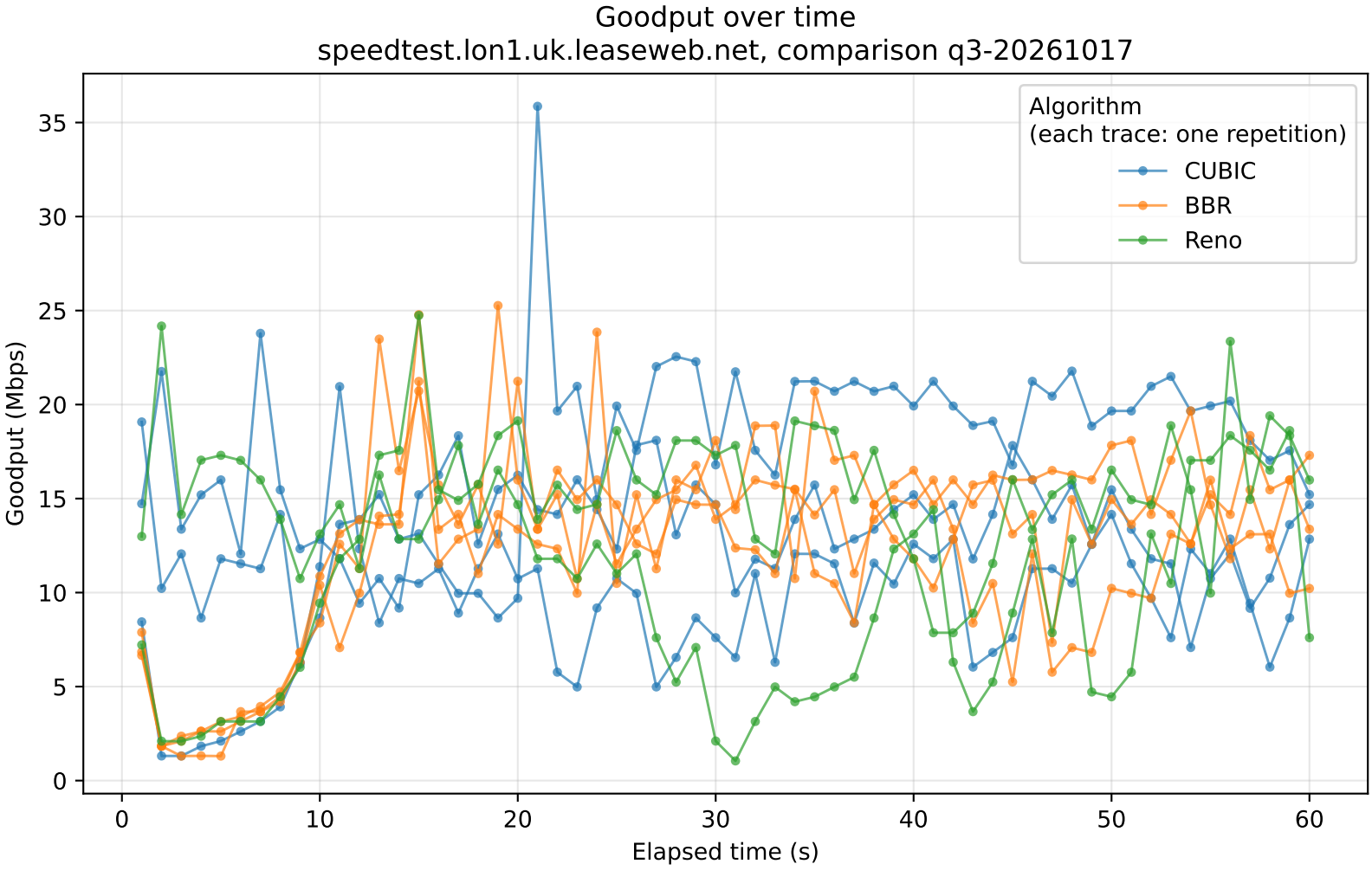

RTT vs goodput
41.110.39.130, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
Reno
BBR
Goodput (Mbps)
Smoothed RTT (ms)
0
5
10
15
20
25
30
35
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5

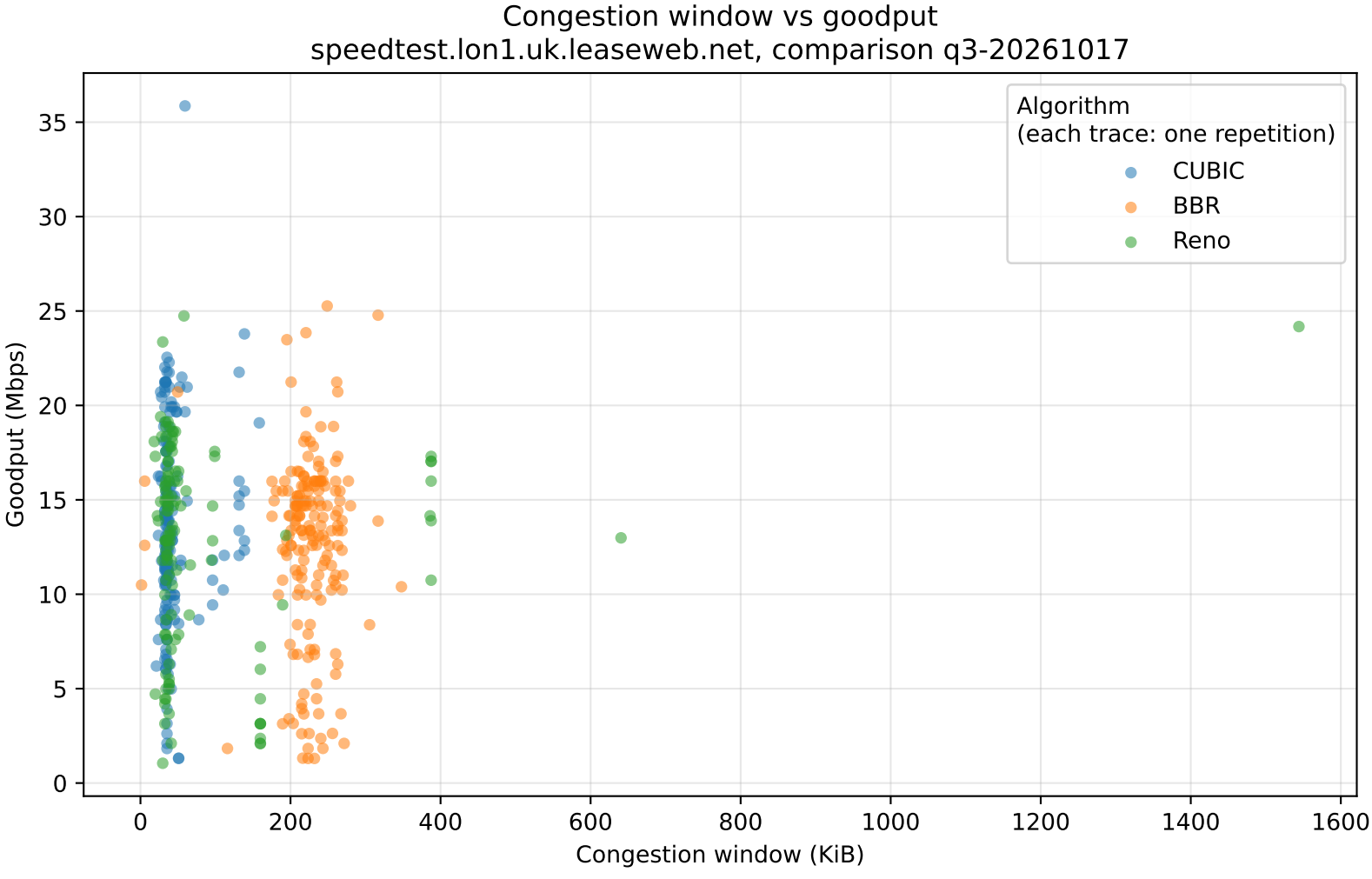

Retransmission proxy vs goodput
41.110.39.130, comparison q3-20261017
Goodput (Mbps)
Retransmissions (packets)
Algorithm
(each trace: one repetition)
CUBIC
Reno
BBR
0
5
10
15
20
25
30
35
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0

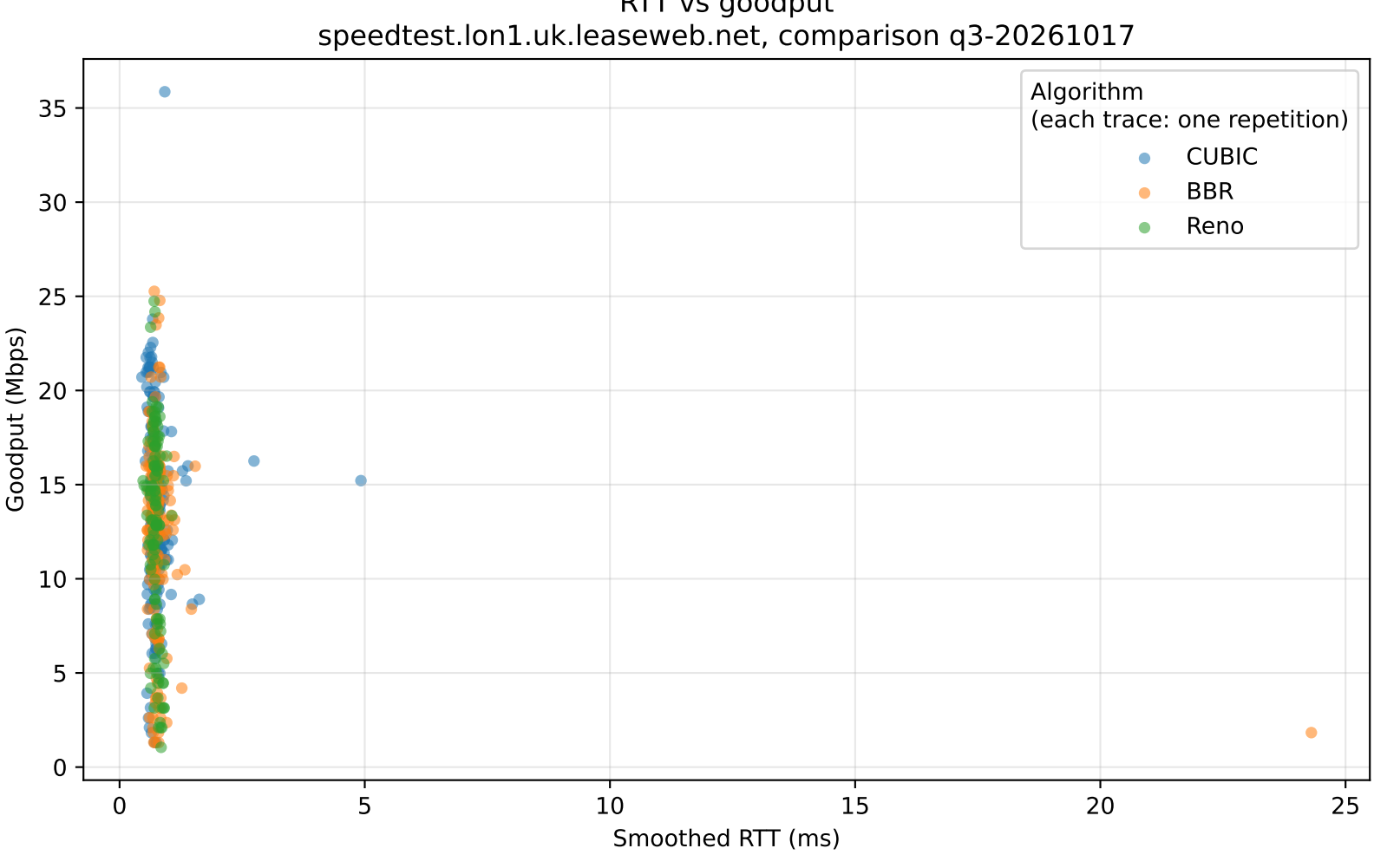

Congestion window over time
speedtest.lon1.uk.leaseweb.net, comparison q3-20261017
Congestion window (KiB)
Elapsed time (s)
Algorithm
(each trace: one repetition)
CUBIC
BBR
Reno
0
200
400
600
800
1000
1200
1400
1600
0
10
20
30
40
50
60

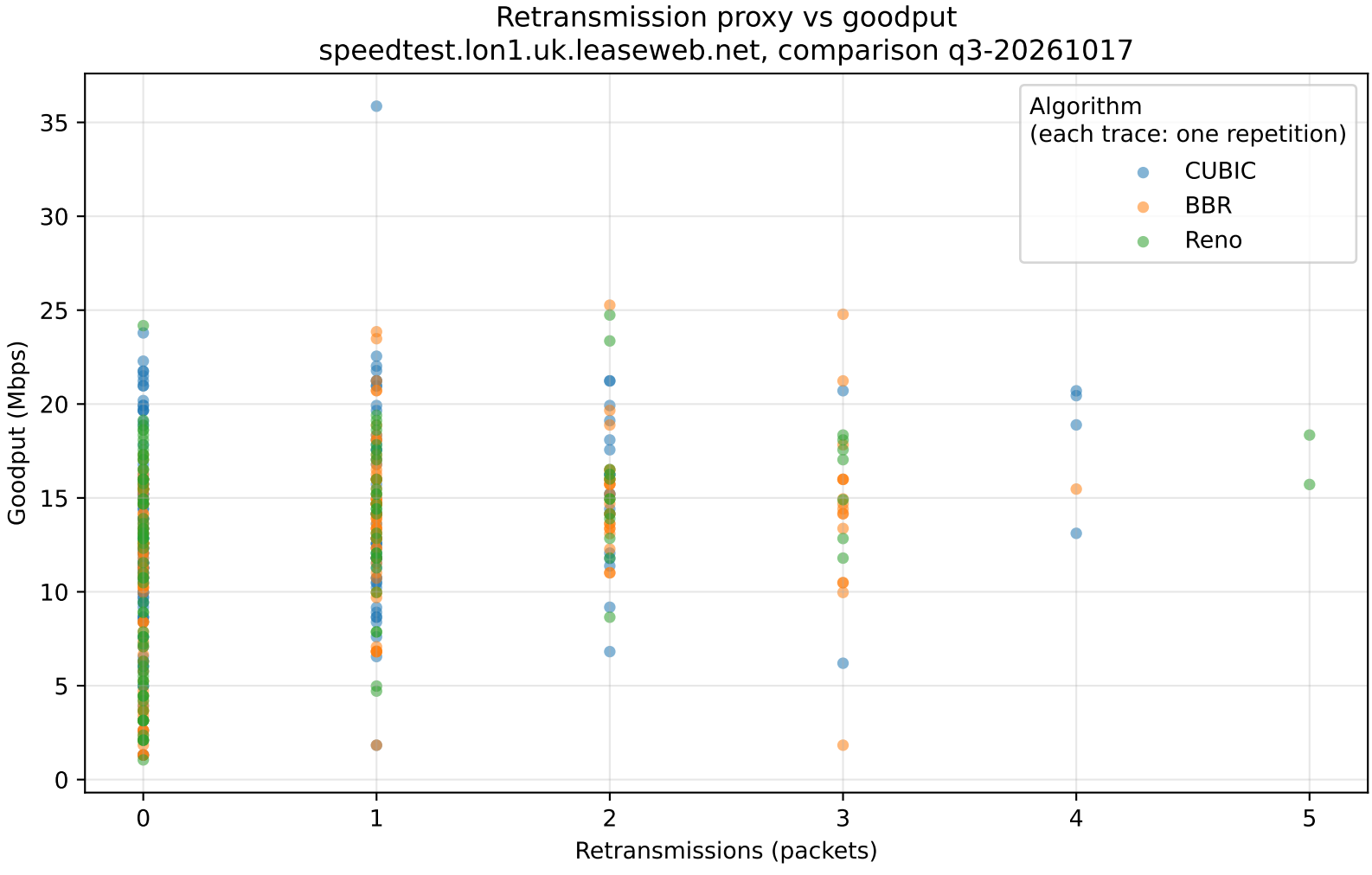

RTT over time
speedtest.lon1.uk.leaseweb.net, comparison q3-20261017
Smoothed RTT (ms)
Elapsed time (s)
Algorithm
(each trace: one repetition)
CUBIC
BBR
Reno
120
100
80
60
40
20
0
0
10
20
30
40
50
60

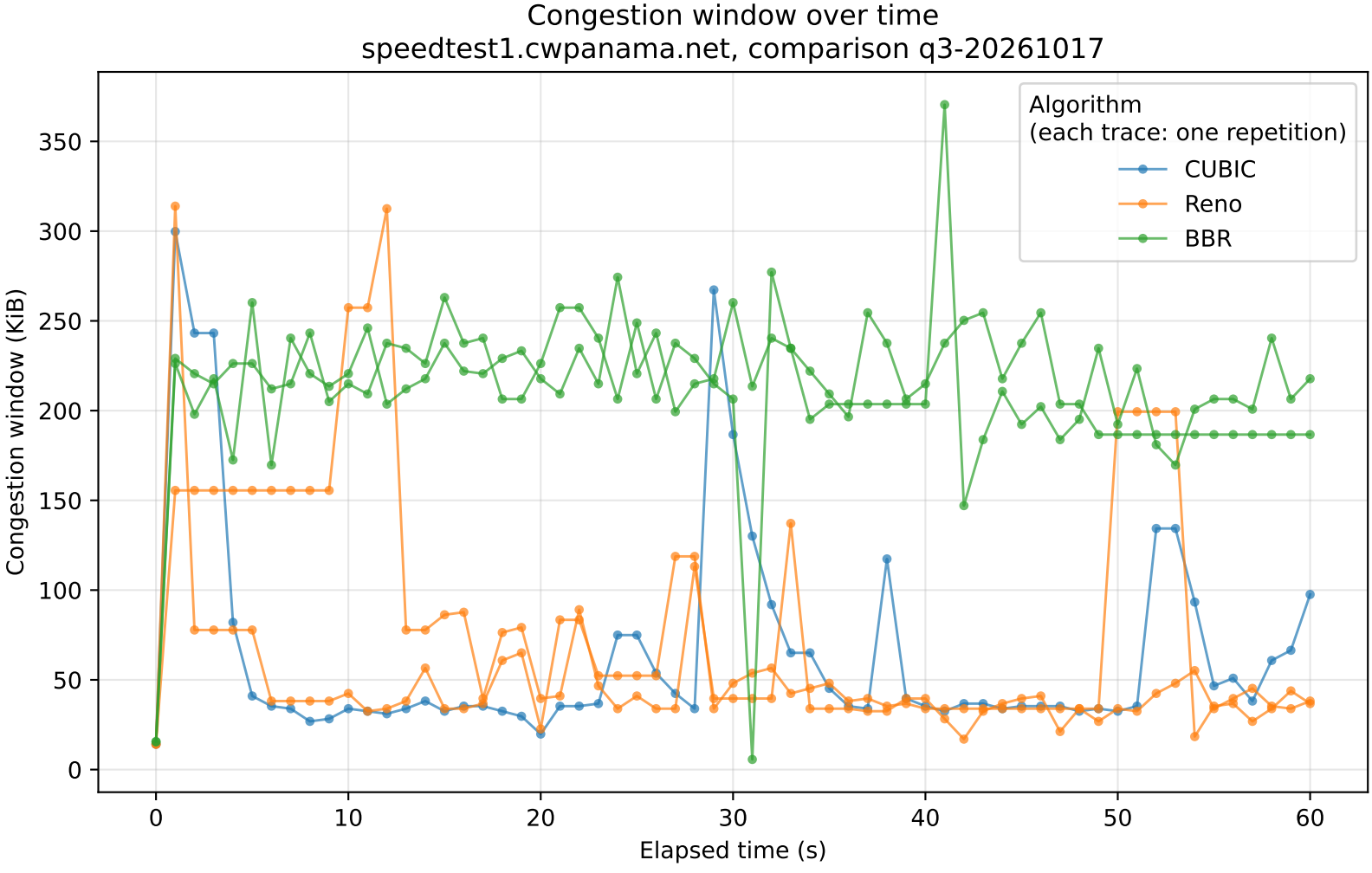

Retransmission proxy over time
speedtest.lon1.uk.leaseweb.net, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
BBR
Reno
Retransmissions (packets)
Elapsed time (s)
0
1
2
3
4
5
0
10
20
30
40
50
60

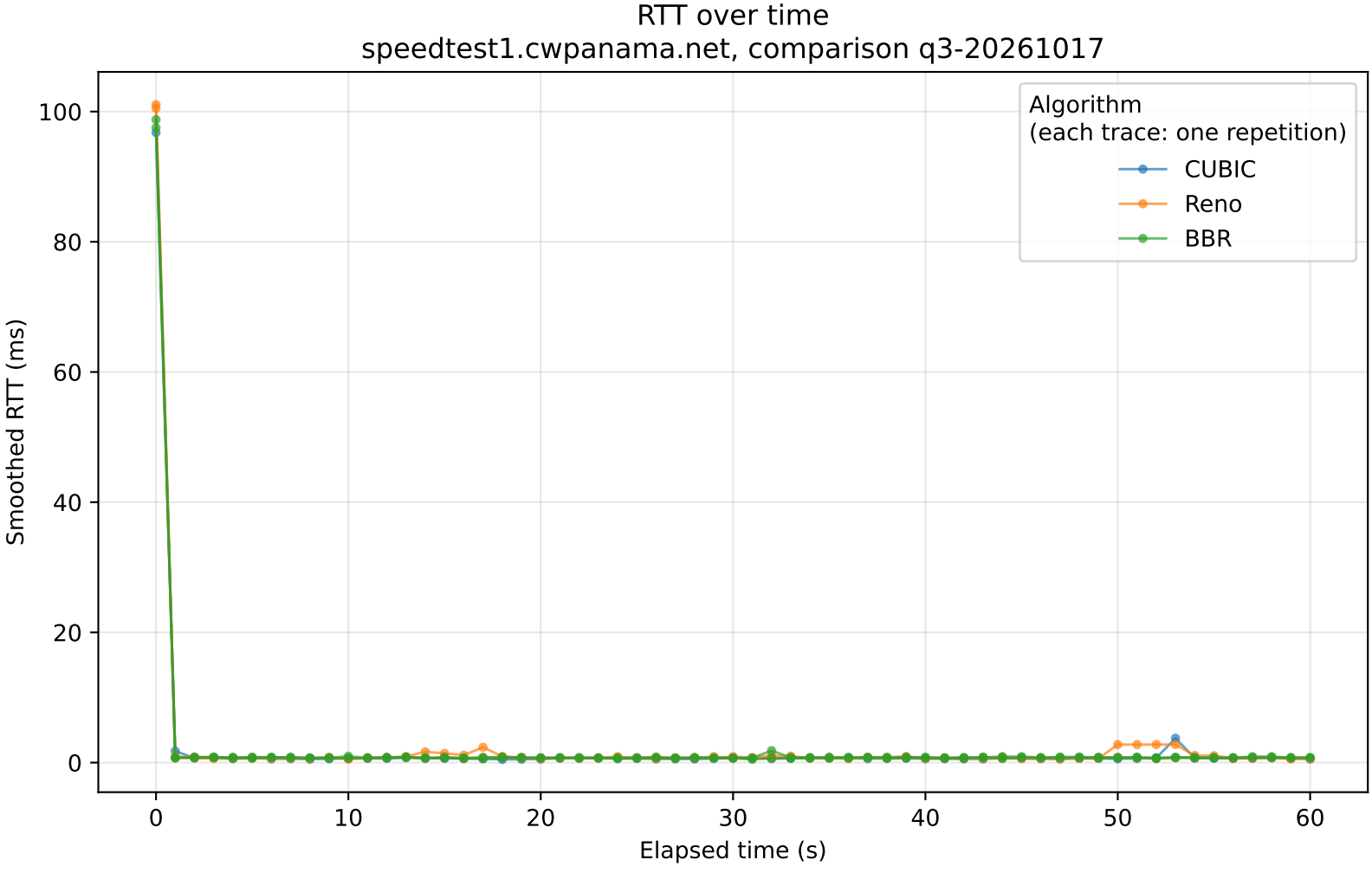

Goodput over time
speedtest.lon1.uk.leaseweb.net, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
BBR
Reno
Goodput (Mbps)
Elapsed time (s)
0
5
10
15
20
25
30
35
0
10
20
30
40
50
60

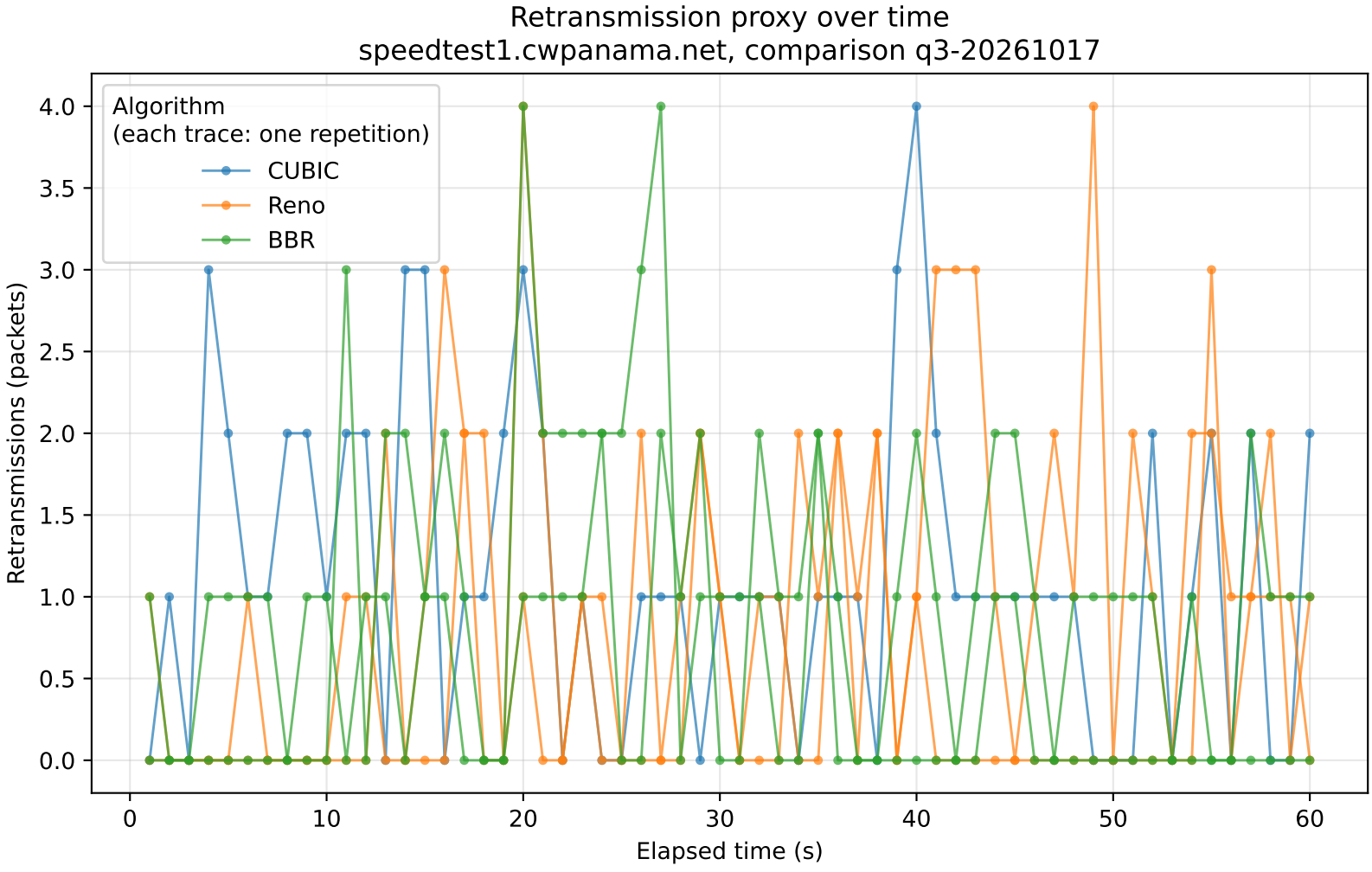

Congestion window vs goodput
speedtest.lon1.uk.leaseweb.net, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
BBR
Reno
Goodput (Mbps)
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200
1400
1600
Congestion window (KiB)

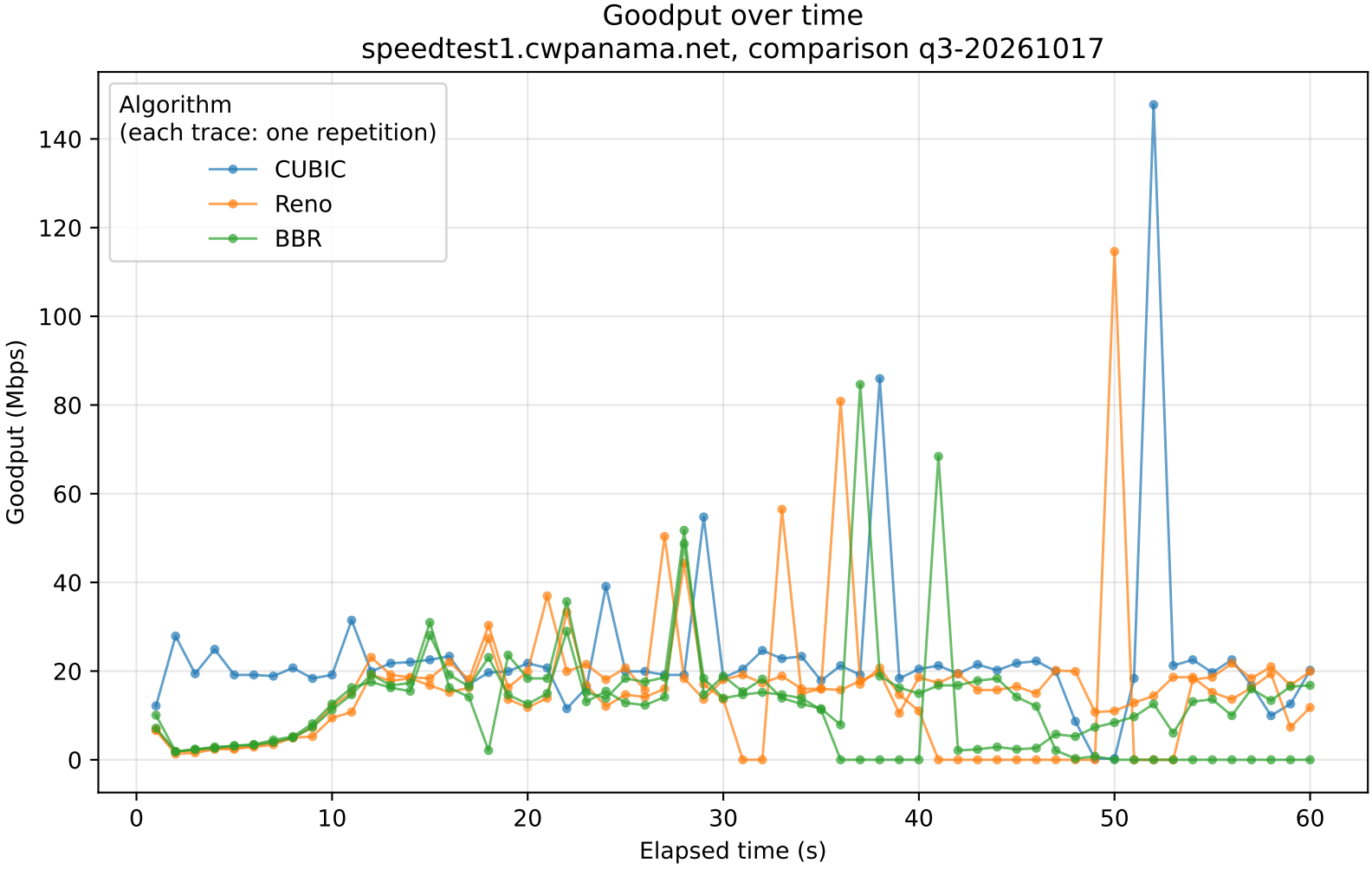

RTT vs goodput
speedtest.lon1.uk.leaseweb.net, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
BBR
Reno
Goodput (Mbps)
0
5
10
15
20
25
30
35
Smoothed RTT (ms)
0
5
10
15
20
25

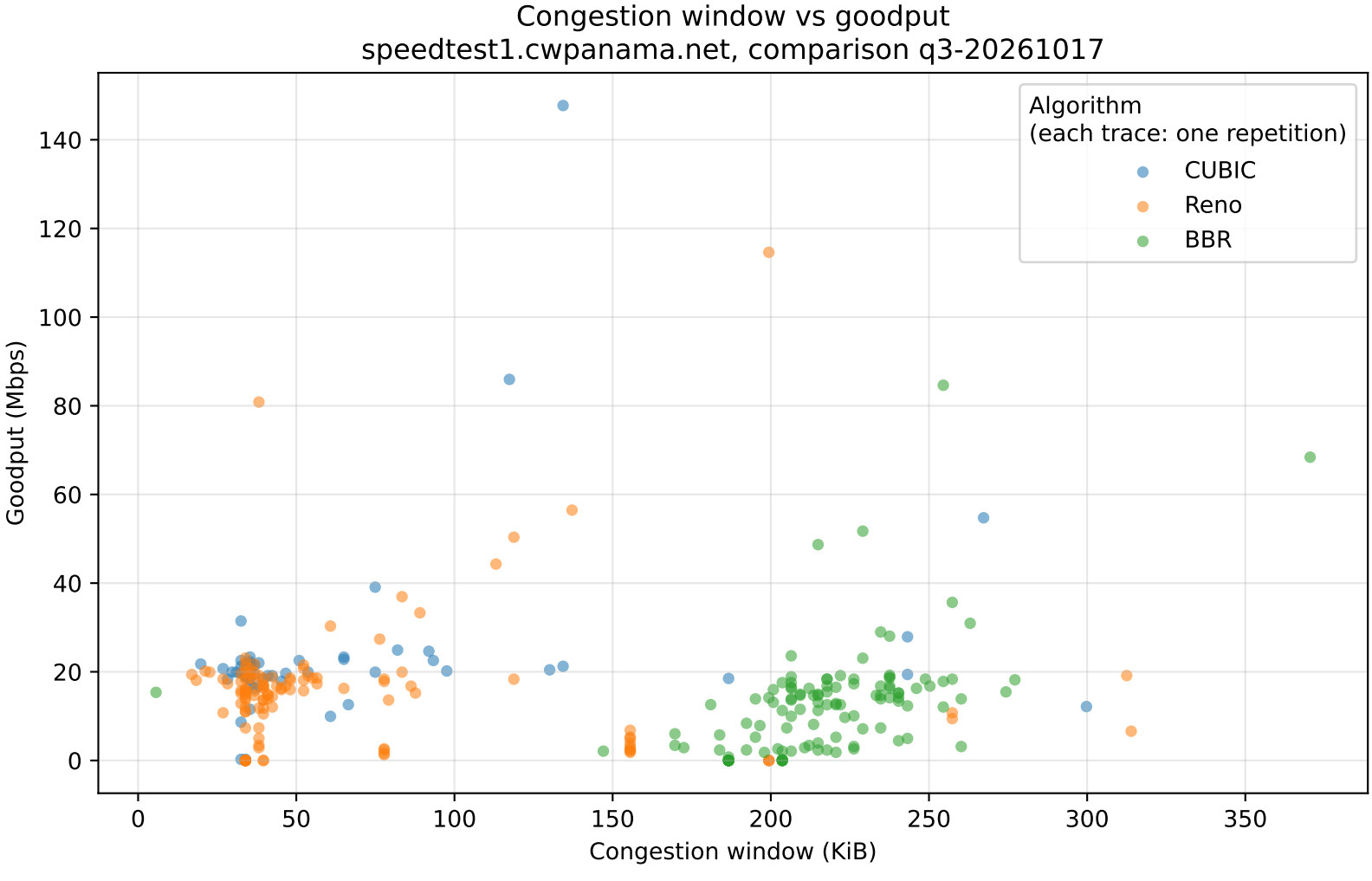

Retransmission proxy vs goodput
speedtest.lon1.uk.leaseweb.net, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
BBR
Reno
Goodput (Mbps)
0
5
10
15
20
25
30
35
0
1
2
3
4
5
Retransmissions (packets)

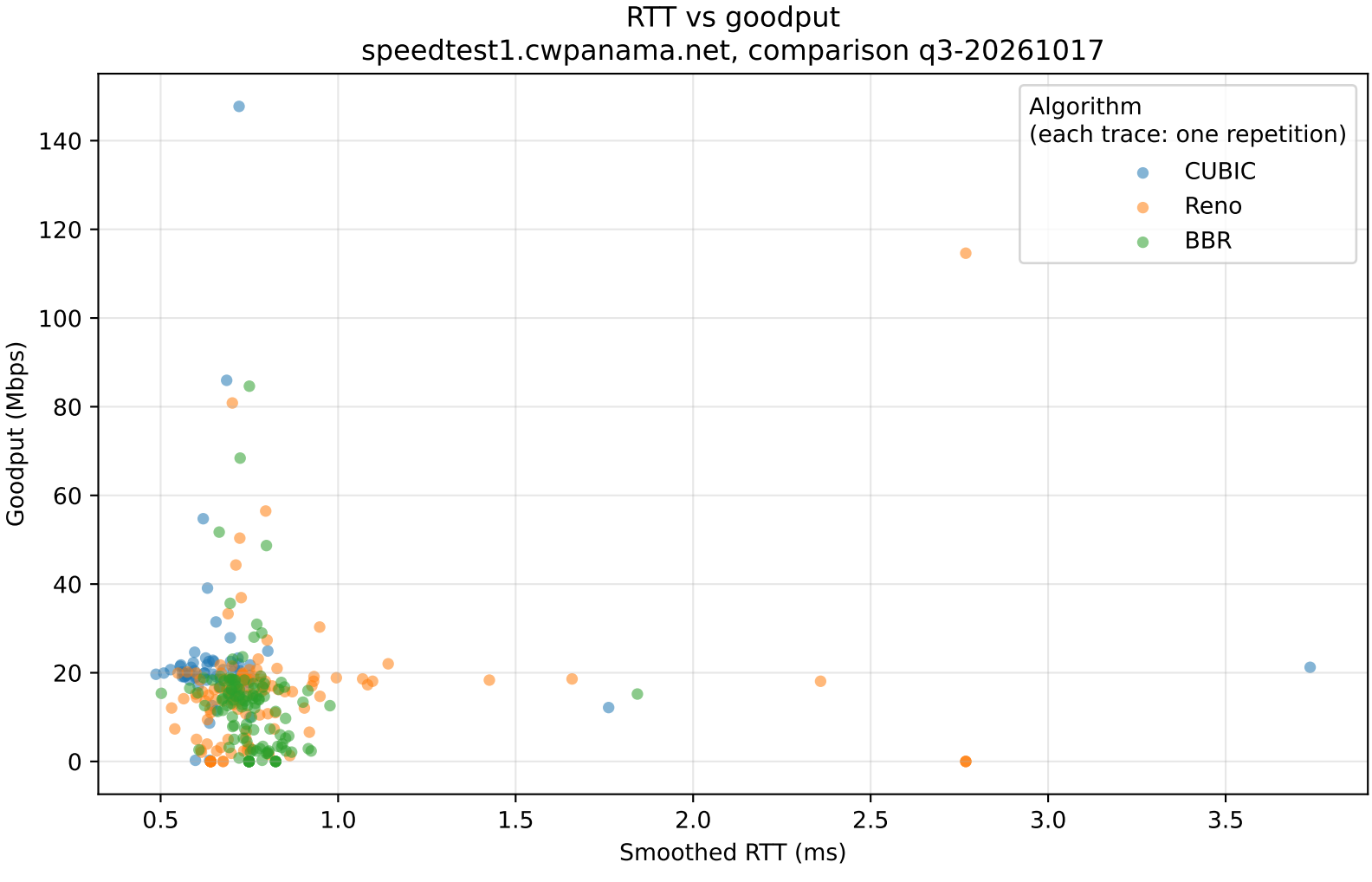

Congestion window over time
speedtest1.cwpanama.net, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
Reno
BBR
Congestion window (KiB)
Elapsed time (s)
0
50
100
150
200
250
300
350
0
10
20
30
40
50
60

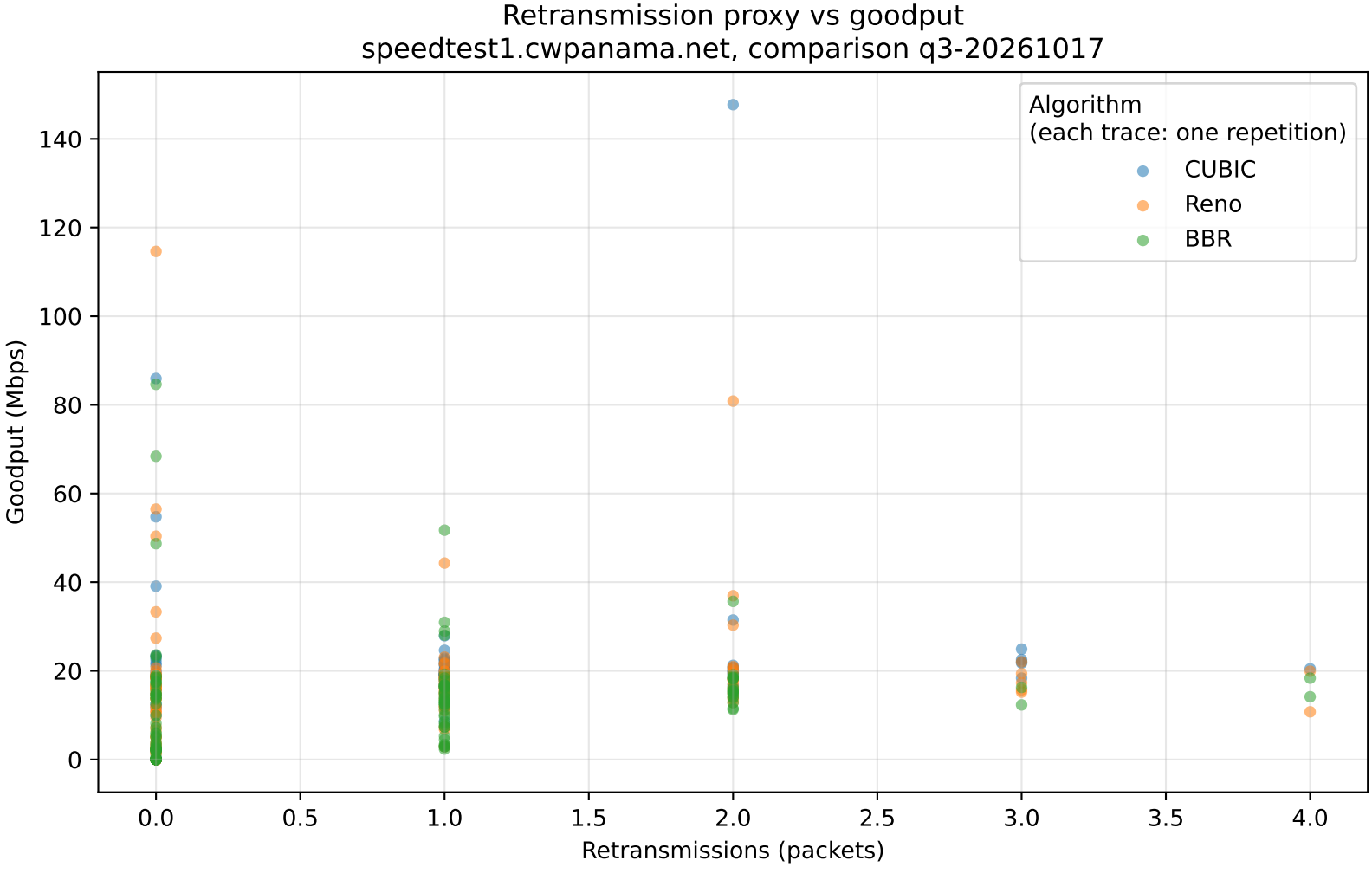

RTT over time
speedtest1.cwpanama.net, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
Reno
BBR
Smoothed RTT (ms)
Elapsed time (s)
0
20
40
60
80
100
0
10
20
30
40
50
60

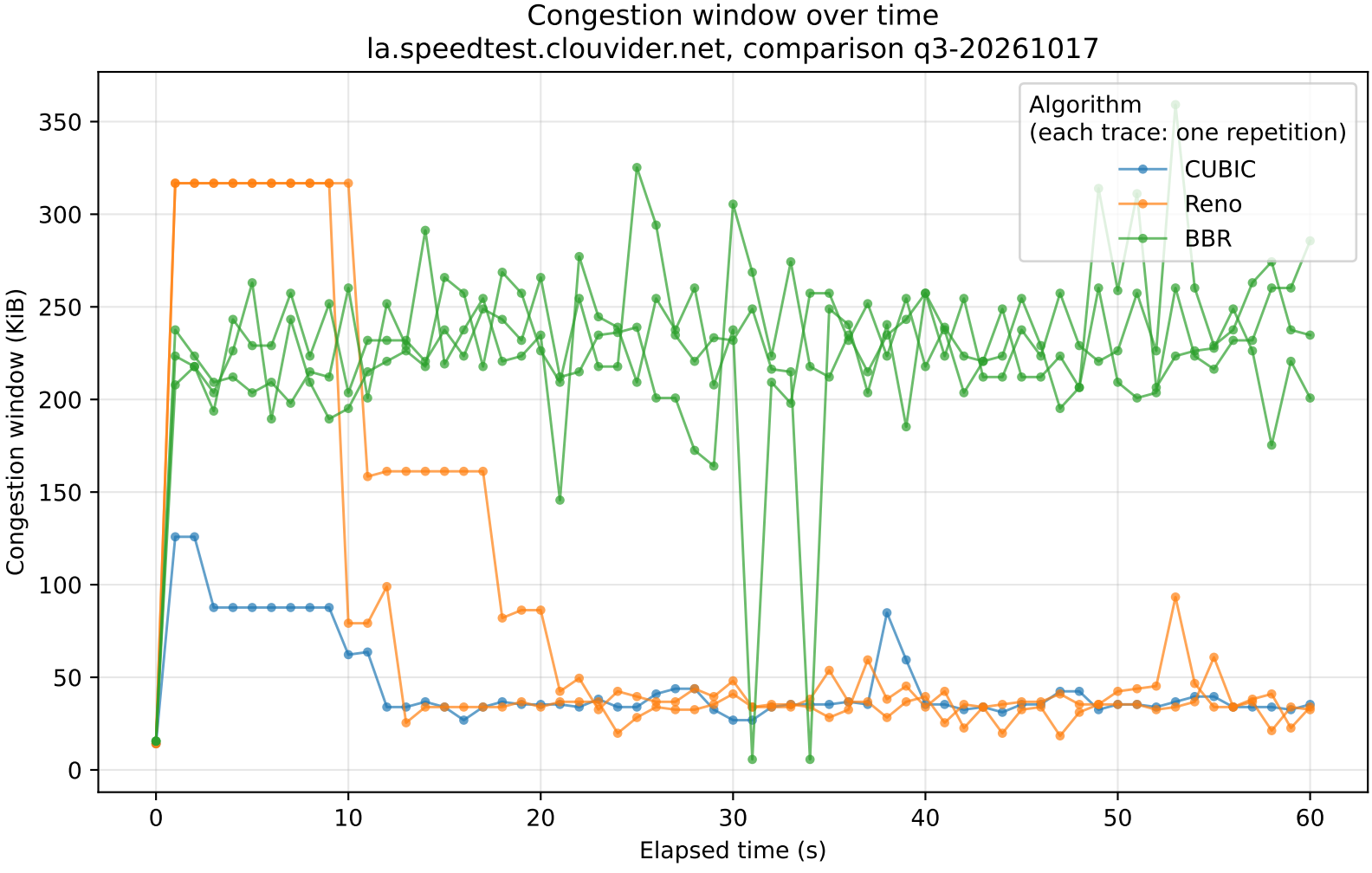

Retransmission proxy over time
speedtest1.cwpanama.net, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
Reno
BBR
Retransmissions (packets)
Elapsed time (s)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0
10
20
30
40
50
60

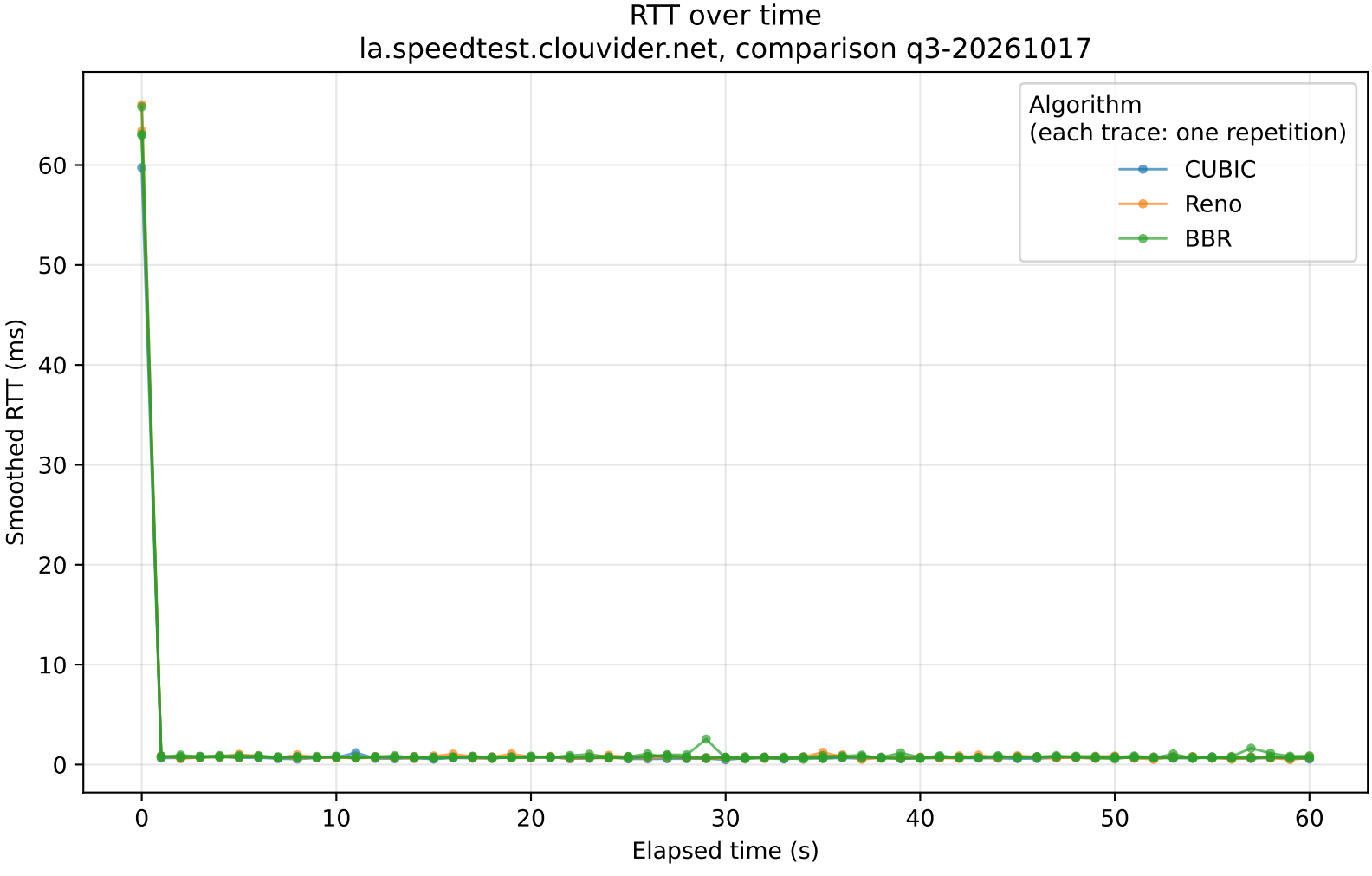

Goodput over time
speedtest1.cwpanama.net, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
Reno
BBR
Goodput (Mbps)
Elapsed time (s)
0
20
40
60
80
100
120
140
0
10
20
30
40
50
60

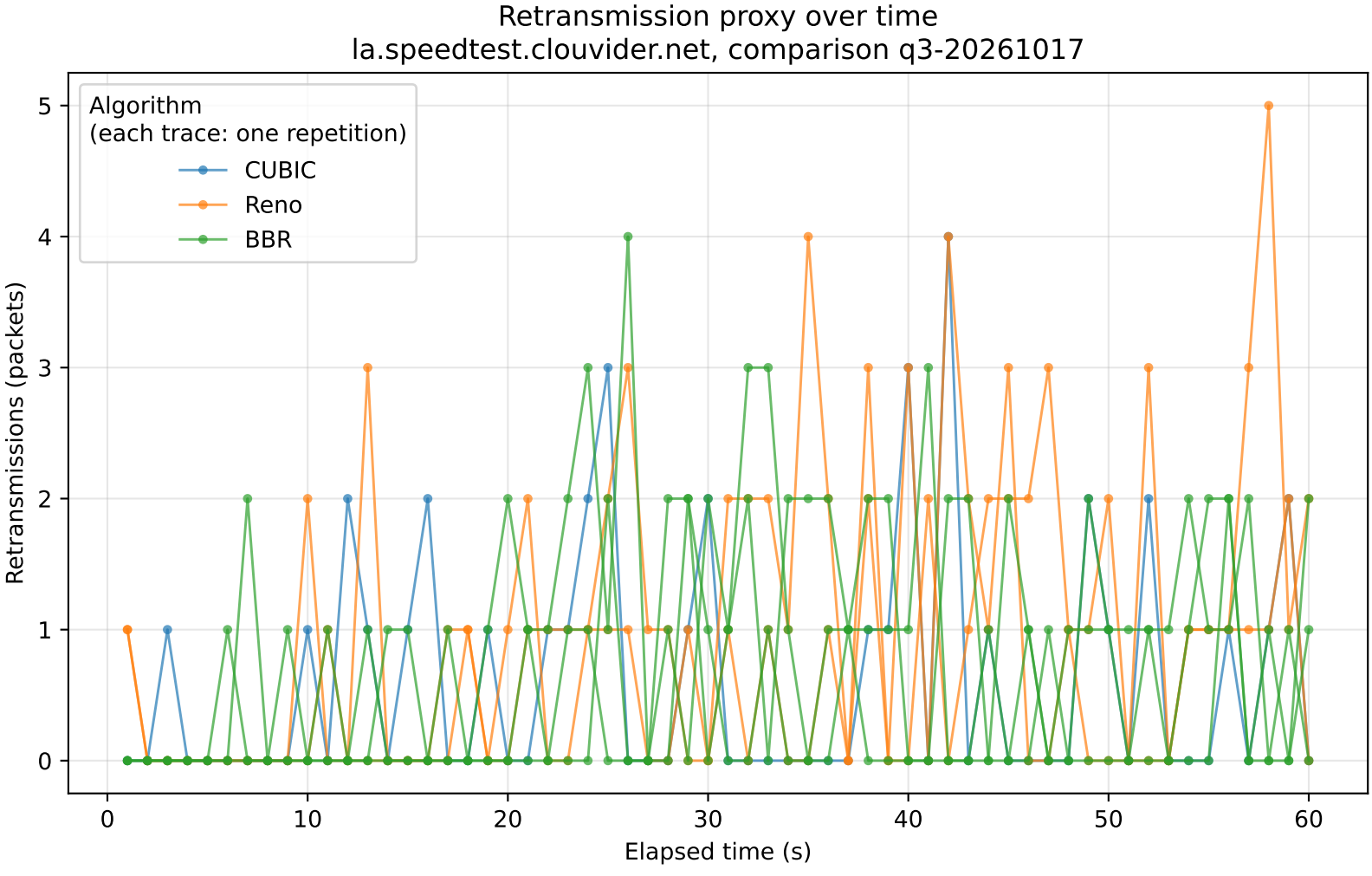

Congestion window vs goodput
speedtest1.cwpanama.net, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
Reno
BBR
Goodput (Mbps)
Congestion window (KiB)
0
20
40
60
80
100
120
140
0
50
100
150
200
250
300
350

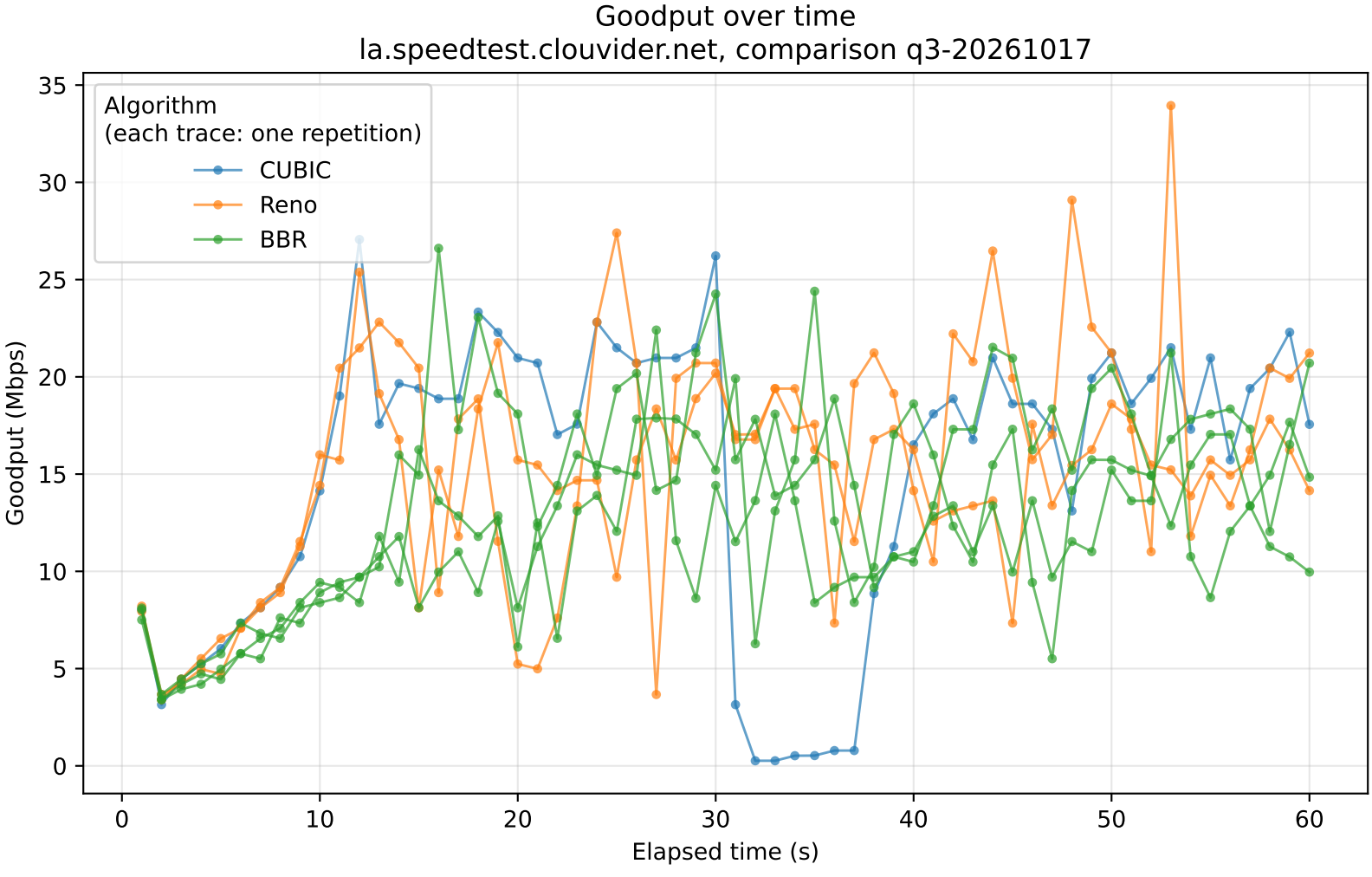

RTT vs goodput
speedtest1.cwpanama.net, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
Reno
BBR
Goodput (Mbps)
0
20
40
60
80
100
120
140
0.5
1.0
1.5
2.0
2.5
3.0
3.5
Smoothed RTT (ms)

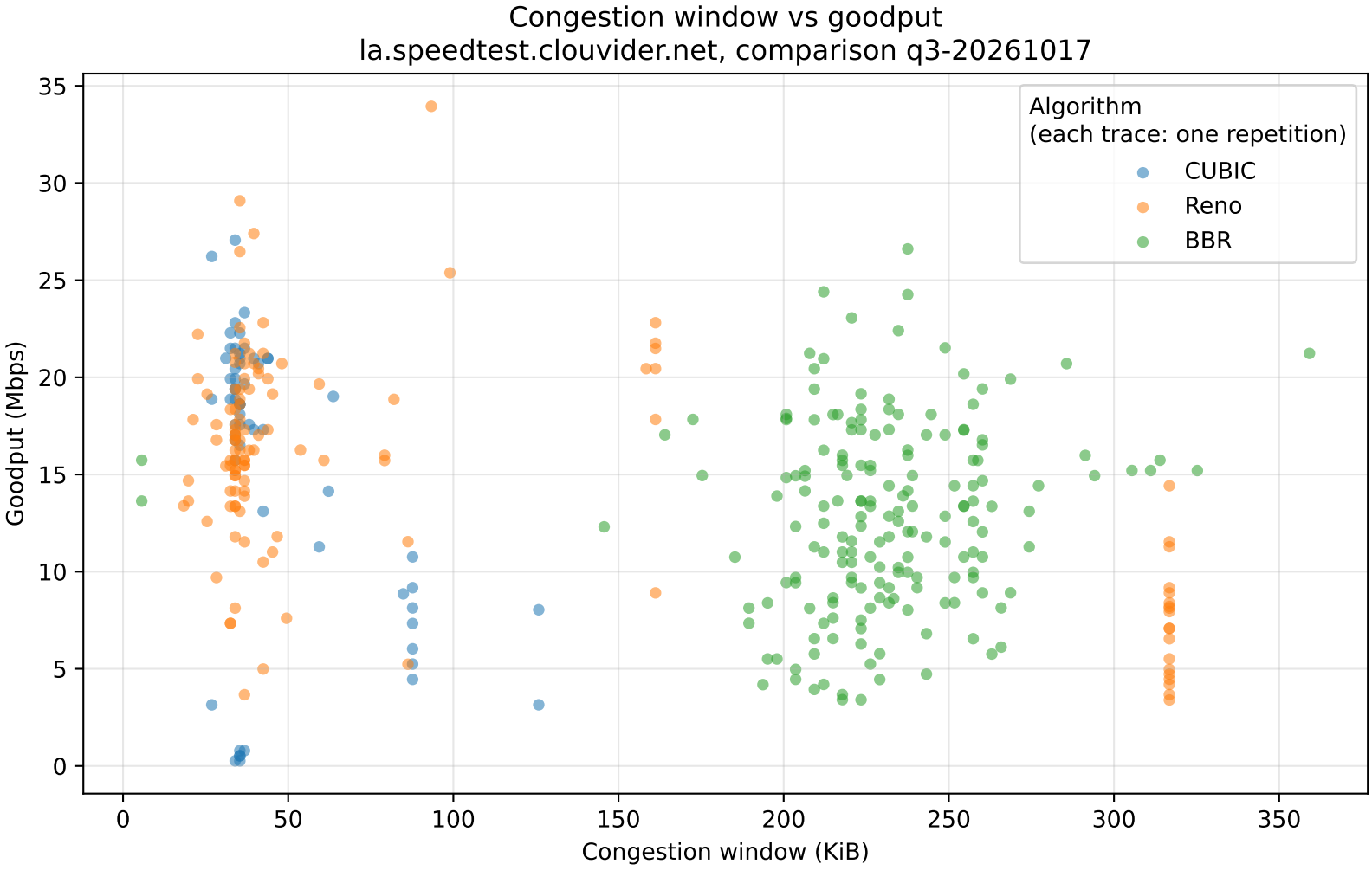

Retransmission proxy vs goodput
speedtest1.cwpanama.net, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
Reno
BBR
Goodput (Mbps)
0
20
40
60
80
100
120
140
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Retransmissions (packets)

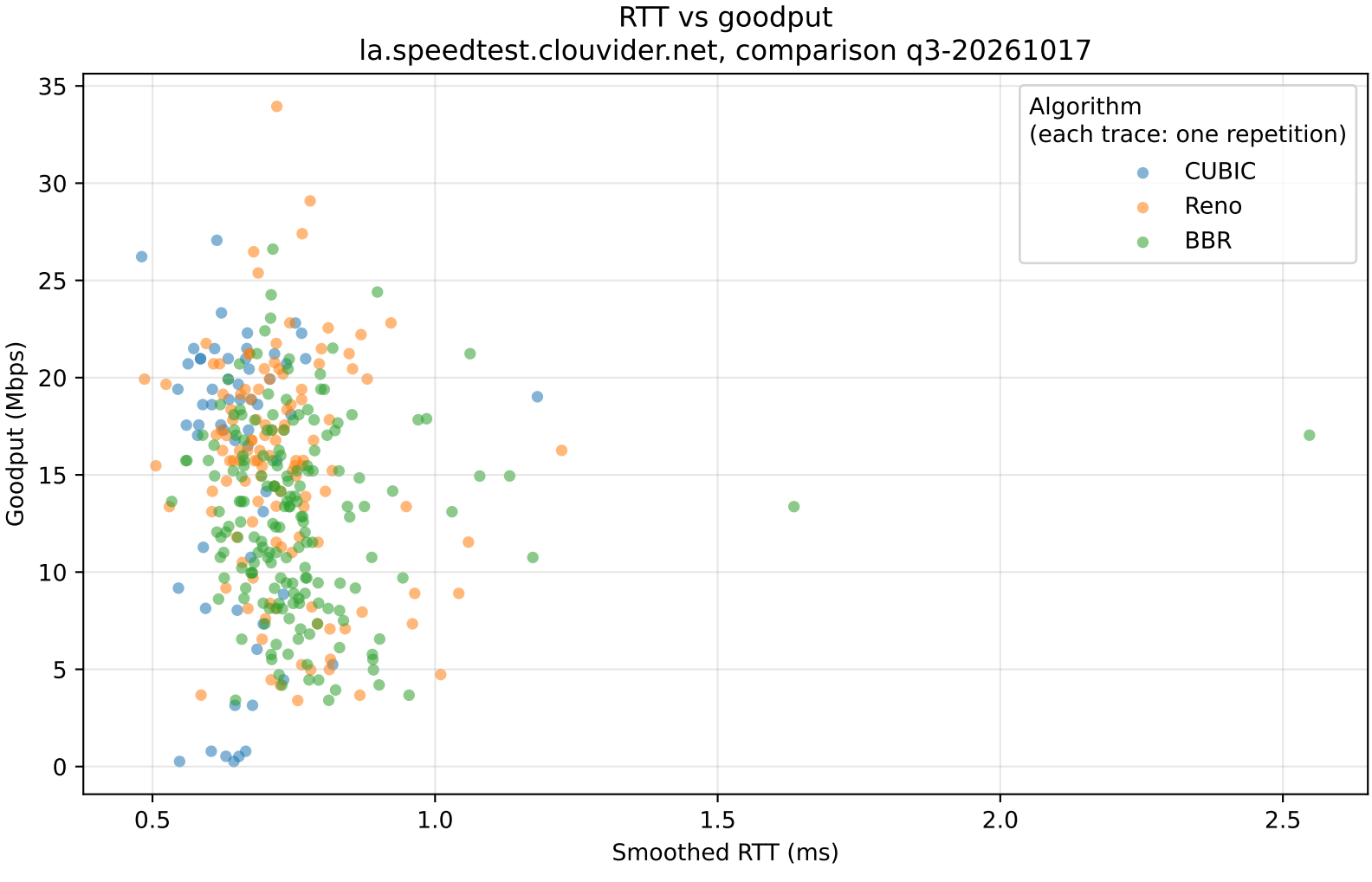

Congestion window over time
la.speedtest.clouvider.net, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
Reno
BBR
Congestion window (KiB)
Elapsed time (s)
0
50
100
150
200
250
300
350
0
10
20
30
40
50
60

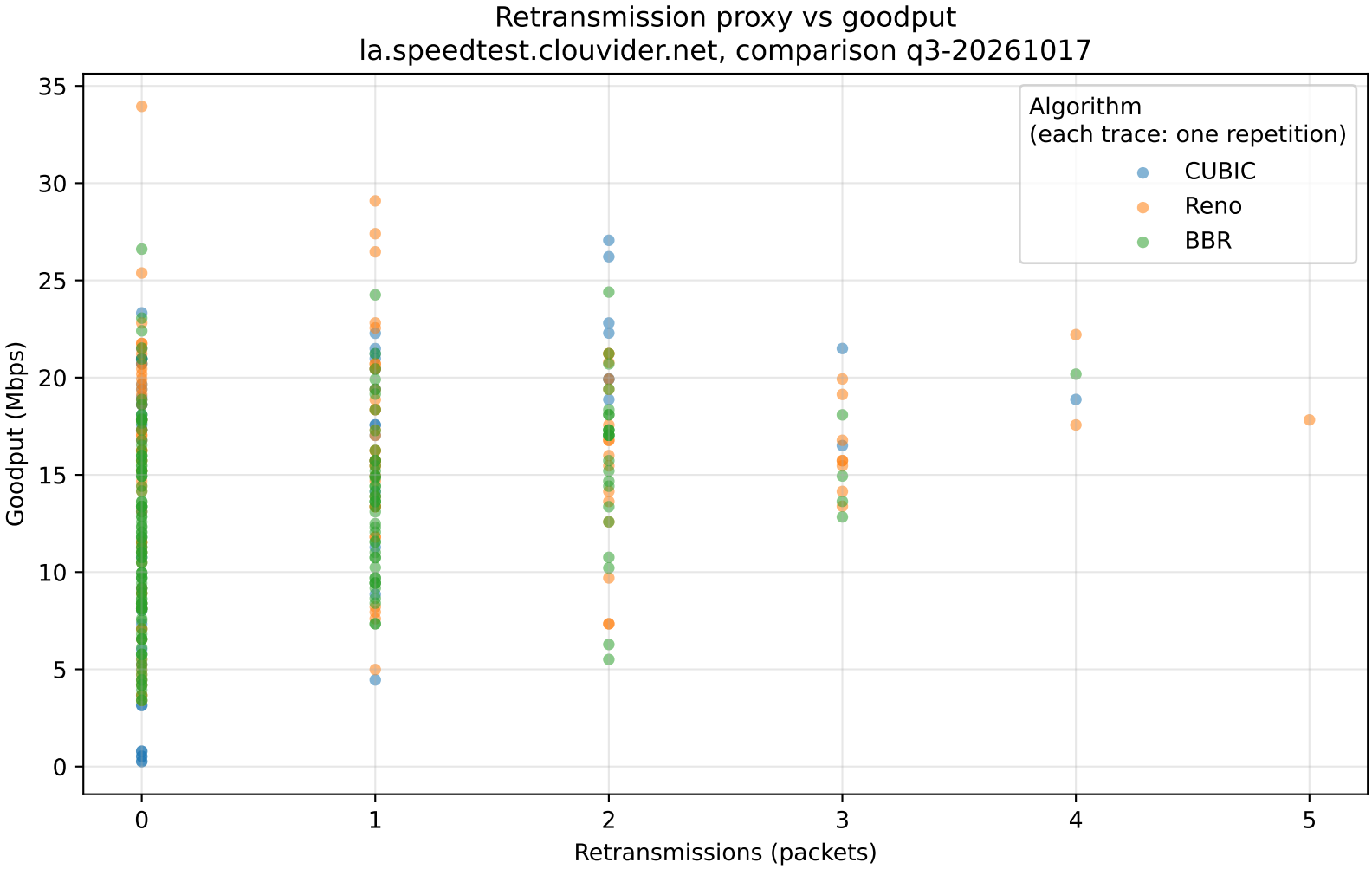

RTT over time
la.speedtest.clouvider.net, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
Reno
BBR
Smoothed RTT (ms)
0
10
20
30
40
50
60
Elapsed time (s)
0
10
20
30
40
50
60

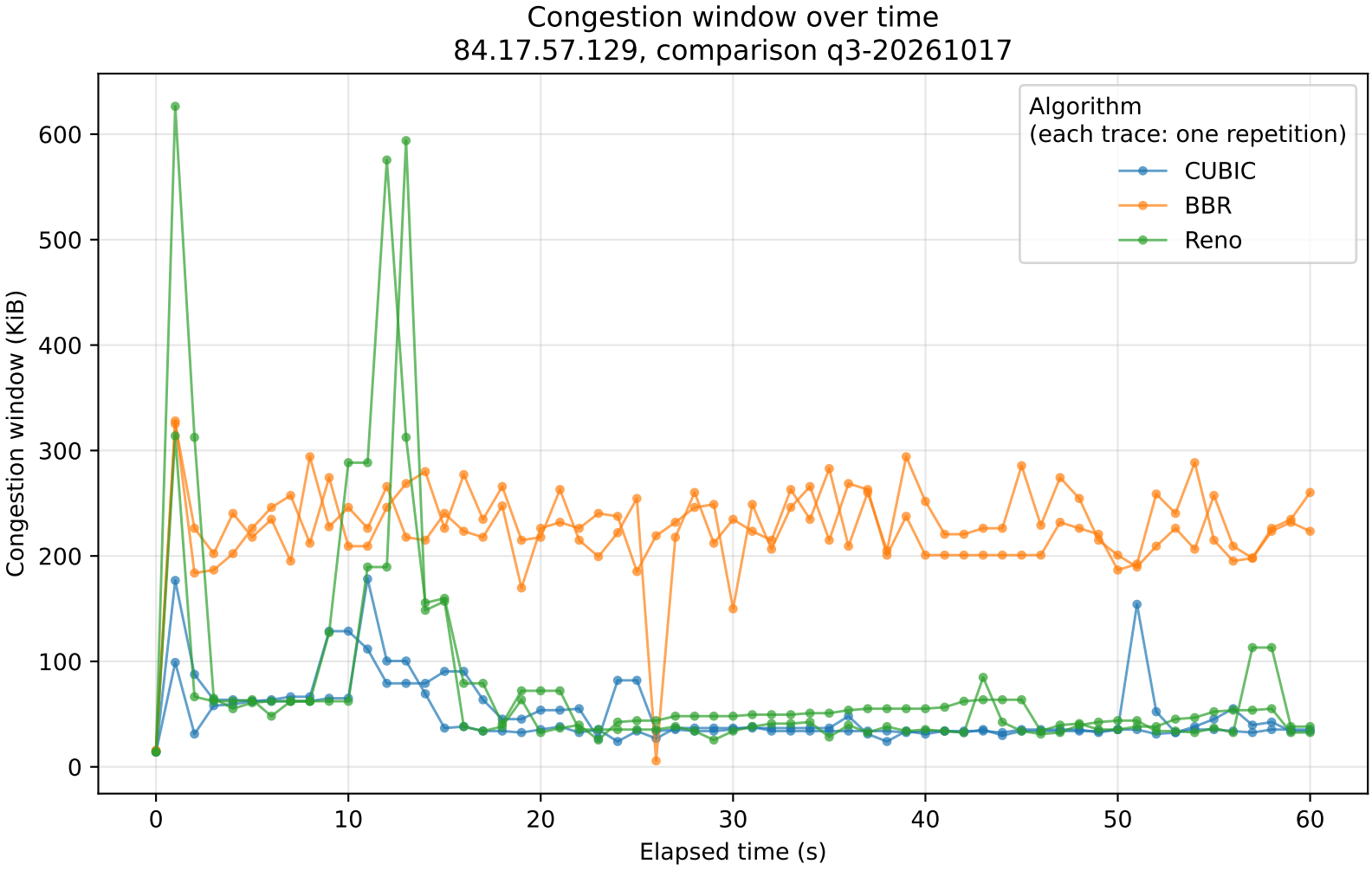

Retransmission proxy over time
la.speedtest.clouvider.net, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
Reno
BBR
Retransmissions (packets)
Elapsed time (s)
0
1
2
3
4
5
0
10
20
30
40
50
60

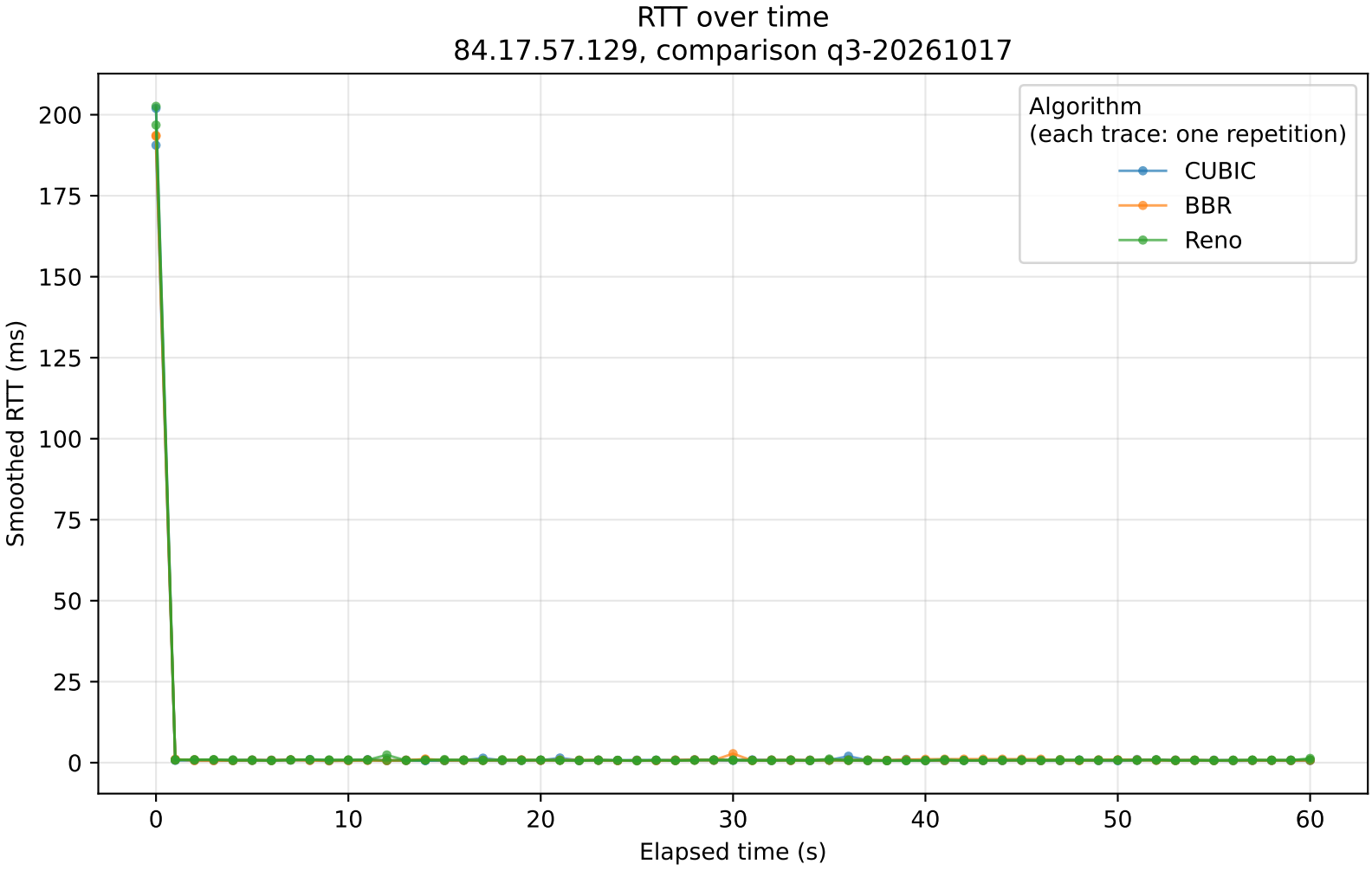

Goodput over time
la.speedtest.clouvider.net, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
Reno
BBR
Goodput (Mbps)
Elapsed time (s)
0
5
10
15
20
25
30
35
0
10
20
30
40
50
60

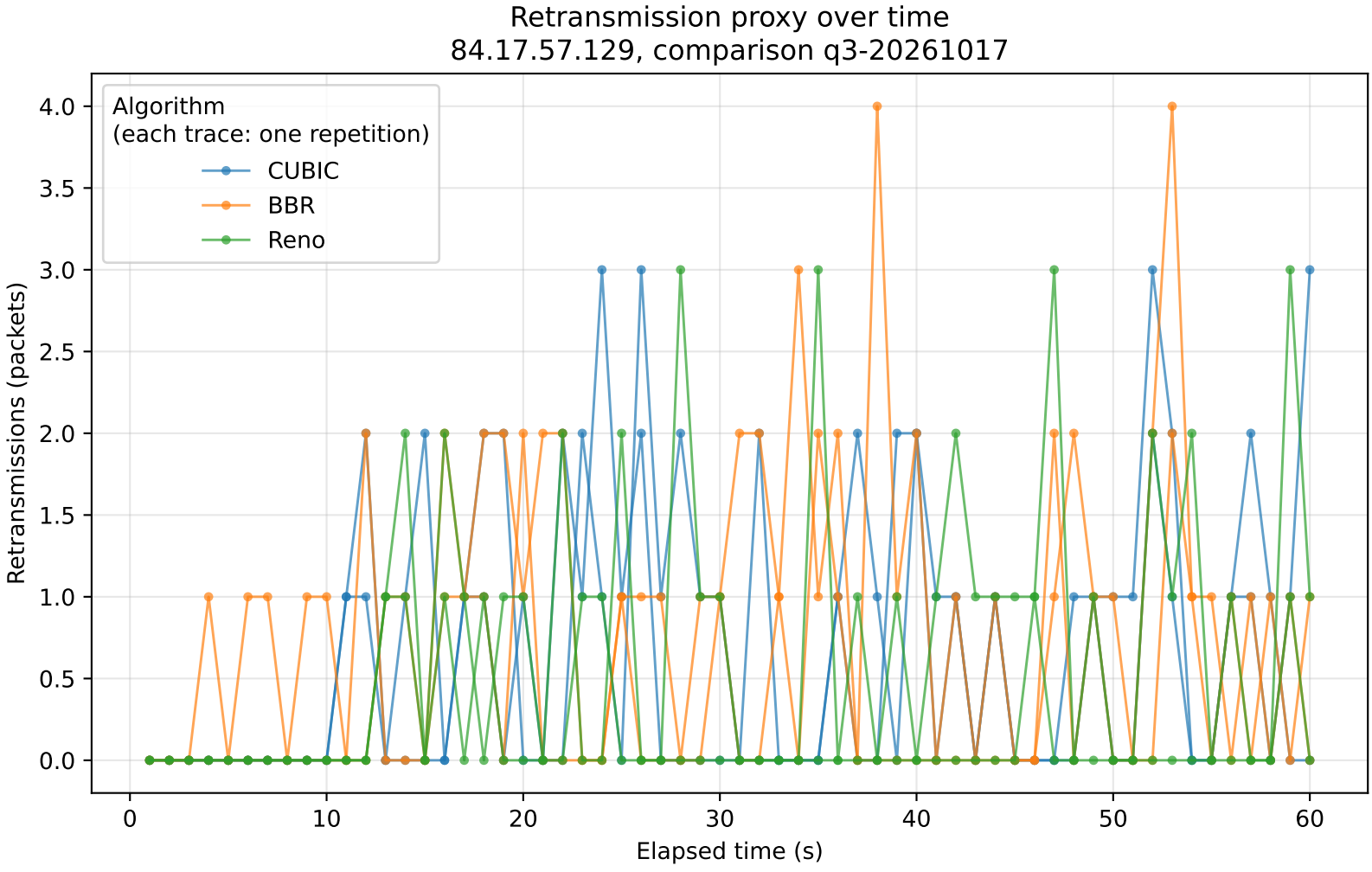

Congestion window vs goodput
la.speedtest.clouvider.net, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
Reno
BBR
Goodput (Mbps)
Congestion window (KiB)
0
5
10
15
20
25
30
35
0
50
100
150
200
250
300
350

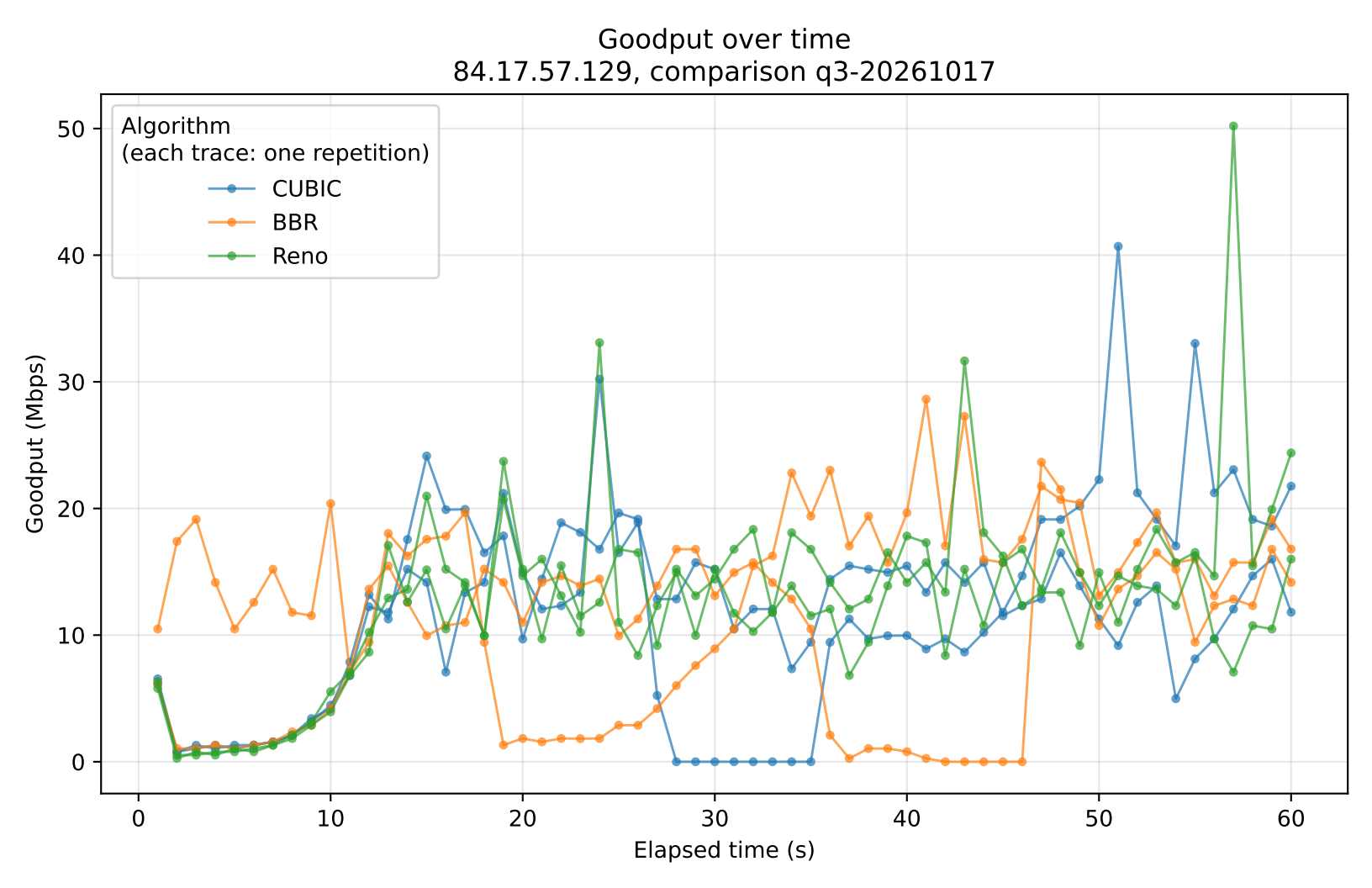

RTT vs goodput
la.speedtest.clouvider.net, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
Reno
BBR
Goodput (Mbps)
Smoothed RTT (ms)
0
5
10
15
20
25
30
35
0.5
1.0
1.5
2.0
2.5

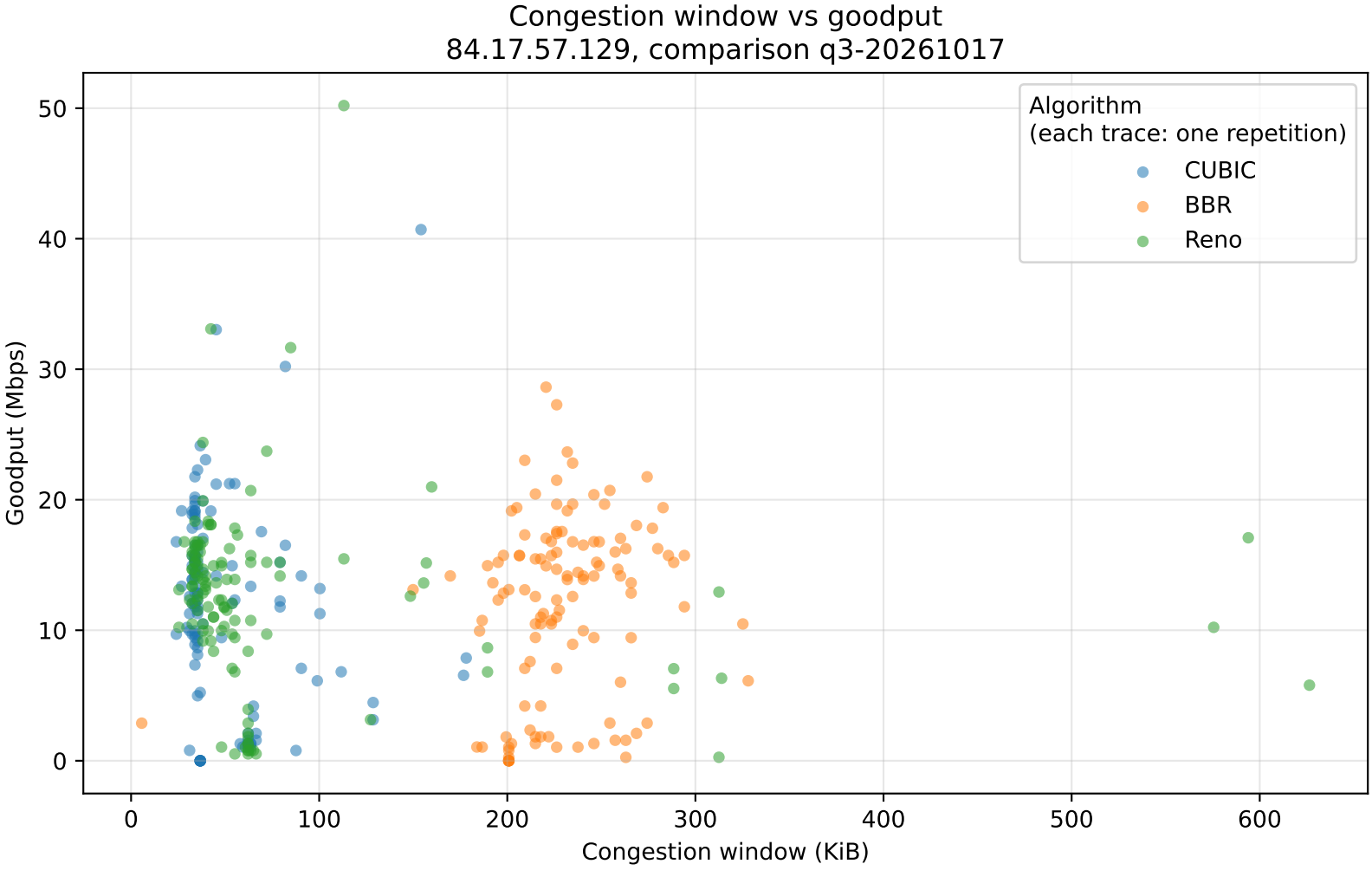

Retransmission proxy vs goodput
la.speedtest.clouvider.net, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
Reno
BBR
Goodput (Mbps)
Retransmissions (packets)
0
5
10
15
20
25
30
35
0
1
2
3
4
5

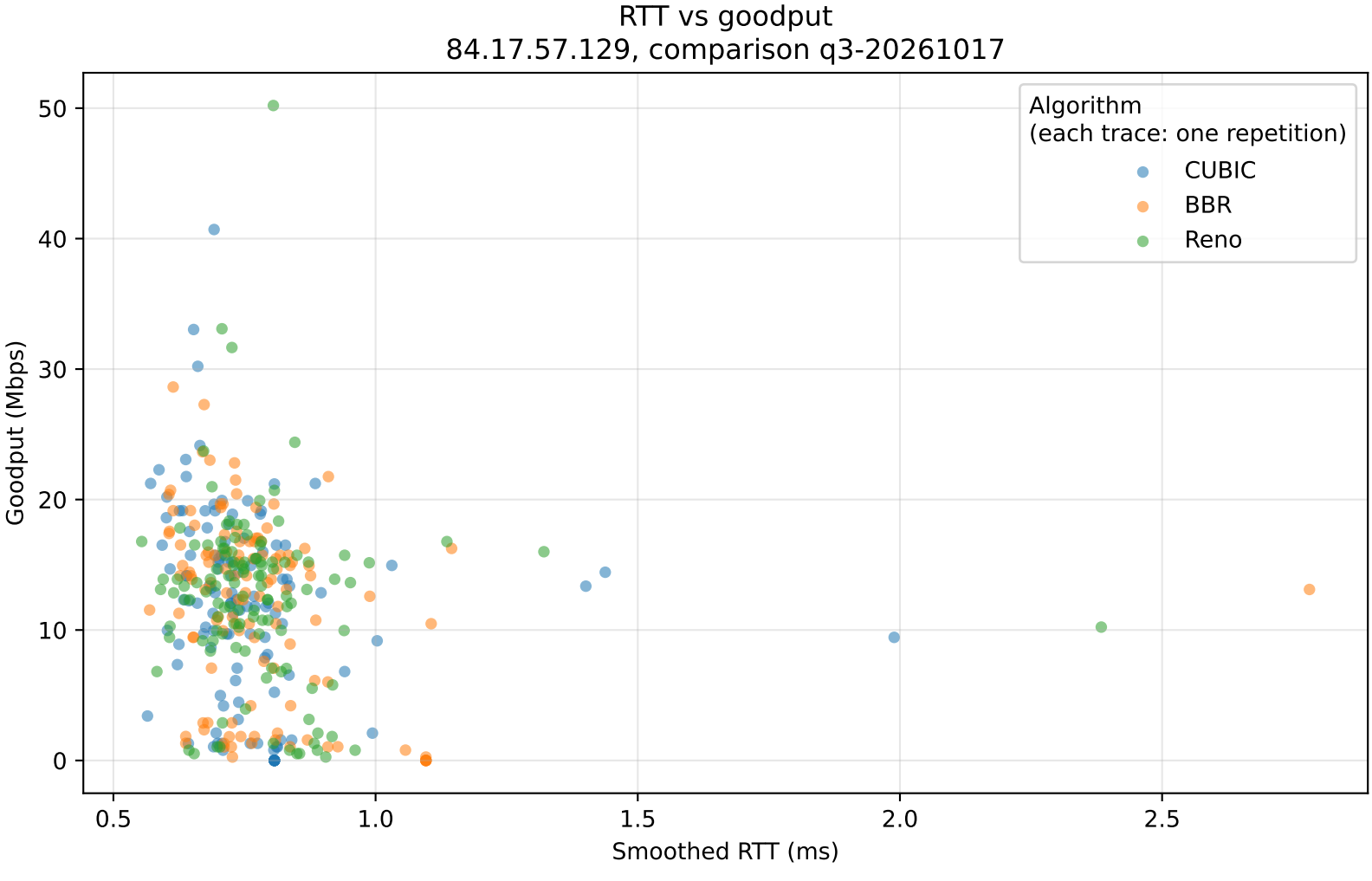

Congestion window over time
84.17.57.129, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
BBR
Reno
Congestion window (KiB)
Elapsed time (s)
0
100
200
300
400
500
600
0
10
20
30
40
50
60

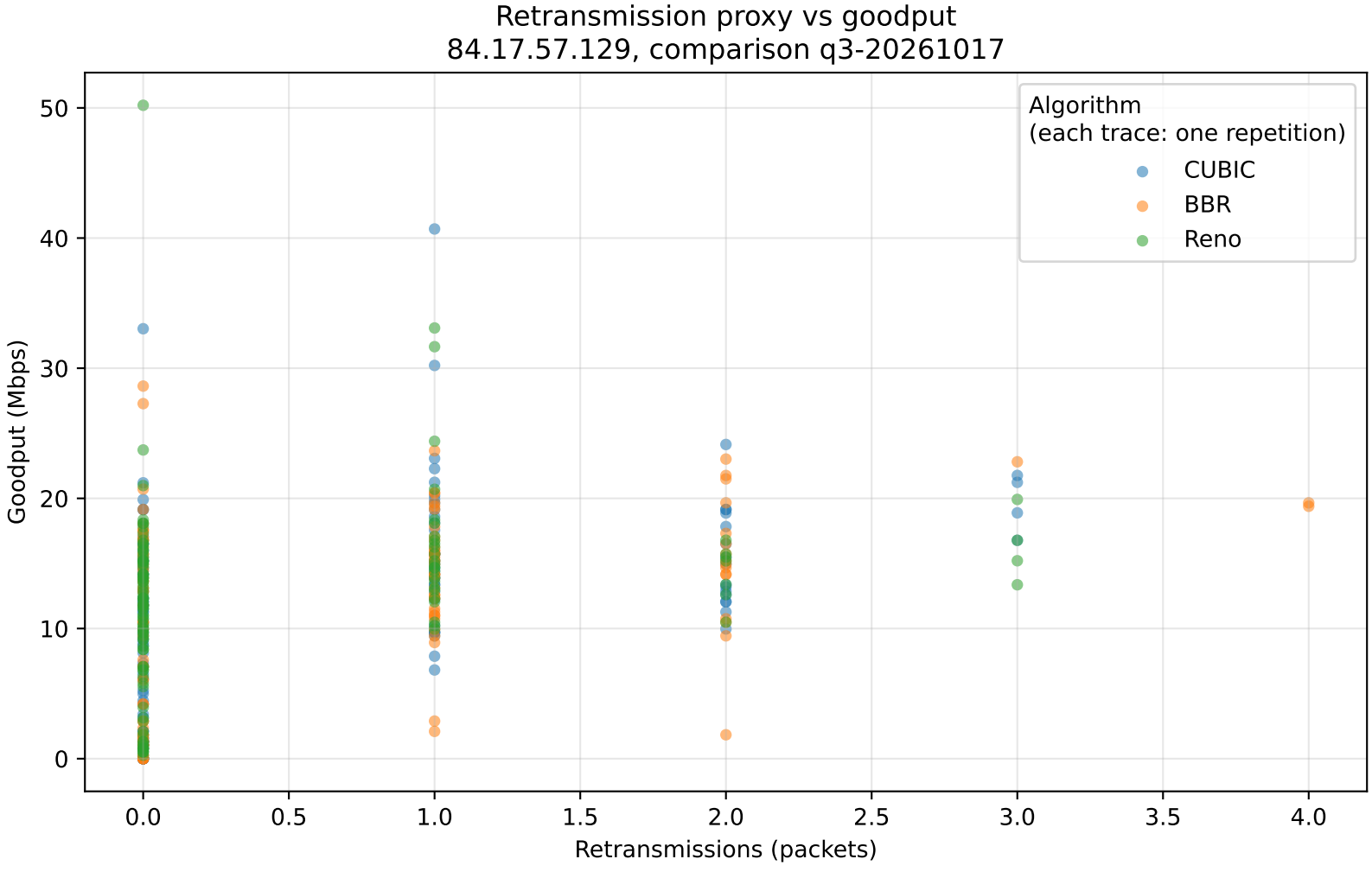

RTT over time
84.17.57.129, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
BBR
Reno
Smoothed RTT (ms)
200
175
150
125
100
75
50
25
0
0
10
20
30
40
50
60
Elapsed time (s)

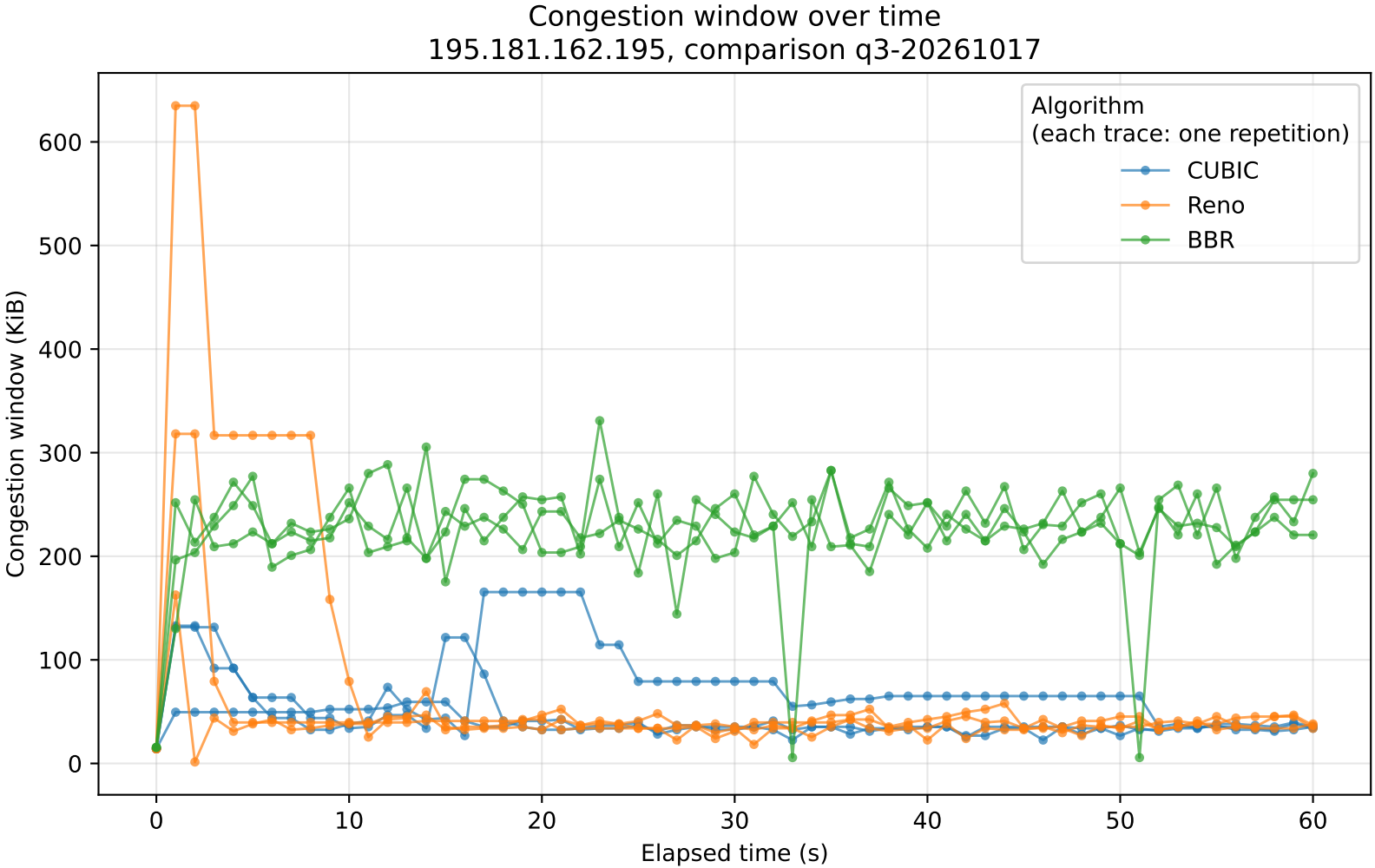

Retransmission proxy over time
84.17.57.129, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
BBR
Reno
Retransmissions (packets)
Elapsed time (s)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0
10
20
30
40
50
60

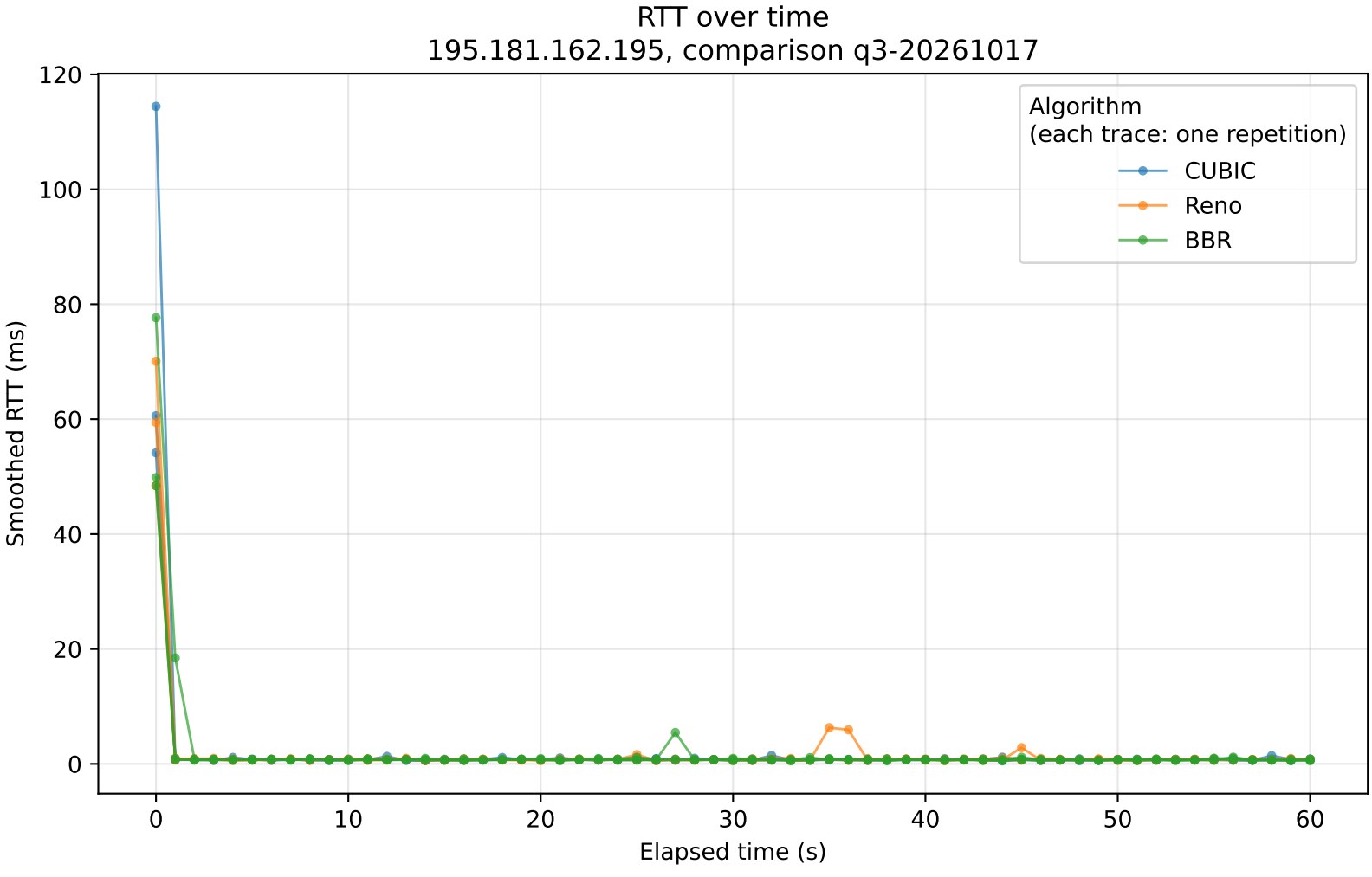

Goodput over time
84.17.57.129, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
BBR
Reno
Goodput (Mbps)
Elapsed time (s)
0
10
20
30
40
50
60

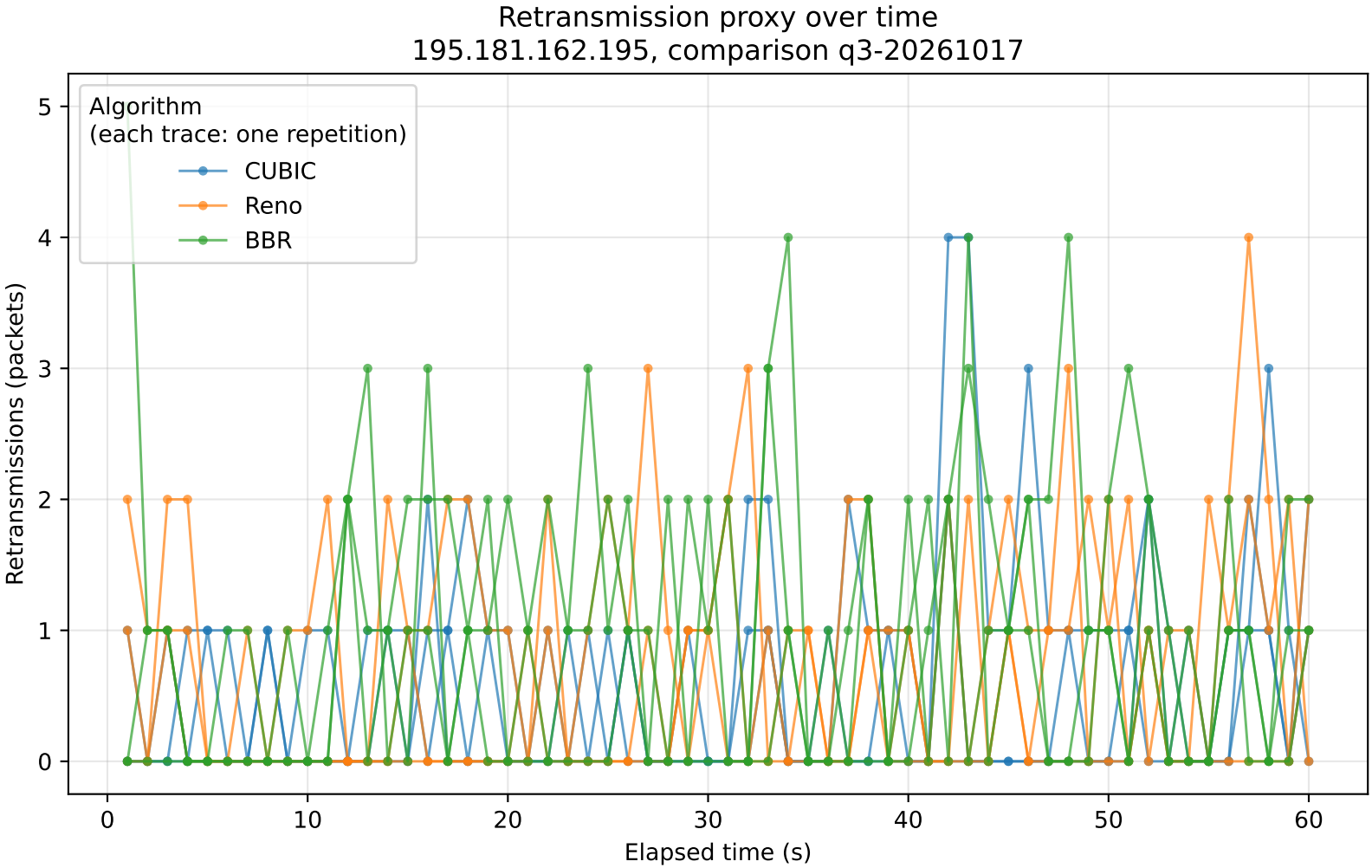

Congestion window vs goodput
84.17.57.129, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
BBR
Reno
Goodput (Mbps)
0
10
20
30
40
50
Congestion window (KiB)
0
100
200
300
400
500
600

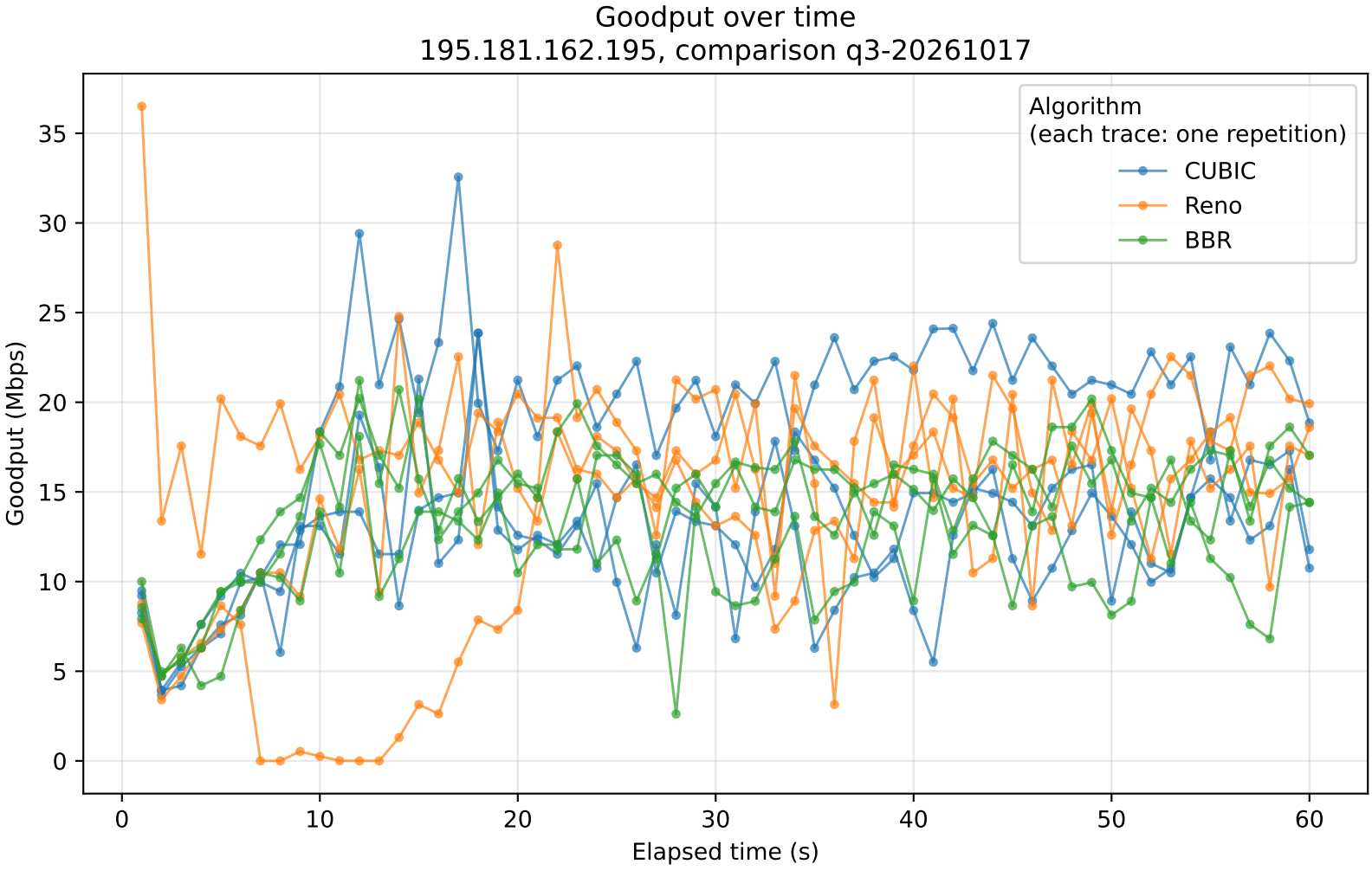

RTT vs goodput
84.17.57.129, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
BBR
Reno
Goodput (Mbps)
0
10
20
30
40
50
0.5
1.0
1.5
2.0
2.5
Smoothed RTT (ms)

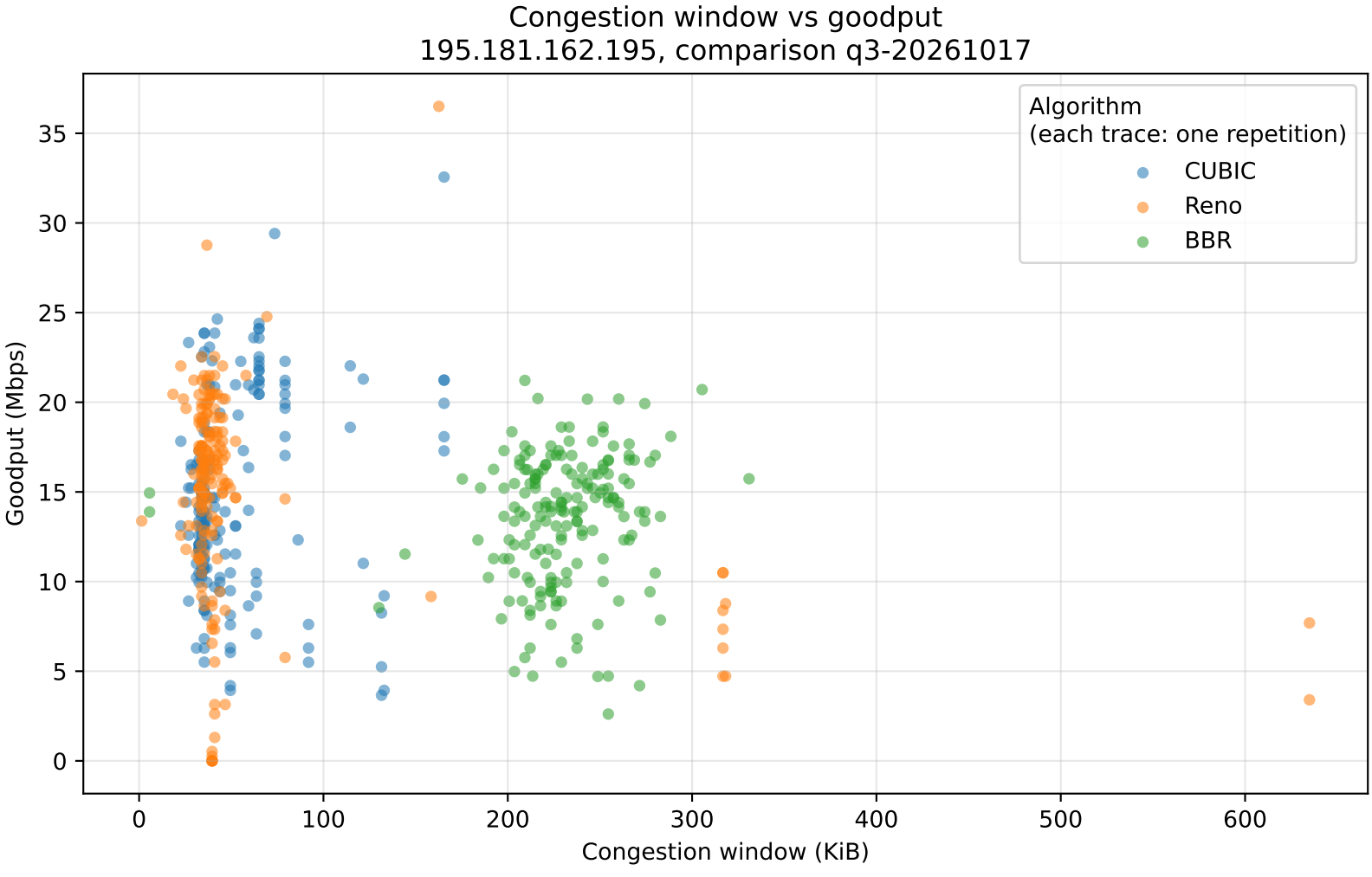

Retransmission proxy vs goodput
84.17.57.129, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
BBR
Reno
Goodput (Mbps)
0
10
20
30
40
50
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Retransmissions (packets)

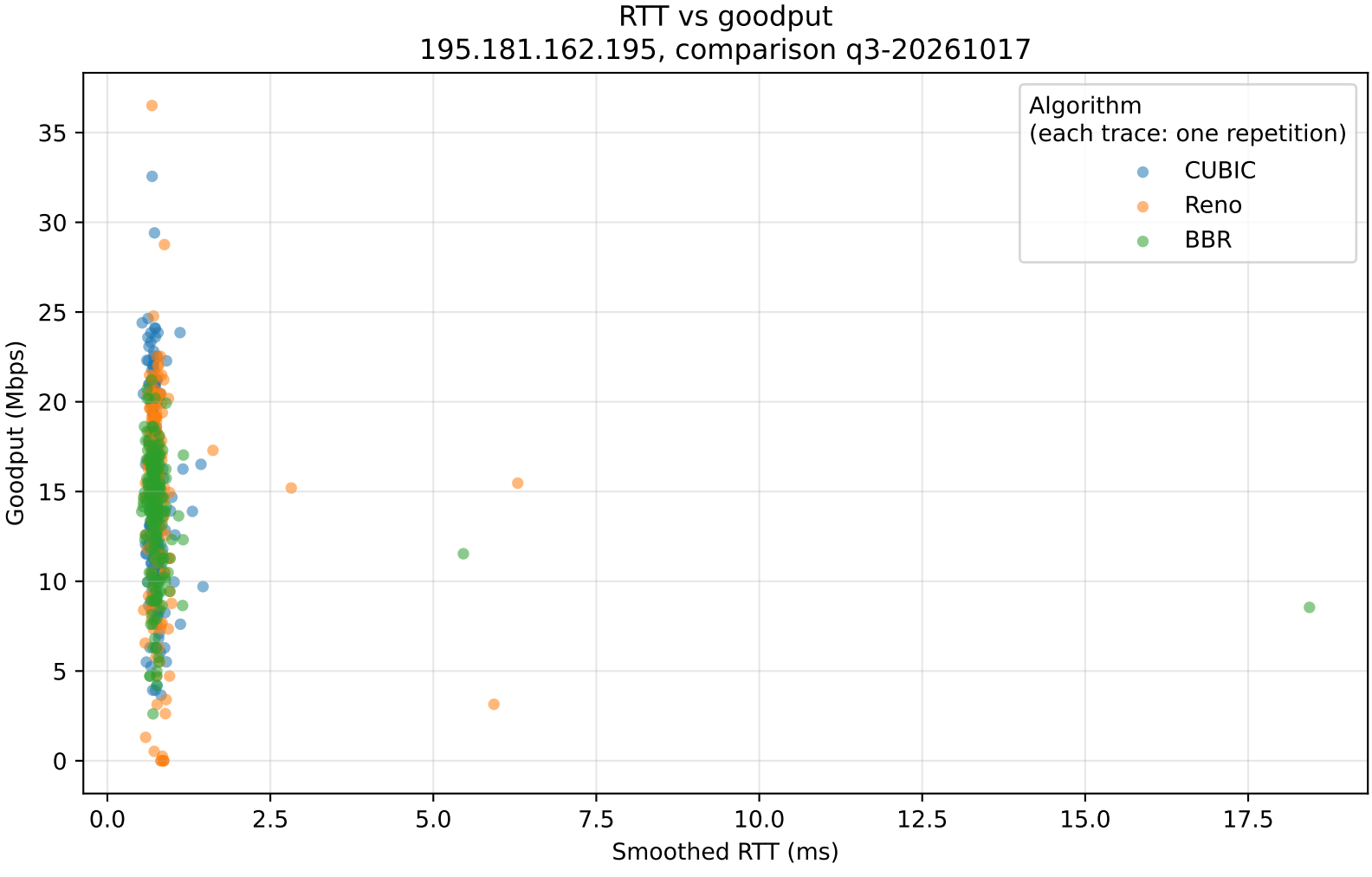

Congestion window over time
195.181.162.195, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
Reno
BBR
Congestion window (KiB)
Elapsed time (s)
0
100
200
300
400
500
600
0
10
20
30
40
50
60

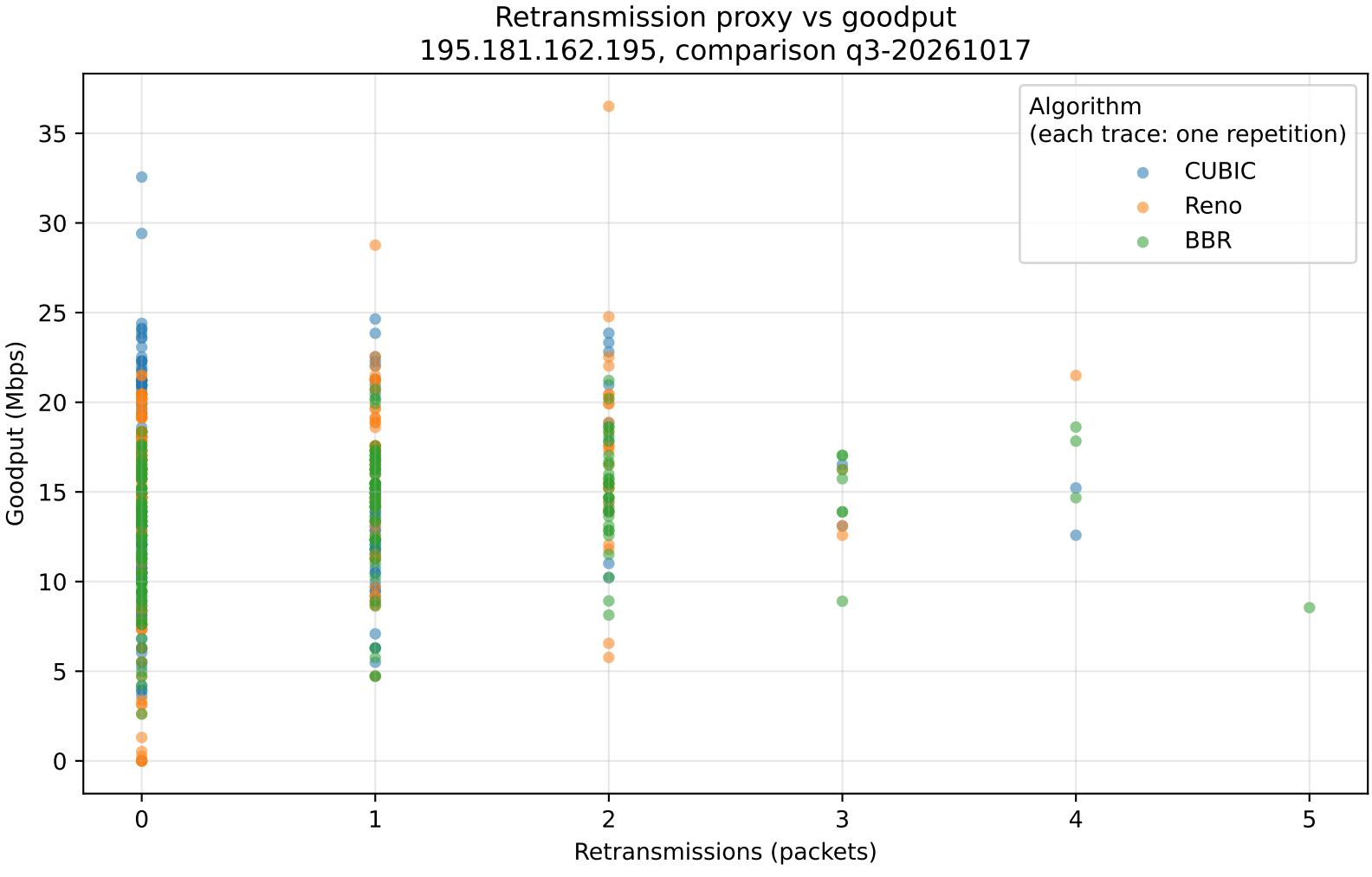

RTT over time
195.181.162.195, comparison q3-20261017
Smoothed RTT (ms)
Elapsed time (s)
Algorithm
(each trace: one repetition)
CUBIC
Reno
BBR
0
20
40
60
80
100
120
0
10
20
30
40
50
60

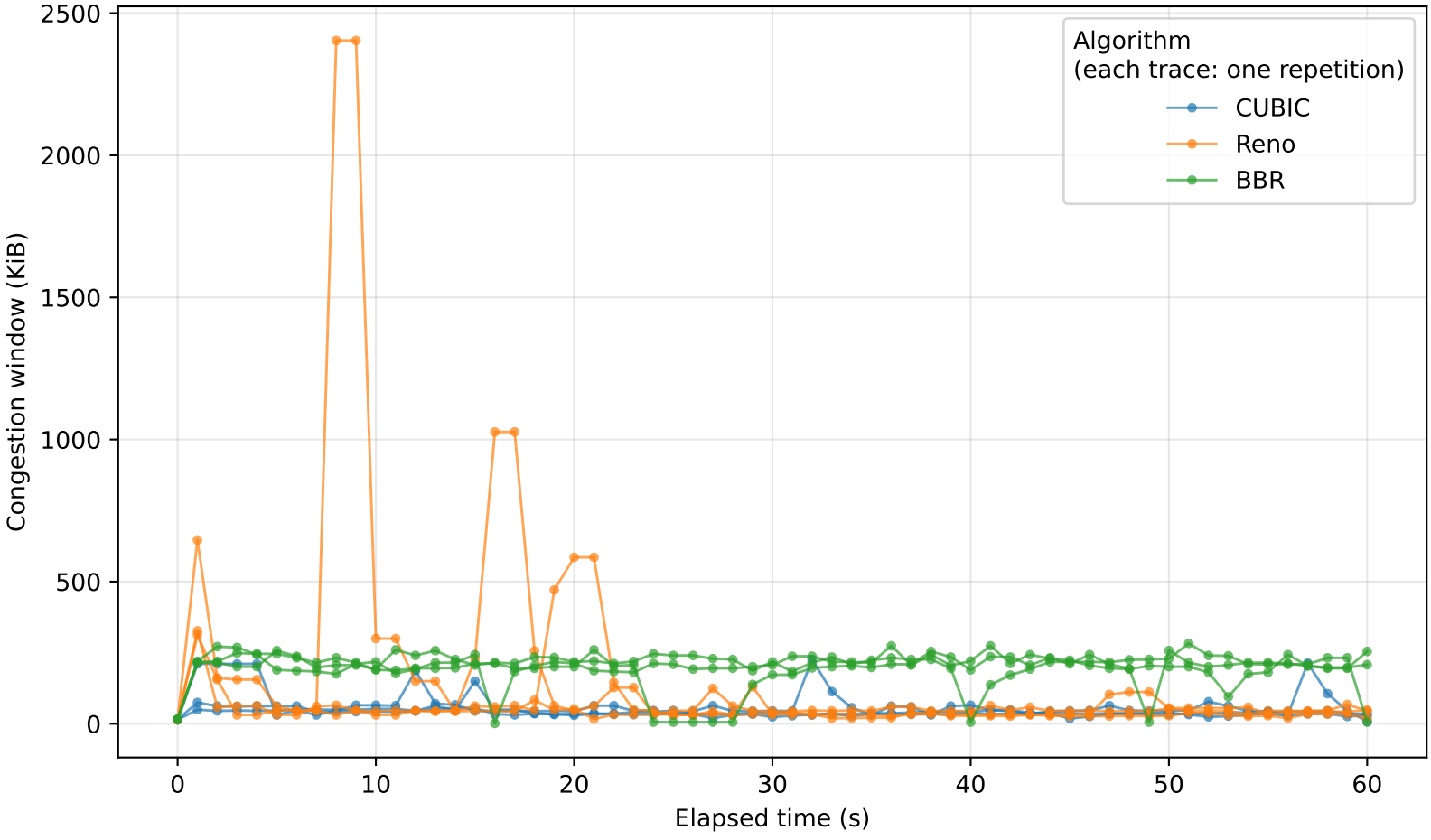

Retransmission proxy over time
195.181.162.195, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
Reno
BBR
Retransmissions (packets)
Elapsed time (s)
0
1
2
3
4
5
0
10
20
30
40
50
60

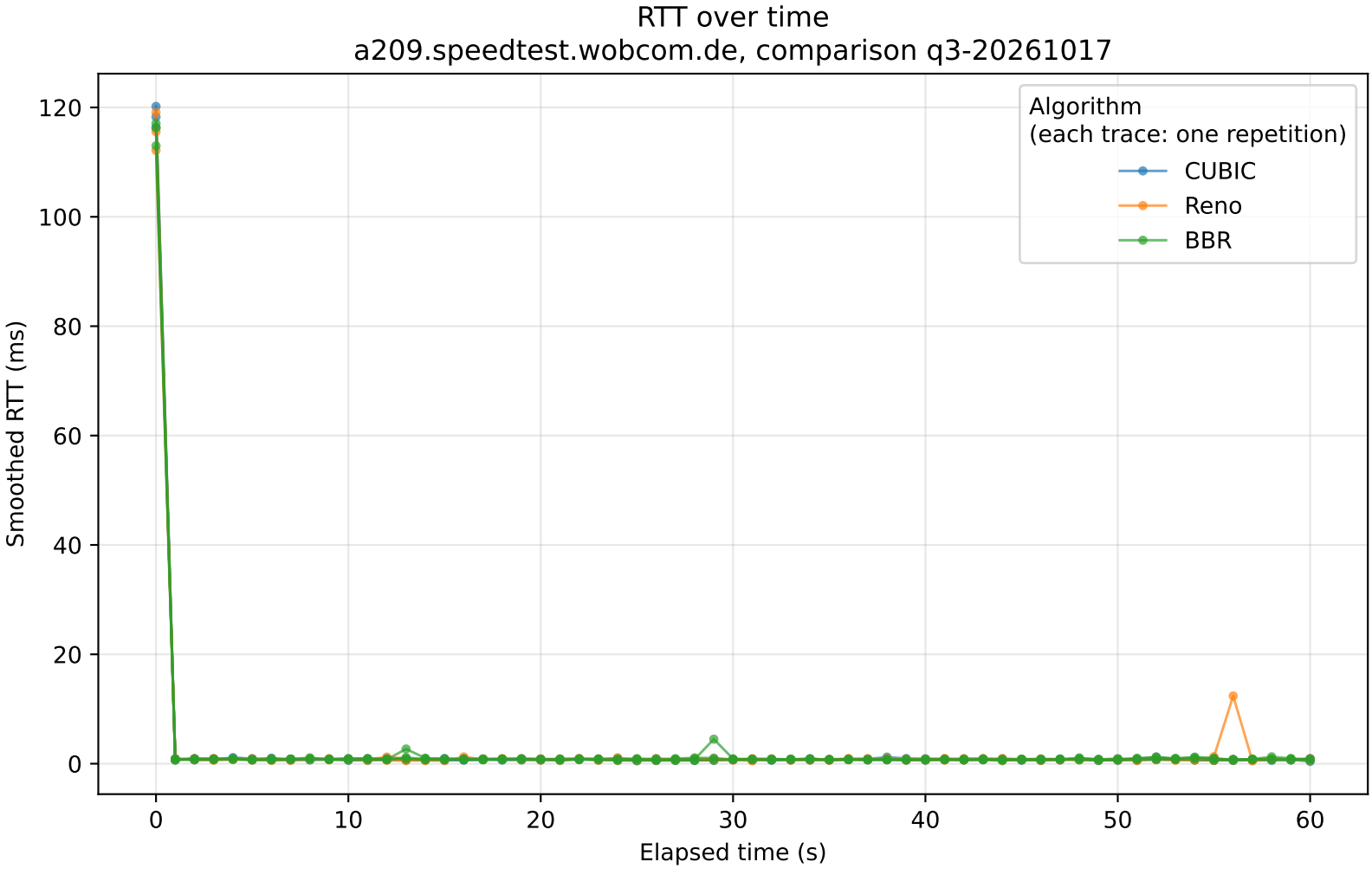

Goodput over time
195.181.162.195, comparison q3-20261017
Goodput (Mbps)
Elapsed time (s)
Algorithm
(each trace: one repetition)
CUBIC
Reno
BBR
0
5
10
15
20
25
30
35
0
10
20
30
40
50
60

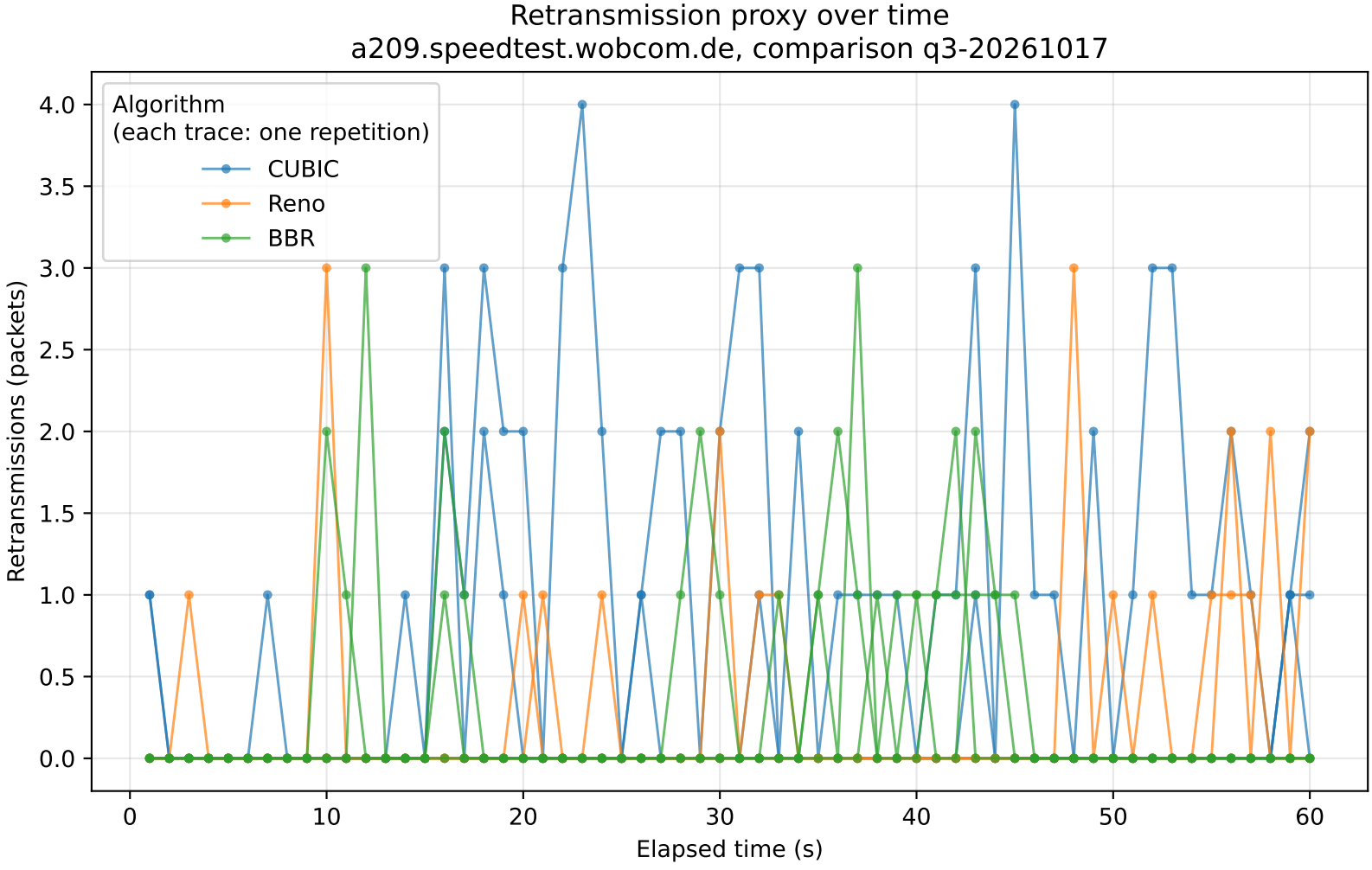

Congestion window vs goodput
195.181.162.195, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
Reno
BBR
Goodput (Mbps)
Congestion window (KiB)
0
5
10
15
20
25
30
35
0
100
200
300
400
500
600

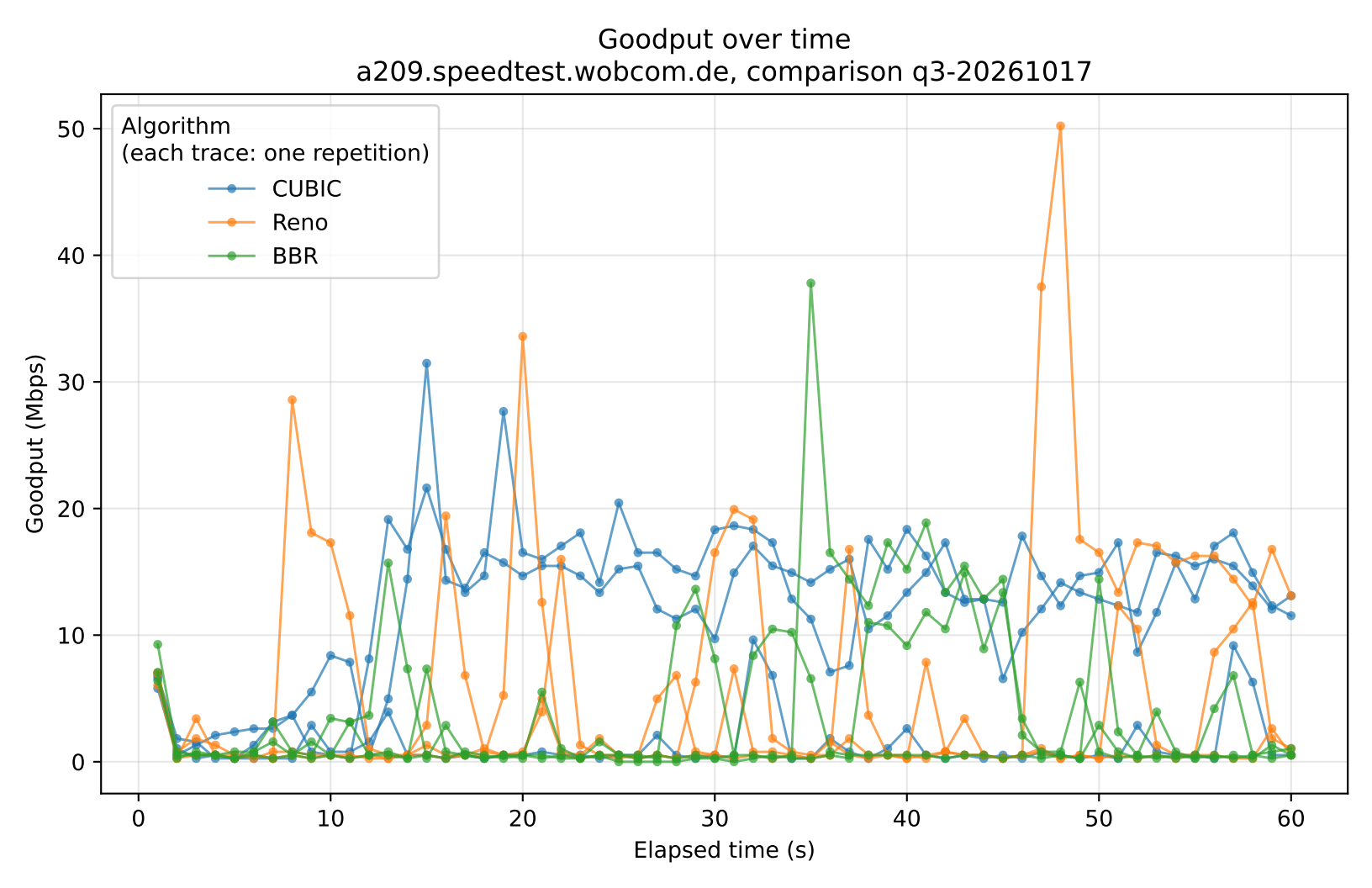

RTT vs goodput
195.181.162.195, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
Reno
BBR
Goodput (Mbps)
0
5
10
15
20
25
30
35
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Smoothed RTT (ms)

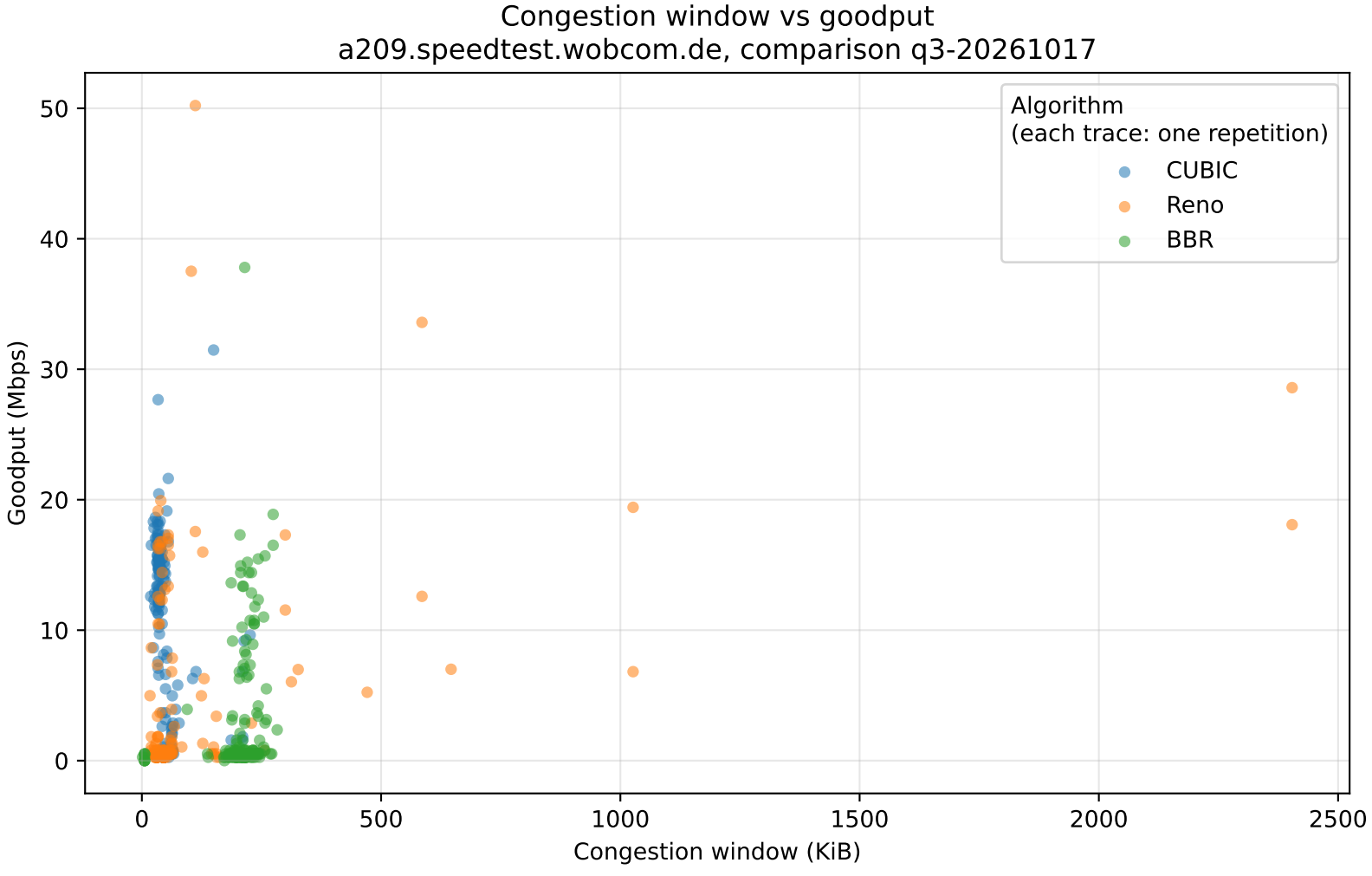

Retransmission proxy vs goodput
195.181.162.195, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
Reno
BBR
Goodput (Mbps)
0
5
10
15
20
25
30
35
0
1
2
3
4
5
Retransmissions (packets)

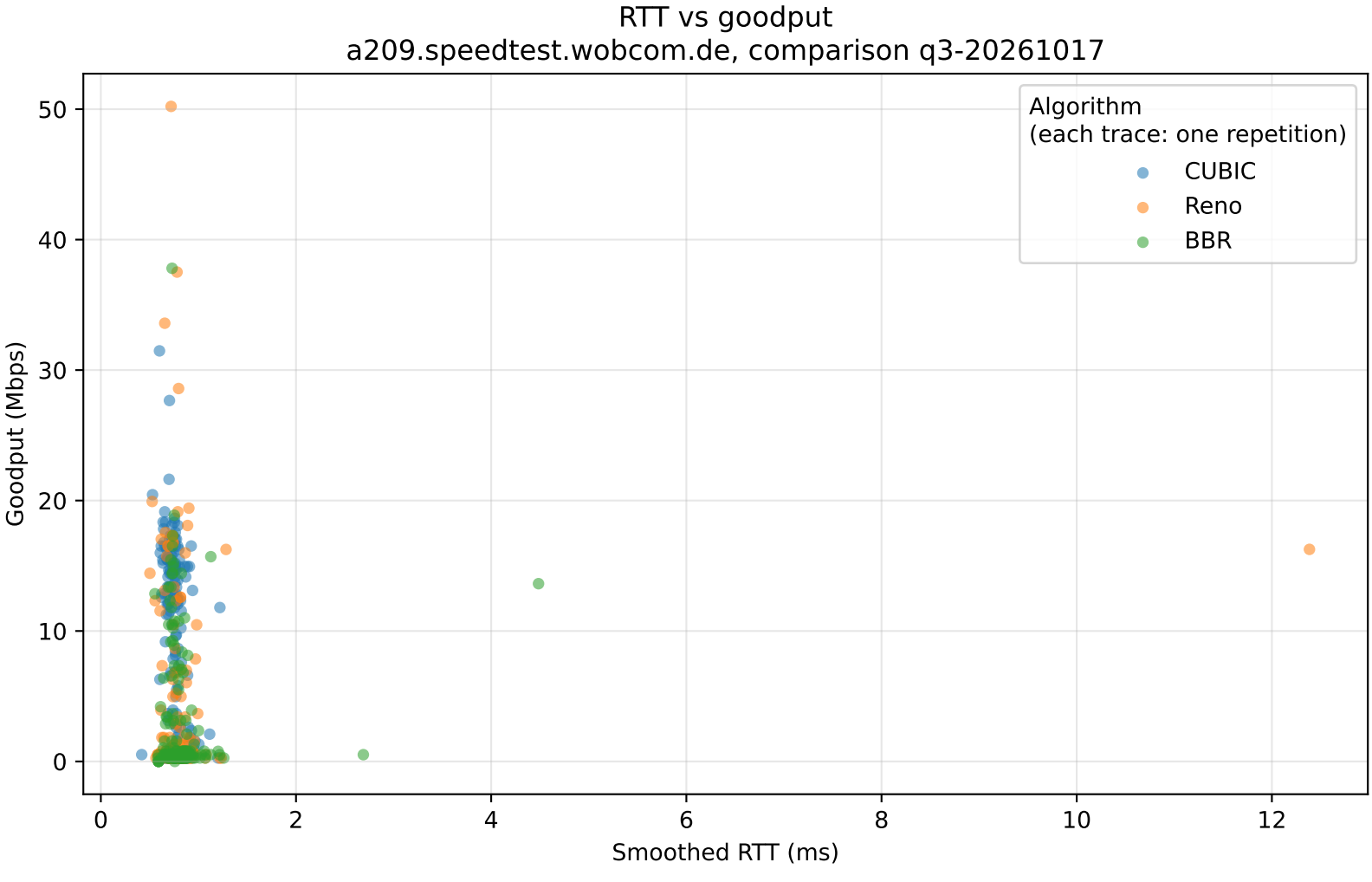

Congestion window over time
a209.speedtest.wobcom.de, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
Reno
BBR
Congestion window (KiB)
2500
2000
1500
1000
500
0
0
10
20
30
40
50
60
Elapsed time (s)

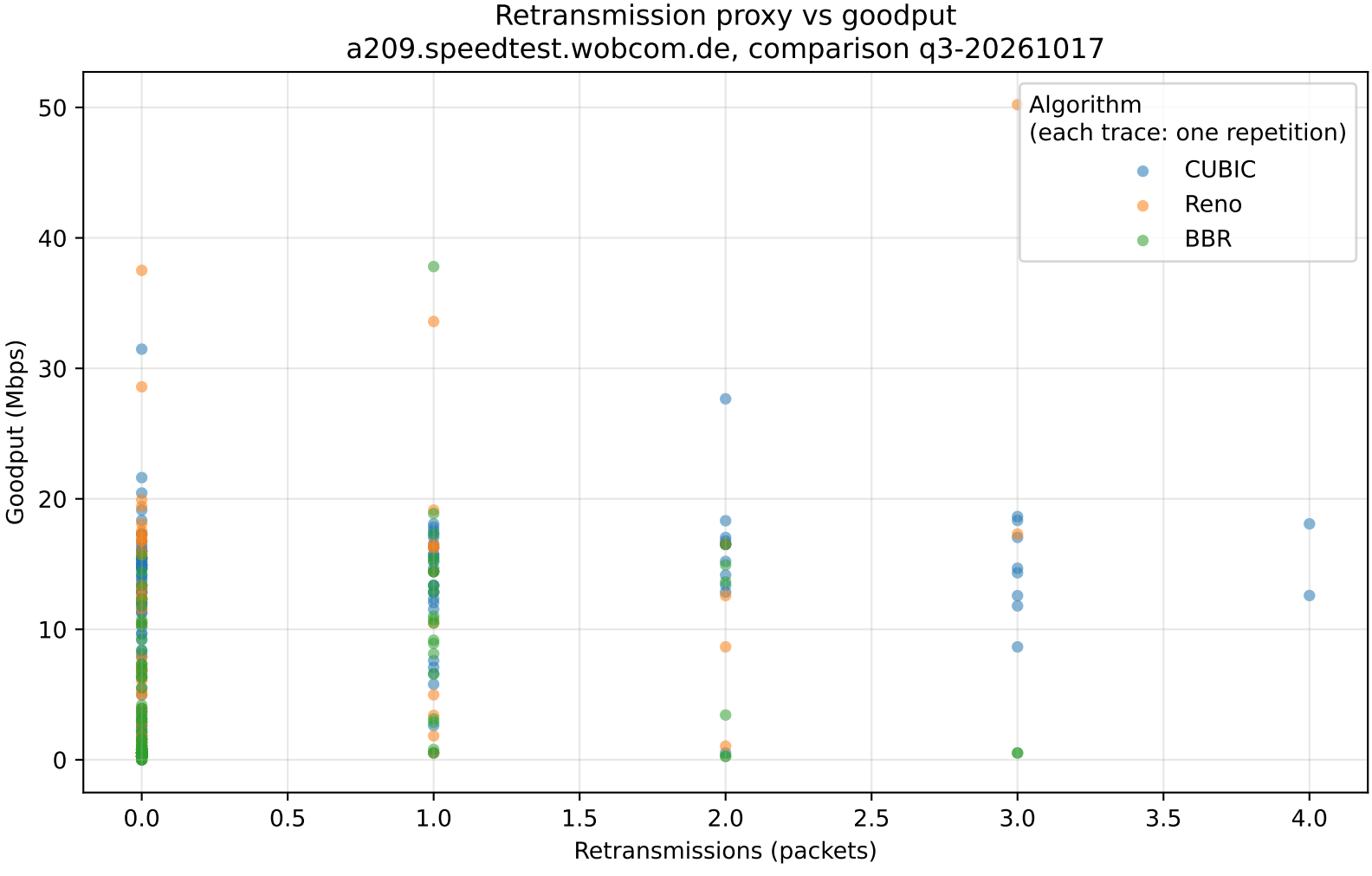

RTT over time
a209.speedtest.wobcom.de, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
Reno
BBR
Smoothed RTT (ms)
Elapsed time (s)
0
20
40
60
80
100
120
0
10
20
30
40
50
60

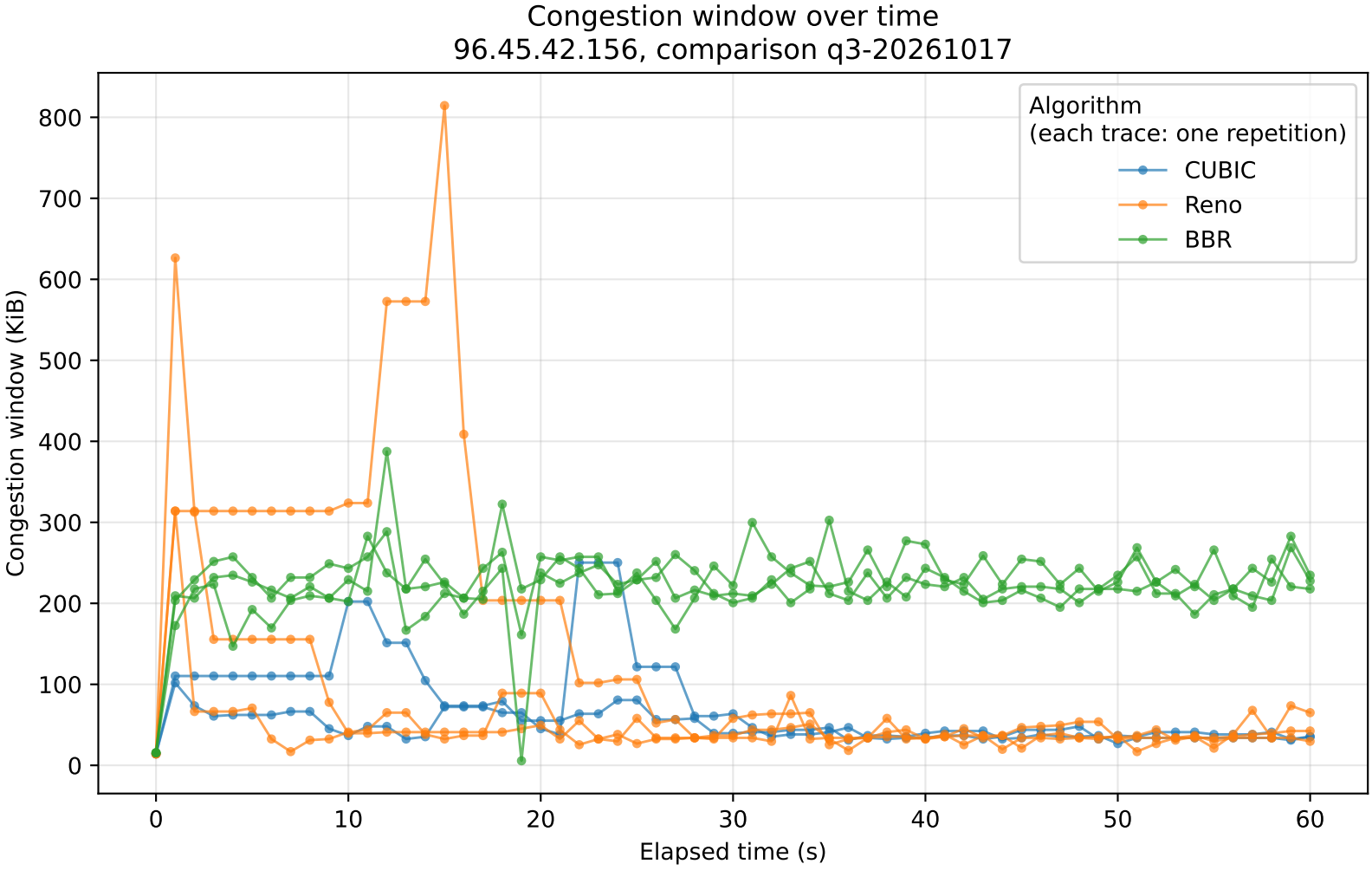

Retransmission proxy over time
a209.speedtest.wobcom.de, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
Reno
BBR
Retransmissions (packets)
Elapsed time (s)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0
10
20
30
40
50
60

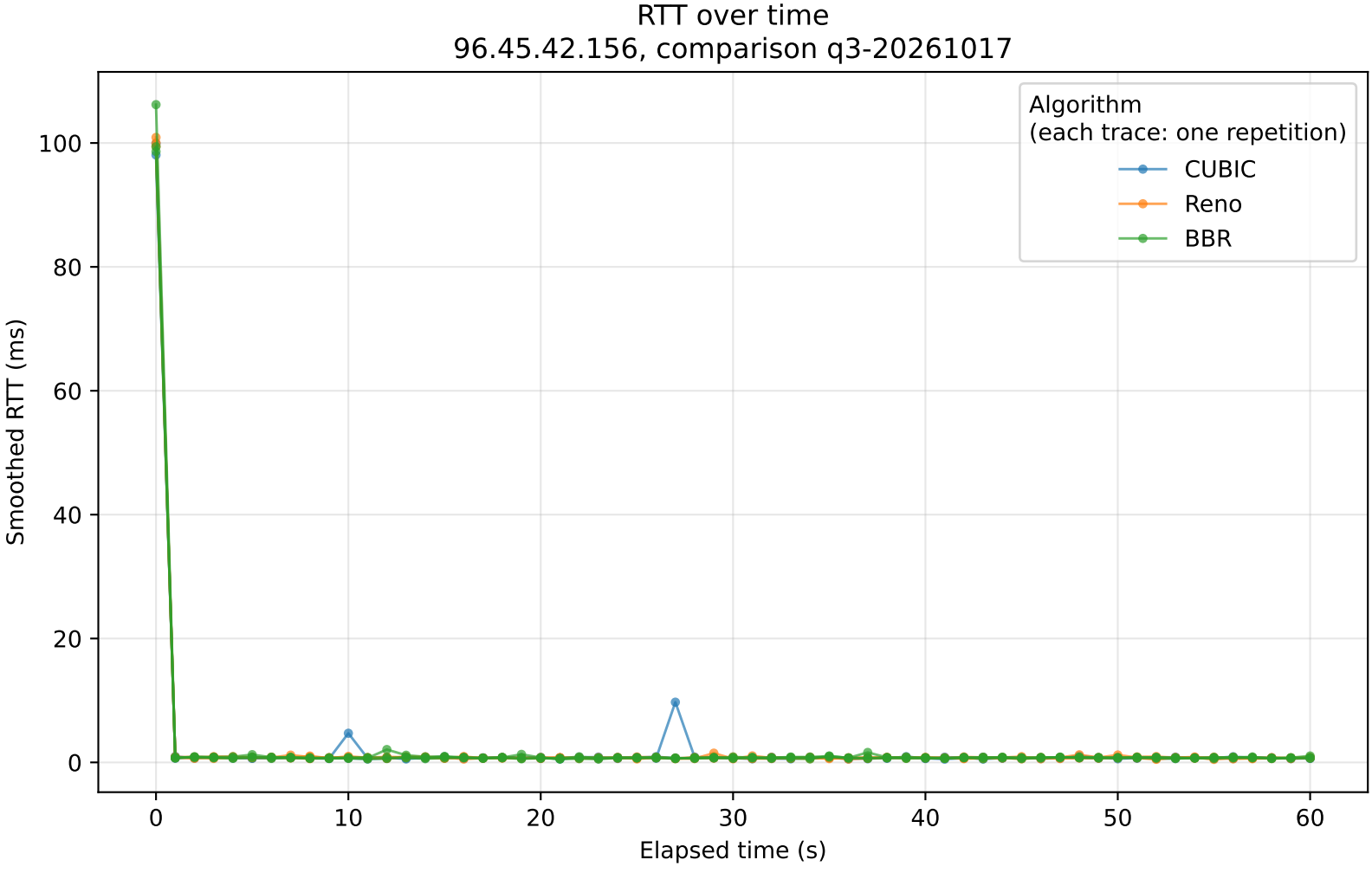

Goodput over time
a209.speedtest.wobcom.de, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
Reno
BBR
Goodput (Mbps)
Elapsed time (s)
0
10
20
30
40
50
60

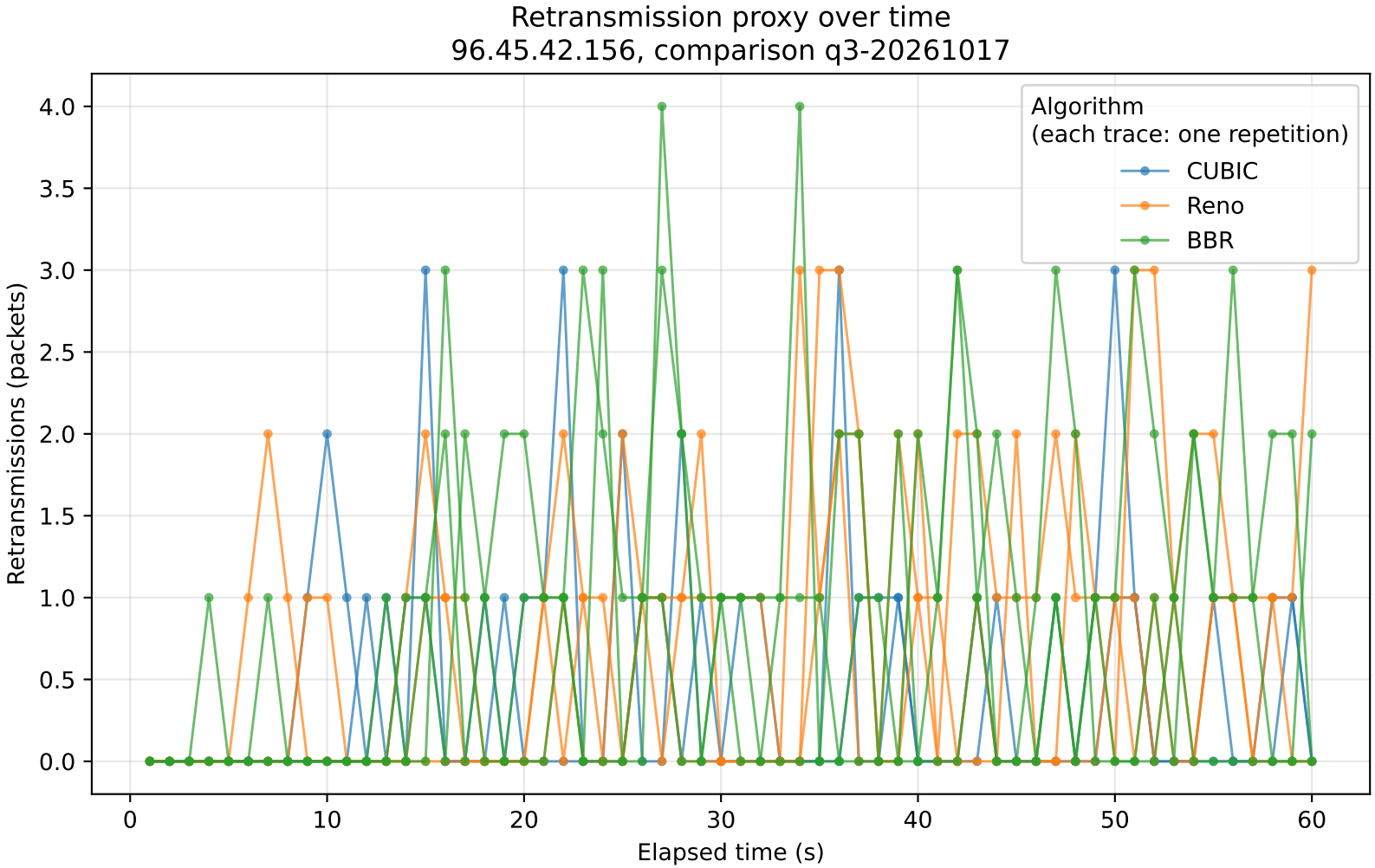

Congestion window vs goodput
a209.speedtest.wobcom.de, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
Reno
BBR
Goodput (Mbps)
0
10
20
30
40
50
0
500
1000
1500
2000
2500
Congestion window (KiB)

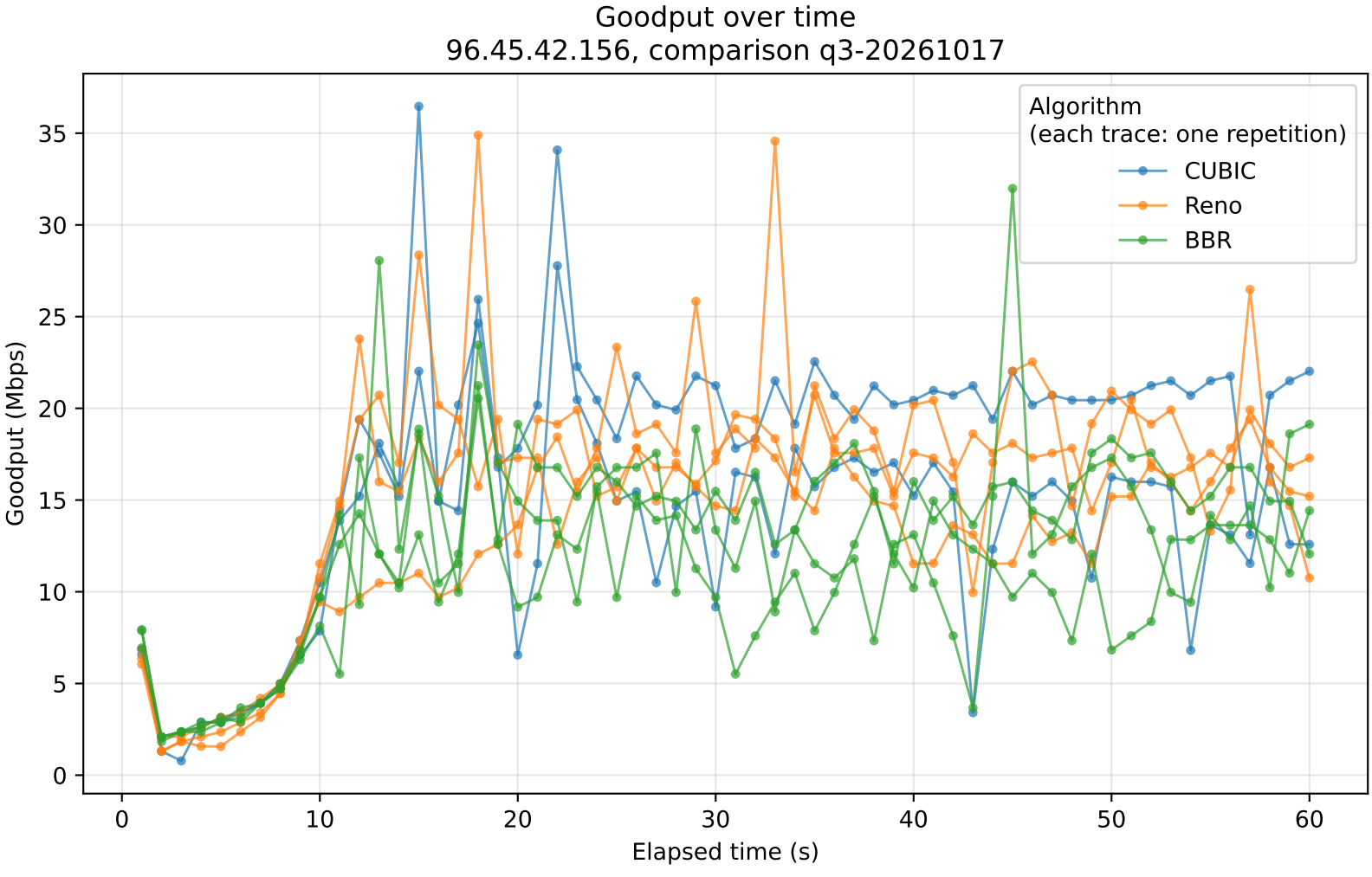

RTT vs goodput
a209.speedtest.wobcom.de, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
Reno
BBR
Goodput (Mbps)
Smoothed RTT (ms)
0
10
20
30
40
50
0
2
4
6
8
10
12

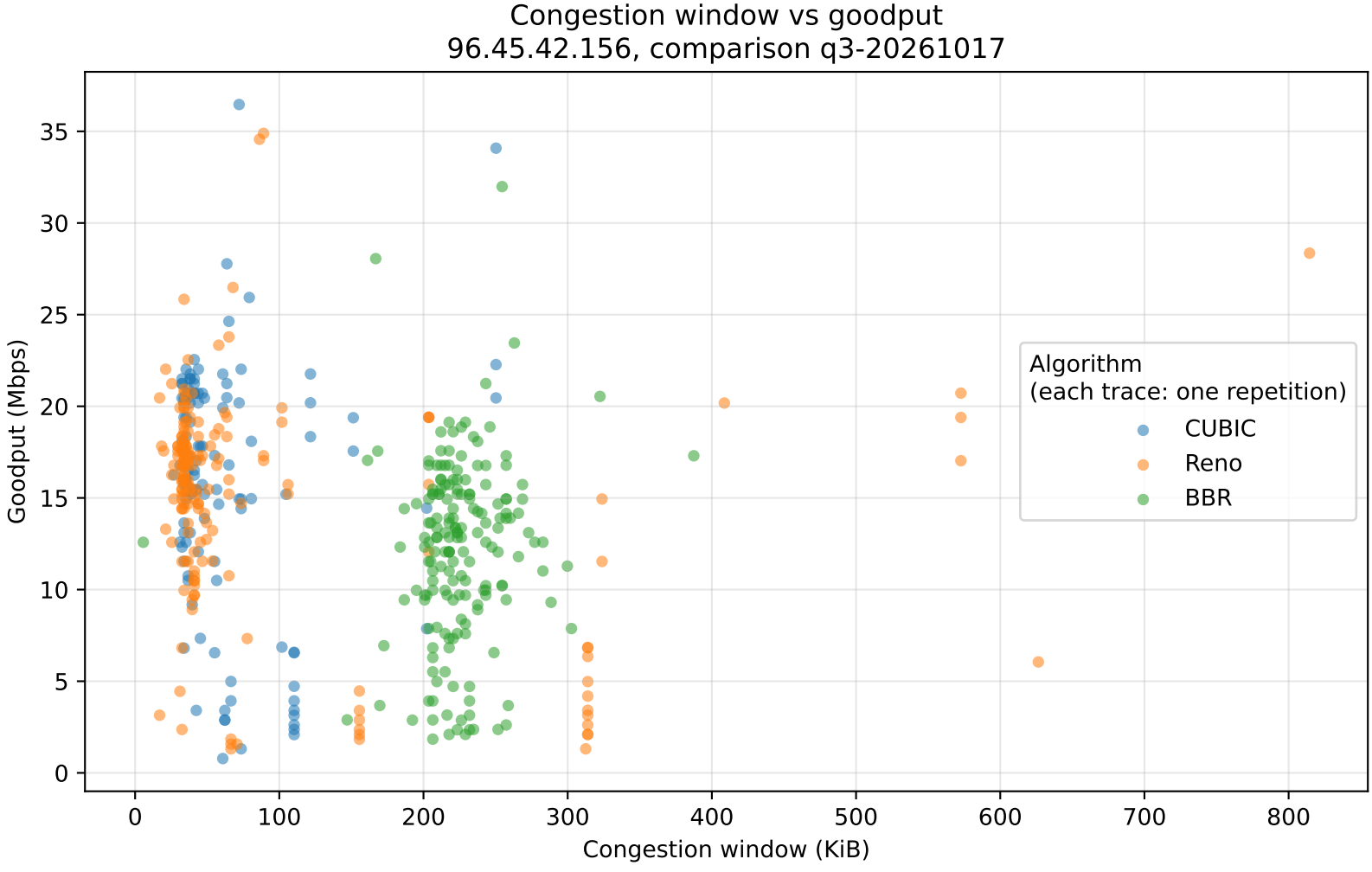

Retransmission proxy vs goodput
a209.speedtest.wobcom.de, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
Reno
BBR
Goodput (Mbps)
0
10
20
30
40
50
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Retransmissions (packets)

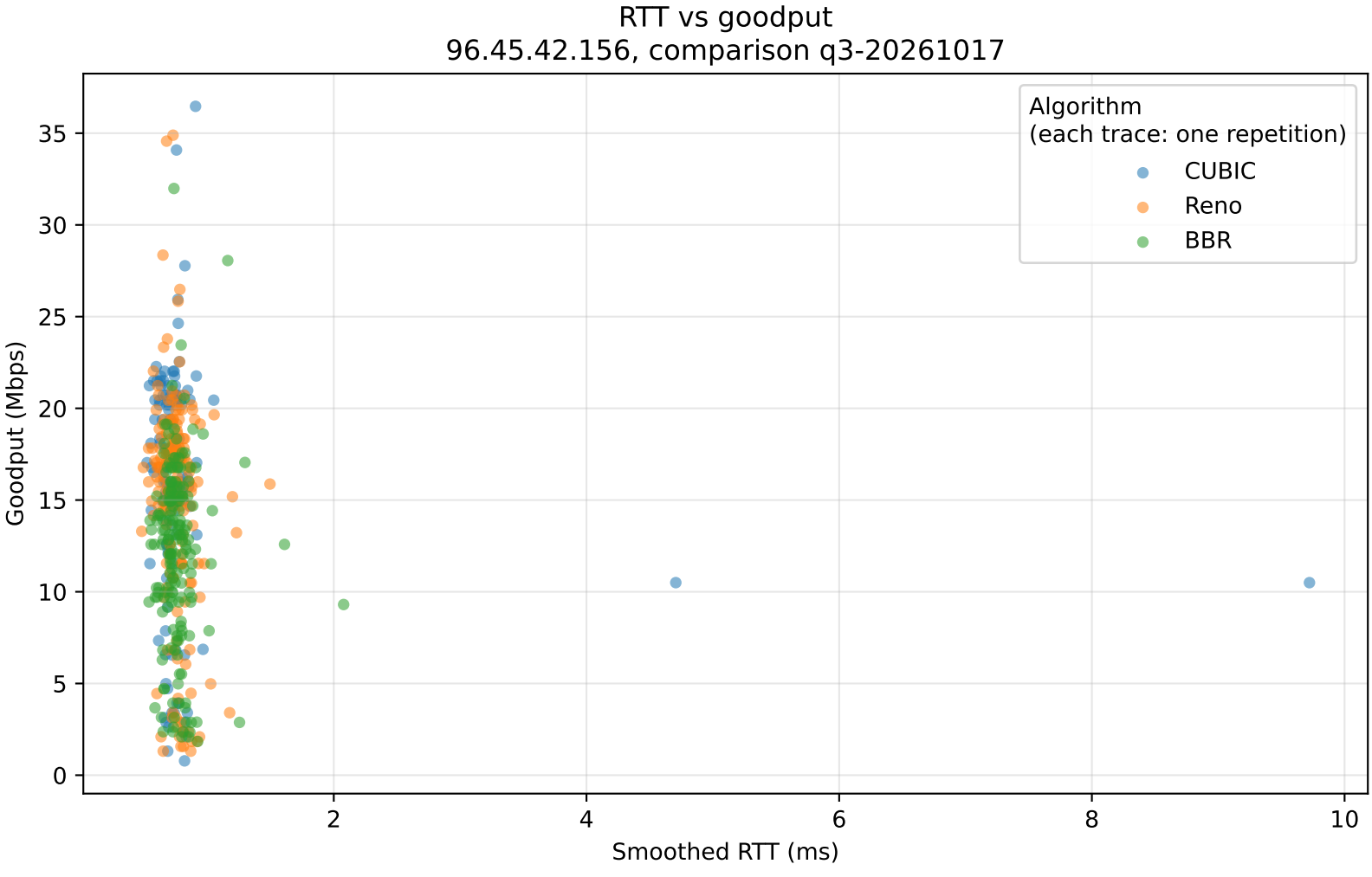

Congestion window over time
96.45.42.156, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
Reno
BBR
Congestion window (KiB)
Elapsed time (s)
0
100
200
300
400
500
600
700
800
0
10
20
30
40
50
60

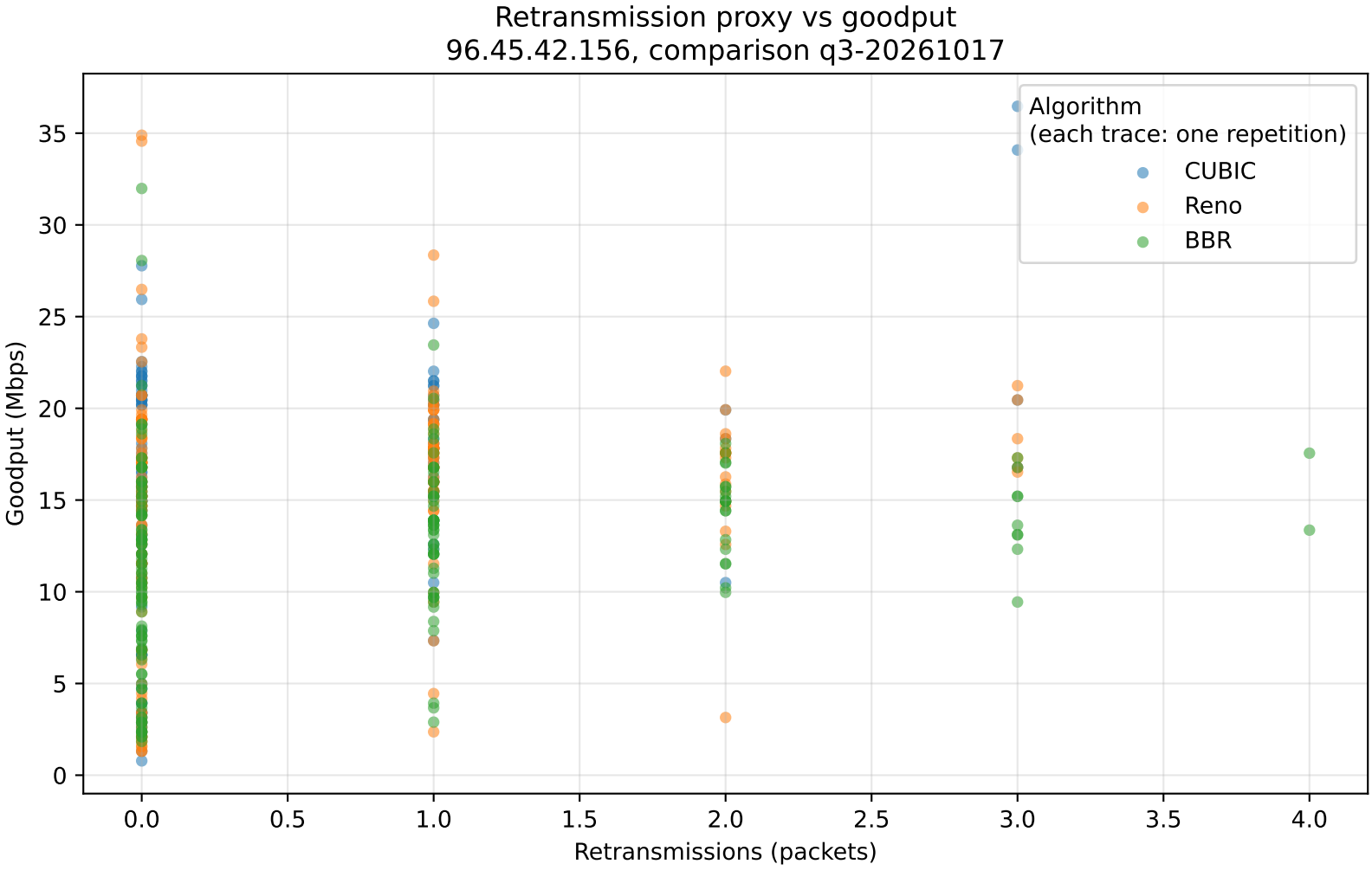

RTT over time
96.45.42.156, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
Reno
BBR
Smoothed RTT (ms)
0
20
40
60
80
100
0
10
20
30
40
50
60
Elapsed time (s)


Retransmission proxy over time
96.45.42.156, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
Reno
BBR
Retransmissions (packets)
Elapsed time (s)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0
10
20
30
40
50
60


Goodput over time
96.45.42.156, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
Reno
BBR
Goodput (Mbps)
Elapsed time (s)
0
5
10
15
20
25
30
35
0
10
20
30
40
50
60


Congestion window vs goodput
96.45.42.156, comparison q3-20261017
Goodput (Mbps)
Congestion window (KiB)
0
5
10
15
20
25
30
35
0
100
200
300
400
500
600
700
800
Algorithm
(each trace: one repetition)
CUBIC
Reno
BBR


RTT vs goodput
96.45.42.156, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
Reno
BBR
Goodput (Mbps)
0
5
10
15
20
25
30
35
Smoothed RTT (ms)
2
4
6
8
10


Retransmission proxy vs goodput
96.45.42.156, comparison q3-20261017
Algorithm
(each trace: one repetition)
CUBIC
Reno
BBR
Goodput (Mbps)
0
5
10
15
20
25
30
35
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Retransmissions (packets)